

Topological Energy Condensate Theory (TECT): 22

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I. DIRECT FINITE EXTRACTION OF Δ_{00} AND L_*

II. SHORT KOREAN EXPLANATION

At the current stage, the carrier-side longitudinal problem is a literal finite matrix-element calculation. The enriched shell carrier is already fixed as

$$\mathcal{H}_D^{(N)} = \text{span}\{|+, 1\rangle, |+, 2\rangle, |-, 1\rangle, |-, 2\rangle\},$$

and the projected core block has the explicit valley form

$$H_{\text{core}}^{(N)}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix}.$$

Therefore the remaining longitudinal audit reduces to:

1. compute the two 2×2 Hermitian matrices $E_\pm(0)$,
2. diagonalize them and extract the normalized low eigenvectors $\eta_\pm$,
3. compute

$$\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_-, \quad \Sigma_3 = \text{diag}(1, -1),$$

4. compute the shell-normal reduced J -matrix

$$U_{\parallel} := \hat{n}^j \mathcal{U}_j(0),$$

5. extract

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}), \quad L_* = m_{\parallel} u_{\parallel}.$$

So the remaining frontier is finite and explicit.

III. 1. EXPLICIT 2×2 VALLEY BLOCKS

Write the two valley-diagonal projected matrices as general Hermitian matrices:

$$E_+(0) = \begin{pmatrix} a_+ & c_+ \\ \bar{c}_+ & b_+ \end{pmatrix}, \quad E_-(0) = \begin{pmatrix} a_- & c_- \\ \bar{c}_- & b_- \end{pmatrix}, \quad (1)$$

with

$$a_{\pm}, b_{\pm} \in \mathbb{R}, \quad c_{\pm} \in \mathbb{C}.$$

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Define the trace/difference data

$$t_{\pm} := \frac{a_{\pm} + b_{\pm}}{2}, \quad d_{\pm} := \frac{a_{\pm} - b_{\pm}}{2}, \quad \rho_{\pm} := |c_{\pm}|. \quad (2)$$

Then

$$E_{\pm}(0) = t_{\pm} \mathbf{1}_2 + d_{\pm} \Sigma_3 + (\Re c_{\pm}) \Sigma_1 - (\Im c_{\pm}) \Sigma_2. \quad (3)$$

Its eigenvalues are

$$\varepsilon_{\pm}^{(\text{low/high})} = t_{\pm} \mp R_{\pm}, \quad R_{\pm} := \sqrt{d_{\pm}^2 + \rho_{\pm}^2}. \quad (4)$$

The low internal carrier state is therefore the normalized eigenvector of $\varepsilon_{\pm}^{(\text{low})} = t_{\pm} - R_{\pm}$.

IV. 2. EXPLICIT NORMALIZED LOW EIGENVECTORS

Let

$$c_{\pm} = \rho_{\pm} e^{i\phi_{\pm}}, \quad \rho_{\pm} \geq 0. \quad (5)$$

Introduce the Bloch-sphere angle $\theta_{\pm}$ by

$$\cos \theta_{\pm} = \frac{d_{\pm}}{R_{\pm}}, \quad \sin \theta_{\pm} = \frac{\rho_{\pm}}{R_{\pm}}, \quad 0 \leq \theta_{\pm} \leq \pi. \quad (6)$$

A convenient normalized low eigenvector choice is

$$\eta_{\pm} = \begin{pmatrix} -e^{-i\phi_{\pm}} \sin(\theta_{\pm}/2) \\ \cos(\theta_{\pm}/2) \end{pmatrix}, \quad (7)$$

up to an irrelevant overall phase. Equivalently, in component notation,

$$\eta_{\pm} = \begin{pmatrix} A_1^{(\pm)} \\ A_2^{(\pm)} \end{pmatrix}, \quad |A_1^{(\pm)}|^2 + |A_2^{(\pm)}|^2 = 1. \quad (8)$$

If one wants a formula directly from the matrix entries without spherical angles, one may also use

$$r_{\pm} := -\frac{c_{\pm}}{a_{\pm} - \varepsilon_{\pm}^{(\text{low})}} = -\frac{c_{\pm}}{d_{\pm} + R_{\pm}}, \quad (9)$$

whenever $d_{\pm} + R_{\pm} \neq 0$, and then

$$\eta_{\pm} = \frac{1}{\sqrt{1 + |r_{\pm}|^2}} \begin{pmatrix} 1 \\ r_{\pm} \end{pmatrix}. \quad (10)$$

V. 3. CLOSED FORMULA FOR Δ_{00}

By definition,

$$\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_- = \overline{A_1^{(+)}} A_1^{(-)} - \overline{A_2^{(+)}} A_2^{(-)}. \quad (11)$$

Using (7), this becomes

$$\Delta_{00} = e^{i(\phi_+ - \phi_-)} \sin \frac{\theta_+}{2} \sin \frac{\theta_-}{2} - \cos \frac{\theta_+}{2} \cos \frac{\theta_-}{2}. \quad (12)$$

Equivalently, with the ratio form (10),

$$\Delta_{00} = \frac{1 - \bar{r}_+ r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}}. \quad (13)$$

Therefore the exact vanishing condition is

$$\boxed{\Delta_{00} = 0 \iff \bar{r}_+ r_- = 1.} \quad (14)$$

This is very useful: Δ_{00} vanishes only when the two low valley eigenvectors satisfy a very specific matching relation. So unless an exact symmetry enforces $\bar{r}_+ r_- = 1$, the generic expectation is

$$\boxed{\Delta_{00} \neq 0.} \quad (15)$$

VI. 4. REDUCED SHELL-NORMAL J -MATRIX AND $u_{\parallel}$

Define

$$U_{\parallel} := \hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}, \quad \mathcal{U}_{j,ab}(0) = \langle -, a | \hat{J}_j(G_*) | +, b \rangle. \quad (16)$$

Then

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2}(u_{11} - u_{22}). \quad (17)$$

Now decompose $U_{\parallel}$ in the Pauli basis:

$$U_{\parallel} = u_0 \mathbf{1}_2 + u_1 \Sigma_1 + u_2 \Sigma_2 + u_3 \Sigma_3. \quad (18)$$

Then immediately

$$\boxed{u_{\parallel} = u_3.} \quad (19)$$

So the carrier-side longitudinal scalar does not depend on the full 2×2 matrix. It depends only on the Σ_3 -component of the shell-normal intervalley J -block.

This is the key simplification.

VII. 5. CLOSED FORMULA FOR L_*

By definition,

$$L_* = m_{\parallel} u_{\parallel}. \quad (20)$$

Using (17),

$$\boxed{L_* = \frac{m_{\parallel}}{2}(u_{11} - u_{22}).} \quad (21)$$

Using (45),

$$\boxed{L_* = m_{\parallel} u_3.} \quad (22)$$

Hence the exact vanishing criterion is

$$\boxed{L_* = 0 \iff m_{\parallel} = 0 \quad \text{or} \quad u_{11} = u_{22} \iff m_{\parallel} = 0 \quad \text{or} \quad u_3 = 0.} \quad (23)$$

Therefore, once $m_{\parallel} \neq 0$, the only way the longitudinal slot dies is that the shell-normal intervalley J -block becomes exactly Σ_3 -symmetric in its diagonal entries.

VIII. 6. MINIMAL EXECUTABLE PROTOCOL

The full carrier-side longitudinal audit is now algorithmically closed.

Step 1

Compute the two valley blocks

$$E_{\pm}(0) = \begin{pmatrix} a_{\pm} & c_{\pm} \\ c_{\pm} & b_{\pm} \end{pmatrix}.$$

Step 2

Compute

$$R_{\pm} = \sqrt{\left(\frac{a_{\pm} - b_{\pm}}{2}\right)^2 + |c_{\pm}|^2}, \quad \varepsilon_{\pm}^{(\text{low})} = \frac{a_{\pm} + b_{\pm}}{2} - R_{\pm}.$$

Step 3

Construct the normalized low eigenvectors $\eta_{\pm}$ from either (7) or (10).

Step 4

Evaluate

$$\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_-.$$

Step 5

Compute the shell-normal intervalley J -matrix

$$U_{\parallel} = \hat{n}^j \mathcal{U}_j(0),$$

and extract

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2}(u_{11} - u_{22}).$$

Step 6

Read off

$$m_{\parallel} = m^3(P_0), \quad L_* = m_{\parallel} u_{\parallel}.$$

Step 7

Apply the practical sufficient condition

$$\boxed{|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N|}. \quad (24)$$

In the ultra-minimal truncation $R_N = 0$, it is enough to verify

$$\boxed{\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0}. \quad (25)$$

IX. 7. HONEST THEOREM-LEVEL STATUS

What has now been fully reduced is not merely the existence statement, but the exact finite audit map:

$$\boxed{(E_+(0), E_-(0), U_{\parallel}, m_{\parallel}) \mapsto (\Delta_{00}, L_*)}. \quad (26)$$

What is still not inserted numerically is the actual microscopic value of the entries

$$a_{\pm}, b_{\pm}, c_{\pm}, u_{11}, u_{22}, m_{\parallel}.$$

So the only remaining honest open part is:

$$\boxed{\text{the final microscopic matrix-element insertion, not the reduction logic.}}$$

X. FINAL SUMMARY

The remaining carrier-side longitudinal problem is a finite 2×2 matrix audit.
Write

$$E_{\pm}(0) = \begin{pmatrix} a_{\pm} & c_{\pm} \\ c_{\pm} & b_{\pm} \end{pmatrix}, \quad U_{\parallel} = \hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}.$$

Then the low eigenvectors $\eta_{\pm}$ of $E_{\pm}(0)$ determine

$$\boxed{\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_-}.$$

Equivalently, if

$$r_{\pm} = -\frac{c_{\pm}}{a_{\pm} - \varepsilon_{\pm}^{(\text{low})}},$$

then

$$\boxed{\Delta_{00} = \frac{1 - \bar{r}_+ r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}}}.$$

The shell-normal J -singlet bias is

$$\boxed{u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2}(u_{11} - u_{22}),}$$

so

$$\boxed{L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2}(u_{11} - u_{22}).}$$

Therefore the carrier-side longitudinal witness is controlled by

$$\boxed{\Delta_{00} \propto_X L_* A_{G_*}},$$

up to the explicit remainder R_N .

Hence the practical sufficient condition is

$$\boxed{|\Delta_{00} \propto_X L_* A_{G_*}| > |R_N|}.$$

In the ultra-minimal truncation $R_N = 0$, it is enough to verify

$$\boxed{\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \propto_X \neq 0, \quad A_{G_*} \neq 0.}$$

So the longitudinal frontier is no longer structural. It is reduced to the final insertion of the actual microscopic entries of $E_{\pm}(0)$, $U_{\parallel}$, and $m_{\parallel}$.

XI. DIRECT FINITE EXTRACTION OF THE REMAINING LONGITUDINAL SCALARS

XII. SHORT KOREAN EXPLANATION

At the present stage, the carrier-side longitudinal frontier is a literal finite 2×2 matrix problem. The enriched shell carrier is already fixed as

$$\mathcal{H}_D^{(N)} = \text{span}\{|+, 1\rangle, |+, 2\rangle, |-, 1\rangle, |-, 2\rangle\},$$

and the projected core block is

$$H_{\text{core}}^{(N)}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix}.$$

Therefore the final longitudinal audit reduces to:

1. compute the two Hermitian 2×2 valley blocks $E_\pm(0)$,
2. diagonalize them to obtain the normalized low eigenvectors $\eta_\pm$,
3. extract

$$\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_-, \quad \Sigma_3 = \text{diag}(1, -1),$$

4. compute the shell-normal intervalley matrix

$$U_{\parallel} := \hat{n}^j \mathcal{U}_j(0),$$

5. extract

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}), \quad L_* = m_{\parallel} u_{\parallel}.$$

So the remaining problem is finite and explicit.

XIII. 1. EXPLICIT 2×2 VALLEY BLOCKS

Write the two valley-diagonal projected blocks as general Hermitian matrices:

$$E_+(0) = \begin{pmatrix} a_+ & c_+ \\ \overline{c_+} & b_+ \end{pmatrix}, \quad E_-(0) = \begin{pmatrix} a_- & c_- \\ \overline{c_-} & b_- \end{pmatrix}, \quad (27)$$

with

$$a_{\pm}, b_{\pm} \in \mathbb{R}, \quad c_{\pm} \in \mathbb{C}.$$

Define

$$t_{\pm} := \frac{a_{\pm} + b_{\pm}}{2}, \quad d_{\pm} := \frac{a_{\pm} - b_{\pm}}{2}, \quad \rho_{\pm} := |c_{\pm}|. \quad (28)$$

Then

$$E_{\pm}(0) = t_{\pm} \mathbf{1}_2 + d_{\pm} \Sigma_3 + (\Re c_{\pm}) \Sigma_1 - (\Im c_{\pm}) \Sigma_2. \quad (29)$$

Thus the nontrivial part of each valley block is encoded in the Bloch vector

$$\mathbf{h}_{\pm} = (\Re c_{\pm}, -\Im c_{\pm}, d_{\pm}).$$

XIV. 2. LOW EIGENVALUES AND NORMALIZED LOW EIGENVECTORS

The eigenvalues of $E_{\pm}(0)$ are

$$\varepsilon_{\pm}^{(\text{low/high})} = t_{\pm} \mp R_{\pm}, \quad R_{\pm} := \sqrt{d_{\pm}^2 + \rho_{\pm}^2}. \quad (30)$$

Let

$$c_{\pm} = \rho_{\pm} e^{i\phi_{\pm}}, \quad \rho_{\pm} \geq 0. \quad (31)$$

Introduce the polar angle $\theta_{\pm}$ by

$$\cos \theta_{\pm} = \frac{d_{\pm}}{R_{\pm}}, \quad \sin \theta_{\pm} = \frac{\rho_{\pm}}{R_{\pm}}. \quad (32)$$

A convenient normalized low eigenvector choice is

$$\eta_{\pm} = \begin{pmatrix} -e^{-i\phi_{\pm}} \sin(\theta_{\pm}/2) \\ \cos(\theta_{\pm}/2) \end{pmatrix}, \quad (33)$$

up to an overall phase.

Equivalently, in component form,

$$\eta_{\pm} = \begin{pmatrix} A_1^{(\pm)} \\ A_2^{(\pm)} \end{pmatrix}, \quad |A_1^{(\pm)}|^2 + |A_2^{(\pm)}|^2 = 1. \quad (34)$$

A completely algebraic expression is obtained by setting

$$r_{\pm} := -\frac{c_{\pm}}{a_{\pm} - \varepsilon_{\pm}^{(\text{low})}} = -\frac{c_{\pm}}{d_{\pm} + R_{\pm}}, \quad (35)$$

whenever $d_{\pm} + R_{\pm} \neq 0$, and then

$$\eta_{\pm} = \frac{1}{\sqrt{1 + |r_{\pm}|^2}} \begin{pmatrix} 1 \\ r_{\pm} \end{pmatrix}. \quad (36)$$

XV. 3. EXACT EXTRACTION FORMULA FOR Δ_{00}

By definition,

$$\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_- = \overline{A_1^{(+)}} A_1^{(-)} - \overline{A_2^{(+)}} A_2^{(-)}. \quad (37)$$

Using (33), one obtains

$$\Delta_{00} = e^{i(\phi_+ - \phi_-)} \sin \frac{\theta_+}{2} \sin \frac{\theta_-}{2} - \cos \frac{\theta_+}{2} \cos \frac{\theta_-}{2}. \quad (38)$$

Using (36), one gets the more algebraic form

$$\Delta_{00} = \frac{1 - \overline{r_+} r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}}. \quad (39)$$

Hence the exact vanishing condition is

$$\Delta_{00} = 0 \iff \overline{r_+} r_- = 1. \quad (40)$$

This is important. It shows that Δ_{00} does *not* vanish generically. Its vanishing requires a very specific matching relation between the two low-valley eigenvectors.

Therefore, unless an exact symmetry or exact microscopic tuning enforces $\overline{r_+} r_- = 1$, the generic expectation is

$$\Delta_{00} \neq 0. \quad (41)$$

XVI. 4. EXACT EXTRACTION FORMULA FOR $u_{\parallel}$

Define the shell-normal reduced intervalley J -matrix

$$U_{\parallel} := \hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}, \quad \mathcal{U}_{j,ab}(0) = \langle -, a | \hat{J}_j(G_*, 0) | +, b \rangle. \quad (42)$$

Then

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2}(u_{11} - u_{22}). \quad (43)$$

Equivalently, expand $U_{\parallel}$ in the Pauli basis:

$$U_{\parallel} = u_0 \mathbf{1}_2 + u_1 \Sigma_1 + u_2 \Sigma_2 + u_3 \Sigma_3. \quad (44)$$

Then immediately

$$u_{\parallel} = u_3. \quad (45)$$

So the longitudinal scalar depends only on the Σ_3 -component of the shell-normal intervalley J -matrix.

XVII. 5. EXACT EXTRACTION FORMULA FOR L_*

By definition,

$$L_* = m_{\parallel} u_{\parallel}. \quad (46)$$

Using (43),

$$L_* = \frac{m_{\parallel}}{2}(u_{11} - u_{22}). \quad (47)$$

Using (45),

$$L_* = m_{\parallel} u_3. \quad (48)$$

Therefore the exact vanishing criterion is

$$L_* = 0 \iff m_{\parallel} = 0 \quad \text{or} \quad u_{11} = u_{22} \iff m_{\parallel} = 0 \quad \text{or} \quad u_3 = 0. \quad (49)$$

Thus, once $m_{\parallel} \neq 0$, the only way the longitudinal slot dies is the exact diagonal cancellation

$$u_{11} = u_{22}.$$

Again, this is not generic. It is a codimension-one cancellation inside the shell-normal intervalley block.

XVIII. 6. COMBINED PRACTICAL SUFFICIENT CONDITION

The carrier-side longitudinal witness is controlled by the dominant product

$$\Delta_{00} \alpha_X L_* A_{G_*}, \quad (50)$$

up to the explicit harmonic remainder R_N .

Therefore the practical sufficient condition is

$$|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N|. \quad (51)$$

In the ultra-minimal truncation $R_N = 0$, it is enough to verify

$$\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0. \quad (52)$$

Substituting (39) and (47), one may rewrite this explicitly as

$$\left| \frac{1 - \bar{r}_+ r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}} \right| \cdot |\alpha_X| \cdot \left| \frac{m_{\parallel}}{2}(u_{11} - u_{22}) \right| \cdot |A_{G_*}| > |R_N|. \quad (53)$$

XIX. 7. HONEST CONCLUSION

The remaining carrier-side longitudinal frontier is no longer structural. It is completely reduced to the finite microscopic insertion of the entries

$$a_{\pm}, \quad b_{\pm}, \quad c_{\pm}, \quad u_{11}, \quad u_{22}, \quad m_{\parallel}.$$

Once those are computed from the actual projected refined locked Bloch operator, the longitudinal decision is immediate.

In particular:

1. Δ_{00} is nonzero unless the two low-valley eigenvectors satisfy the exact tuned relation

$$\overline{r_+} r_- = 1;$$

2. L_* is nonzero unless either $m_{\parallel} = 0$ or the shell-normal intervalley diagonal asymmetry cancels exactly:

$$u_{11} = u_{22}.$$

So the remaining open part is not a theorem-level reduction. It is only the explicit symbolic or numerical evaluation of these finite matrix elements.

XX. FINAL SUMMARY

The carrier-side longitudinal frontier is reduced to the finite extraction of two scalars:

$$\Delta_{00}, \quad L_* = m_{\parallel} u_{\parallel}.$$

Write the two valley blocks as

$$E_{\pm}(0) = \begin{pmatrix} a_{\pm} & c_{\pm} \\ \overline{c_{\pm}} & b_{\pm} \end{pmatrix}, \quad \varepsilon_{\pm}^{(\text{low})} = \frac{a_{\pm} + b_{\pm}}{2} - \sqrt{\left(\frac{a_{\pm} - b_{\pm}}{2}\right)^2 + |c_{\pm}|^2}.$$

Define

$$r_{\pm} = -\frac{c_{\pm}}{a_{\pm} - \varepsilon_{\pm}^{(\text{low})}}.$$

Then the normalized low eigenvectors $\eta_{\pm}$ give

$$\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_- = \frac{1 - \overline{r_+} r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}}.$$

Hence

$$\Delta_{00} = 0 \iff \overline{r_+} r_- = 1.$$

Next define the shell-normal intervalley J -matrix

$$U_{\parallel} = \hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}.$$

Then

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2}(u_{11} - u_{22}),$$

and therefore

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2}(u_{11} - u_{22}).$$

Hence

$$L_* = 0 \iff m_{\parallel} = 0 \quad \text{or} \quad u_{11} = u_{22}.$$

The practical sufficient condition for longitudinal survival is

$$|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N|.$$

In the ultra-minimal truncation $R_N = 0$, it is enough to verify

$$\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0.$$

Thus no structural ambiguity remains. Only the explicit symbolic or numerical insertion of the finite projected matrix entries

$$a_{\pm}, b_{\pm}, c_{\pm}, u_{11}, u_{22}, m_{\parallel}$$

remains.

XXI. SHORT KOREAN EXPLANATION

Yes. The reduced longitudinal coefficient can already be computed explicitly from the uploaded derivations. What is already closed is the exact coefficient formula

$$\lambda_{\parallel} = I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \Gamma_X = \frac{\mu_X \nu_X m^2}{M_X^2},$$

together with the explicit primitive overlap value

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.$$

Therefore

$$\lambda_{\parallel} = I_{\theta} \frac{\mu_X \nu_X m^2}{M_X^2} c_{\parallel} - \frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3.$$

This is already a genuine explicit finite formula, not just a structural reduction.

So the longitudinal channel is closed at the reduced coefficient level. What is still not fully inserted is the stronger final carrier-side Bloch audit through

$$\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_-, \quad L_* = m_{\parallel} u_{\parallel},$$

because that requires the actual projected valley matrices $E_{\pm}(0)$ or the actual harmonic coefficients $A_a^{(\pm)}$.

XXII. 1. EXPLICIT REDUCED LONGITUDINAL COEFFICIENT

From the mediated coefficient extraction in the canonical Class II seed basis, one already has

$$\lambda_{\parallel} = I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \Gamma_X = \frac{\mu_X \nu_X m^2}{M_X^2}. \quad (54)$$

This is not a conjecture anymore. It is the already-derived reduced coefficient formula. Moreover, the primitive overlap is explicitly known:

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (55)$$

Substituting (55) into (54) gives

$$\lambda_{\parallel} = I_{\theta} \frac{\mu_X \nu_X m^2}{M_X^2} c_{\parallel} - \frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3. \quad (56)$$

This is already a fully explicit finite expression.

XXIII. 2. IMMEDIATE BRANCHWISE CLOSURE

Equation (56) immediately gives two exact subcases.

A. A. Pure primitive branch

If the mediated longitudinal part is switched off,

$$c_{\parallel} = 0,$$

then

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3. \quad (57)$$

Therefore

$$\lambda_c \neq 0, \quad \theta_0 \kappa \neq 0 \implies \lambda_{\parallel} \neq 0. \quad (58)$$

So the primitive branch is already closed.

B. B. Pure mediated branch

If the primitive coefficient is switched off,

$$\lambda_c = 0,$$

then

$$\lambda_{\parallel} = I_{\theta} \frac{\mu_X \nu_X m^2}{M_X^2} c_{\parallel}. \quad (59)$$

Hence

$$I_{\theta} \neq 0, \quad \mu_X \nu_X m^2 \neq 0, \quad c_{\parallel} \neq 0 \implies \lambda_{\parallel} \neq 0. \quad (60)$$

So the mediated branch is also closed.

XXIV. 3. EXACT CANCELLATION HYPERSURFACE

The only way $\lambda_{\parallel}$ vanishes in the combined branch is the exact cancellation

$$I_{\theta} \frac{\mu_X \nu_X m^2}{M_X^2} c_{\parallel} = \frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3. \quad (61)$$

Thus the zero set is a single codimension-one tuning relation in parameter space. Generically,

$$\lambda_{\parallel} \neq 0. \quad (62)$$

So at the reduced longitudinal-coefficient level, the honest situation is already much stronger than “structure only”: the coefficient is explicitly computed, and its vanishing locus is exactly known.

XXV. 4. CROSS-CHECK FROM THE CANONICAL BLOCH CALIBRATION

The uploaded Bloch-calibration derivation also gives, in canonical normalization,

$$\boxed{C_{\parallel} = C_E = C_O = 1,} \quad (63)$$

and

$$\boxed{\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} I_3.} \quad (64)$$

So the same conclusion appears in the Bloch-calibrated language:

$$\boxed{\alpha_X \beta_X I_3 \neq 0 \implies \lambda_{\parallel} \neq 0.} \quad (65)$$

This is consistent with (56) after identifying the mediated/primitive coefficient dictionary.

XXVI. 5. WHAT IS CLOSED AND WHAT IS NOT

What is now truly closed:

$$\boxed{\text{the reduced longitudinal coefficient } \lambda_{\parallel} \text{ has an explicit closed formula.}} \quad (66)$$

What is not yet fully inserted:

$$\boxed{\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_-, \quad L_* = m_{\parallel} u_{\parallel}} \quad (67)$$

from the actual projected refined locked Bloch operator.

Indeed, the uploaded notes explicitly state that the remaining frontier is the actual finite overlap extraction: the actual projected Bloch basis, the actual longitudinal basis-overlap audit $\exists A : \ell_{\parallel A} \neq 0$, and the actual finite matrix elements of the refined locked operator still need to be evaluated. So the final Bloch-side scalar insertion is not yet numerically/symbolically completed from the currently uploaded data alone.

XXVII. 6. HONEST FINAL VERDICT

Therefore the correct statement is:

$$\boxed{\text{Yes: the longitudinal coefficient } \lambda_{\parallel} \text{ itself can already be computed and closed.}} \quad (68)$$

More precisely,

$$\boxed{\lambda_{\parallel} = I_{\theta} \frac{\mu_X \nu_X m^2}{M_X^2} c_{\parallel} - \frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3.} \quad (69)$$

But

$$\boxed{\text{No: the stronger final actual-Bloch scalar audit } (\Delta_{00}, L_*) \text{ is not yet fully closed from the uploaded formulas alone.}} \quad (70)$$

XXVIII. FINAL SUMMARY

The reduced longitudinal coefficient is already explicitly closed:

$$\lambda_{\parallel} = I_{\theta} \frac{\mu_X \nu_X m^2}{M_X^2} c_{\parallel} - \frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3.$$

Therefore:

$$c_{\parallel} = 0, \lambda_c \neq 0, \theta_0 \kappa \neq 0 \implies \lambda_{\parallel} \neq 0,$$

and

$$\lambda_c = 0, I_{\theta} \neq 0, \mu_X \nu_X m^2 \neq 0, c_{\parallel} \neq 0 \implies \lambda_{\parallel} \neq 0.$$

The only exact zero condition is the cancellation hypersurface

$$I_{\theta} \frac{\mu_X \nu_X m^2}{M_X^2} c_{\parallel} = \frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3.$$

Hence the reduced longitudinal coefficient is generically nonzero.

However, the stronger final actual-Bloch audit through

$$\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_-, \quad L_* = m_{\parallel} u_{\parallel}$$

is not yet fully inserted from the currently uploaded formulas alone, because the actual projected matrices $E_{\pm}(0)$ and $\mathcal{U}_j(0)$ have not yet been explicitly evaluated.

So the honest status is:

$$\text{the reduced longitudinal coefficient } \lambda_{\parallel} \text{ is already closed,}$$

but

$$\text{the final actual projected-Bloch scalar extraction } (\Delta_{00}, L_*) \text{ is still pending.}$$

XXIX. STRATEGIC SUMMARY FOR BREAKING THE FULL-PDE BOTTLENECK

The correct lesson from the current notes is that the obstruction is not a single monolithic “full PDE” problem. It splits into three distinct bottlenecks:

1. the carrier-side longitudinal primitive survival test,
2. the derivative-anisotropy / kinetic-flow problem for M_Z ,
3. the dynamical existence and global matching of the defect background.

Therefore the orthodox strategy is not to delay everything until a fully unique microscopic PDE is derived. Instead, one should adopt the strongest explicit working branch and elevate only the unresolved pieces to sharply formulated assumptions with finite verification targets.

A minimal working package is:

1. refined locked + Class II microscopic branch,
2. existence of a shell-/defect-adapted Callias-admissible background,

3. actual longitudinal carrier survival:

$$\exists A : \ell_{\parallel A} \neq 0, \quad \ell_{\parallel A} = \hat{n}_i \hat{n}_j E_{A,ij}^3,$$

4. optional background-field FRG completion for derivative isotropy:

$$\text{spec}(M_Z|_{\text{traceless}}) < 0.$$

The highest-return immediate move is not another abstract PDE theorem, but the explicit 12-mode BCC Bloch / shell construction:

$$H_{\text{eff}}(k) \rightarrow H_{\text{shell}}(p) \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

In particular, the carrier-side bottleneck is already reduced to the finite longitudinal basis-overlap audit

$$\exists A : \ell_{\parallel A} \neq 0.$$

Thus the right pivot is:

replace the monolithic full-PDE demand by a working-branch theorem package plus finite audits.

XXX. CAN WORKING ASSUMPTIONS BECOME NATURALLY DISCHARGED LATER?

XXXI. SHORT KOREAN EXPLANATION

Yes, but not automatically.

A working assumption is useful only if, after pushing the derivation forward, it is transformed into one of the following:

1. a finite coefficient-extraction problem,
2. a codimension-one non-cancellation condition,
3. a variational existence problem,
4. an RG/fixed-point stability condition.

If this happens, then the assumption is not merely a permanent postulate. It becomes a *dischargeable hypothesis*.

If, on the contrary, the assumption remains a broad structural declaration even after the entire derivation is pushed through, then it has not been reduced and remains a genuine independent postulate.

Thus the right question is not:

“Can we assume it for now?”

but rather:

“Does this assumption reduce, later, to an explicit verification problem?”

XXXII. 1. GENERAL PRINCIPLE

Suppose a theorem package is built from a set of working assumptions

$$\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n.$$

There are three logically distinct outcomes.

Type I: reducible assumptions

An assumption $\mathcal{A}_k$ is reducible if the forward derivation converts it into a finite or sharply formulated check:

$$\mathcal{A}_k \implies \text{explicit finite test / spectral test / overlap test.}$$

Such an assumption is often only provisional. Later it may be discharged by direct calculation.

Type II: genericity assumptions

An assumption is of genericity type if the forward derivation shows that its failure requires exact tuning:

$$\mathcal{A}_k \text{ fails} \iff \text{a codimension-one or higher cancellation condition holds.}$$

Then the assumption is not fully proved, but it is strengthened: its negation is non-generic rather than structurally forced.

Type III: irreducible postulates

An assumption is irreducible if, even after the whole derivation is completed, it remains a large undecomposed statement with no reduction to a finite audit. Such a hypothesis remains a genuine postulate of the framework.

XXXIII. 2. APPLICATION TO THE CURRENT TECT BOTTLENECKS

(A) Microscopic branch assumption

The working branch

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi, P] + \beta_X X_i^a K_i^a[P, \psi]$$

is currently a *working microscopic ansatz*.

This assumption is only partially dischargeable. It may be weakened if one later proves one of the following:

1. this branch is dynamically generated under RG from a broader UV class;
2. this branch is the minimal one compatible with the required carrier coefficients;
3. alternative branches fail to generate the same transport data.

So this assumption is not yet naturally solved. At present it remains a genuine branch-level postulate.

(B) Defect-background existence assumption

The assumption that there exists a shell-/defect-adapted finite-energy background is much more likely to become dischargeable.

Why? Because it can be reduced to:

1. radial or profile existence equations,
2. topological boundary conditions,
3. finite-energy coercivity,
4. asymptotic matching inequalities.

So this is a good assumption: it is large at first, but it has a clear route toward becoming a genuine existence theorem.

(C) Longitudinal carrier survival assumption

This is the best example of an assumption that naturally reduces.
Initially one may say:

“the longitudinal carrier channel survives.”

But after the transport classification and basis reduction, this becomes

$$\exists A : \ell_{\parallel A} \neq 0, \quad \ell_{\parallel A} = \hat{n}_i \hat{n}_j E_{A,ij}^3.$$

This is no longer a philosophical assumption. It is a finite basis-overlap audit.

So this assumption is not merely being postponed; it is being *compressed* into a concrete calculable statement.

(D) IR isotropy / Lorentz restoration assumption

This one is intermediate.

It is not as irreducible as the branch ansatz, because it can in principle be tested by background-field FRG, stability matrices, and anisotropy eigenvalues. But it is not as directly dischargeable as the longitudinal carrier audit either.

So this assumption is potentially reducible, but only after a separate RG module is constructed.

XXXIV. 3. WHAT IT MEANS FOR AN ASSUMPTION TO BECOME “NATURAL”

There are two different meanings of “natural resolution.”

(i) Logical naturality

An assumption becomes logically natural if the theory shows that its failure is only possible under exact tuning.
For example, if one proves

$$\lambda_{\parallel} = 0 \iff \text{exact cancellation hypersurface,}$$

then nonzero longitudinal transport is not yet a theorem from first principles, but it becomes the generic outcome of the working branch.

(ii) Constructive naturality

An assumption becomes constructively natural if it reduces to a concrete existence or computation problem.
For example,

$$\exists A : \ell_{\parallel A} \neq 0$$

is constructively natural because it can be checked by explicit projected matrix elements.

XXXV. 4. THE CORRECT STRATEGY

Therefore the right strategy is not to avoid assumptions altogether. The right strategy is to admit only those assumptions that satisfy:

$$\boxed{\text{forward derivation} \implies \text{explicit discharge map.}} \tag{71}$$

In practice, each working assumption should come with one of the following tags:

1. **Finite audit:** explicit matrix element, overlap, or coefficient extraction;
2. **Variational existence:** direct method / profile equation / asymptotic matching;
3. **Genericity:** vanishing requires exact tuning;
4. **RG completion:** requires a separate flow/stability module.

Only such assumptions are scientifically productive.

XXXVI. 5. HONEST VERDICT FOR THE CURRENT PROJECT

For the present TECT program:

1. the microscopic branch ansatz is still a genuine working postulate;
2. the defect-background assumption has a real chance of becoming a theorem;
3. the longitudinal carrier survival assumption is already close to being discharged, because it has reduced to a finite audit;
4. the IR isotropy/Lorentz assumption is potentially dischargeable, but only through a dedicated background-field RG analysis.

Hence the answer is:

Yes, some assumptions can later dissolve naturally, but only if they reduce to explicit audits or existence problems.

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They do not disappear by philosophy. They disappear only by reduction.

XXXVII. FINAL SUMMARY

A working assumption can later become naturally discharged, but only in a precise sense. This happens when the forward derivation transforms the assumption into one of the following:

1. a finite matrix-element or overlap audit,
2. a variational/profile existence problem,
3. a genericity statement whose failure requires exact tuning,
4. a separate RG stability condition.

For the present TECT program:

1. the microscopic branch ansatz remains a genuine working postulate;
2. the defect-background existence assumption is plausibly reducible to a theorem;
3. the longitudinal carrier-survival assumption is already close to discharge, since it has reduced to the finite condition

$$\exists A : \ell_{\parallel A} \neq 0;$$

4. the IR isotropy/Lorentz assumption is potentially reducible, but only through an additional background-field RG module.

Hence assumptions do not disappear automatically. They become natural only when they are reduced. The correct research rule is therefore:

Admit only assumptions that come with an explicit discharge map.

XXXVIII. THREE-GRADE CLASSIFICATION OF THE CURRENT WORKING ASSUMPTIONS

XXXIX. SHORT KOREAN EXPLANATION

The current TECT assumptions should not be treated as one undifferentiated block. They split into three sharply different classes:

1. Grade A: assumptions/conditions already compressed to finite executable audits;
2. Grade B: assumptions reducible only through a separate module (existence/matching/FRG);
3. Grade C: genuine working postulates that must currently be retained honestly.

This grading is operational rather than philosophical. Its purpose is to determine where the derivation should start once a given assumption package is adopted.

XL. GRADING PRINCIPLE

We define:

Grade A := finite audit now,	Grade B := reducible, but via a separate module,	Grade C := current genuine working postulates
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XLI. MASTER TABLE

Grade	Code	Assumption / condition	Example / consequence
A	A1	Longitudinal carrier survival	$\exists A : \ell_{\parallel A} \neq 0$
A	A2	Bloch longitudinal scalar audit	$\Delta_{00}, \quad L_* = \frac{m_{\parallel}}{2} \text{Tr}(\Sigma_3 \hat{n}^j \hat{u}^j)$
A	A3	Remote-gap finite audit	$\Delta_{\text{ber}}^{\text{fin}}$
B	B1	Actual defect background realizes a Callias pair	The microscopic defect PDE yields smooth P , $A_i =$
B	B2	Mirror lifting support hypothesis	$a_{\infty} \neq 0$
B	B3	IR isotropy / Lorentz completion	$\text{spec}(M_Z _{\mathcal{U}_{\text{tr}}}) \subset (-\infty, 0)$
C	C1	Working microscopic branch ansatz	$L_{\text{mic}} = L_{\Psi}^{\text{ref}} + \frac{M_X^2}{2} X_i^{\text{ref}}$
C	C2	Locked 12-mode BCC working background	$k_* = \frac{G_*}{2}, \quad \Psi_0 = z_0 \phi$
C	C3	Residual-valley symmetry as a sufficient Gate-4 assumption	Exact $U(1)_V$ or $\mathbb{Z}_2^V$ survives on t

XLII. START PACKAGES

The assumptions naturally organize into three operational starting packages.

Package S1: carrier-first

Adopt

$$C1 + C2,$$

then begin with

$$A1 \rightarrow A2 \rightarrow A3.$$

This is currently the highest-return route, because the carrier bottleneck has already been reduced to finite overlap and matrix-element audits.

Package S2: defect-first

Adopt

$$C1 + B1,$$

then begin with

$$B2.$$

This route is appropriate if the primary objective is chirality / mirror lifting rather than carrier acceptance.

Package S3: IR-completion

Adopt

$$C1 + B3,$$

then begin with the explicit computation of

$$\eta_{OD}, \quad \eta_{gf}, \quad \mu_{good}.$$

This is the correct route for Lorentz/isotropy completion, but it is not the immediate first priority.

XLIII. HONEST OPERATIONAL VERDICT

The best current operating policy is:

$$\boxed{\text{fix } (C1, C2) \text{ as the working platform, and discharge } A1 \text{ first.}} \quad (74)$$

This removes the last honest open carrier-side slot at the highest leverage point, while keeping the heavier defect and FRG modules sharply separated.

XLIV. FINAL SUMMARY

The current TECT assumptions split into three strict grades:

1. Grade A: already compressed to finite executable audits;
2. Grade B: reducible, but only through separate existence/matching/FRG modules;
3. Grade C: current genuine working postulates.

The most important Grade-A item is the longitudinal carrier survival audit

$$\exists A : \ell_{\parallel A} \neq 0, \quad \ell_{\parallel A} = \hat{n}_i \hat{n}_j E_{A,ij}^3,$$

which is the correct immediate next executable step.

The defect-side open hypotheses are no longer vague: they reduce to

$$\text{Callias-background realization} \quad \text{and} \quad (a_\infty, b_\infty) \neq (0, 0)$$

through a core-to-infinity matching problem.

The IR isotropy/Lorentz problem is also sharply isolated: the strict symmetric-point truncation is insufficient, so one must move to a background-field FRG computation and verify

$$\text{spec}(M_Z|_{\mathcal{U}_{\text{tr}}}) < 0 \quad \text{or} \quad \mu_{\text{good}} > \eta_{OD} + \eta_{\text{gf}}.$$

Therefore the correct current operating policy is:

fix the refined locked + Class II branch and the locked 12-mode BCC background as the working platform,

then discharge the Grade-A carrier audit first.

This gives the cleanest route toward reducing the remaining hypotheses into explicit verifiable computations.

XLV. DIRECT GATE-2 LONGITUDINAL CLOSURE THEOREM

XLVI. SHORT KOREAN EXPLANATION

At the present stage, the strongest honest theorem that can be proved directly is not yet the full unconditional numerical microscopic closure of the final projected Bloch data. What *is* already provable is the exact Gate-2 closure statement on the current refined locked + Class II working branch:

$$\exists A : \ell_{\parallel A} \neq 0 \implies \lambda_{\parallel} \neq 0.$$

Moreover, in the current Branch-B realization, the primitive longitudinal seed is fixed to be

$$V_{(\parallel)}^{\mu}(\xi) = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3,$$

and the reduced witness coefficient is

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}.$$

Since $I_{\theta'^3} \neq 0$ is already analytically closed for the explicit defect profile, the entire longitudinal question reduces to whether the microscopic transport space contains a nonzero projection onto the primitive seed.

Thus Gate 2 is now a direct finite theorem, not an open conceptual PDE problem.

XLVII. 1. FIXED SHELL-ADAPTED LONGITUDINAL TARGET

Fix the shell-adapted frame

$$\hat{n} = \frac{G_*}{|G_*|}, \quad \{e_1, e_2, \hat{n}\},$$

and define the primitive longitudinal seed tensor

$$B_{\parallel} : \quad (B_{\parallel})_{ij}^3 = \hat{n}_i \hat{n}_j, \quad (B_{\parallel})_{ij}^1 = (B_{\parallel})_{ij}^2 = 0. \quad (75)$$

Equivalently, the dual longitudinal witness functional is

$$L_{\parallel}(E) = \hat{n}_i \hat{n}_j E_{ij}^3. \quad (76)$$

For any basis $\{E_A\}_{A=1}^d$ of the actual microscopic transport space

$$V_{\text{TRECT}} \subset \mathbb{R}_{\text{out}}^3 \otimes \mathbb{R}_{\text{grad}}^3 \otimes U_{\text{int}},$$

define

$$\ell_{\parallel A} := L_{\parallel}(E_A) = \hat{n}_i \hat{n}_j E_{A,ij}^3. \quad (77)$$

XLVIII. 2. BASIS DECOMPOSITION AND EXACT AUDIT CRITERION

Let

$$V_{\min} = \text{span}\{B_{\parallel}, B_I, B_J\}$$

be the shell seed space. Each microscopic basis element admits the shell-adapted decomposition

$$E_A = a_A B_{\parallel} + b_A B_I + c_A B_J + R_A, \quad R_A \perp V_{\min}. \quad (78)$$

Because $L_{\parallel}$ annihilates B_I, B_J and R_A , while $L_{\parallel}(B_{\parallel}) \neq 0$, one gets

$$\boxed{\ell_{\parallel A} = L_{\parallel}(E_A) = \mu_{\parallel} a_A, \quad \mu_{\parallel} := L_{\parallel}(B_{\parallel}) \neq 0.} \quad (79)$$

Therefore

$$\boxed{\exists A : \ell_{\parallel A} \neq 0 \iff \exists A : a_A \neq 0.} \quad (80)$$

Equivalently, if N_{mic} denotes the full microscopic shell-relevant transport tensor and

$$P_{\text{seed}} N_{\text{mic}} = c_{\parallel}^{(\text{mic})} B_{\parallel} + c_I^{(\text{mic})} B_I + c_J^{(\text{mic})} B_J, \quad (81)$$

then

$$\boxed{\exists A : \ell_{\parallel A} \neq 0 \iff c_{\parallel}^{(\text{mic})} \neq 0.} \quad (82)$$

This is the exact finite Gate-2 criterion.

XLIX. 3. PRIMITIVE BRANCH-B LONGITUDINAL SOURCE

In the exact concrete Class II Branch-B realization, the longitudinal primitive source is fixed by the defect geometry. Indeed,

$$\boxed{(\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3,} \quad (83)$$

hence the canonical odd first-order longitudinal transport seed is

$$\boxed{V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu} = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3.} \quad (84)$$

Therefore the reduced longitudinal witness is exactly

$$\boxed{\lambda_{\parallel} = \frac{1}{2} c_{\parallel}^{(\text{mic})} I_{\theta'^3}.} \quad (85)$$

L. 4. NONVANISHING OF THE OVERLAP INTEGRAL

For the explicit reflection-symmetric defect profile

$$\theta(\xi) = \theta_0 \text{sech}(\kappa \xi),$$

with the normalized low modes

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}} \text{sech}(\kappa \xi), \quad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa \xi) \text{sech}(\kappa \xi),$$

the primitive longitudinal overlap is

$$I_{\theta'^3} = \int_{-\infty}^{\infty} d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3 = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (86)$$

Hence

$$\boxed{\theta_0 \kappa \neq 0 \implies I_{\theta'^3} \neq 0.} \quad (87)$$

LI. 5. DIRECT GATE-2 CLOSURE THEOREM

[Direct Gate-2 longitudinal closure] Fix the current refined locked + Class II working branch and suppose the concrete longitudinal realization is on Branch B, so that

$$V_{(\parallel)}^\mu(\xi) = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3$$

and

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel}^{(\text{mic})} I_{\theta'^3}.$$

Assume $\theta_0 \kappa \neq 0$. Then the following are equivalent:

1. there exists a microscopic basis vector $E_A \in V_{\text{TECT}}$ such that

$$\ell_{\parallel A} \neq 0;$$

2. the primitive longitudinal seed coefficient is nonzero:

$$c_{\parallel}^{(\text{mic})} \neq 0.$$

Moreover, either condition implies

$$\boxed{\lambda_{\parallel} \neq 0.} \tag{88}$$

By (82),

$$\exists A : \ell_{\parallel A} \neq 0 \iff c_{\parallel}^{(\text{mic})} \neq 0.$$

By (85),

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel}^{(\text{mic})} I_{\theta'^3}.$$

By (87), the overlap factor $I_{\theta'^3}$ is nonzero whenever $\theta_0 \kappa \neq 0$. Therefore

$$c_{\parallel}^{(\text{mic})} \neq 0 \implies \lambda_{\parallel} \neq 0.$$

Combining the two implications yields (88).

LII. 6. PRACTICAL ONE-MEDIATOR SUFFICIENT FORM

Inside the finite coefficient-matching dictionary of the current Class II branch, one has

$$\boxed{\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}, \quad L_* := m_{\parallel} u_{\parallel}.} \tag{89}$$

Equivalently, in the practical Branch-B overlap form,

$$\boxed{\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3.} \tag{90}$$

Hence the practical sufficient conditions

$$\boxed{\alpha_X \neq 0, \quad L_* \neq 0, \quad I_{\theta'^3} \neq 0} \tag{91}$$

or equivalently

$$\boxed{\alpha_X \beta_X \neq 0, \quad C_{\parallel} \neq 0, \quad I_3 \neq 0} \tag{92}$$

imply

$$\boxed{\lambda_{\parallel} \neq 0.} \tag{93}$$

LIII. 7. HONEST STATUS

This proves the exact direct theorem that is available *now*.
What is proved:

$$\boxed{\exists A : \ell_{\parallel A} \neq 0 \iff c_{\parallel}^{(\text{mic})} \neq 0 \implies \lambda_{\parallel} \neq 0.} \quad (94)$$

What is not yet proved from the currently inserted microscopic data alone is the unconditional statement

$$\exists A : \ell_{\parallel A} \neq 0$$

for the final complete projected Bloch operator without any remaining finite insertion. That still requires the actual evaluation of the projected microscopic data

$$E_{\pm}(0), \quad \mathcal{U}_j(0), \quad C_{\parallel}$$

or, equivalently, the explicit final insertion of L_* or $\lambda_{c;OO}$.

LIV. FINAL SUMMARY

In the current refined locked + Class II working branch, the primitive longitudinal channel is uniquely fixed to

$$V_{(\parallel)}^{\mu}(\xi) = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3.$$

Its reduced witness is exactly

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel}^{(\text{mic})} I_{\theta'^3}.$$

For the explicit defect profile, the overlap integral is analytically nonzero:

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \neq 0 \quad (\theta_0 \kappa \neq 0).$$

On the microscopic transport space V_{TECT} , the longitudinal audit is

$$\ell_{\parallel A} = \hat{n}_i \hat{n}_j E_{A,ij}^3.$$

Using the shell-seed decomposition,

$$\exists A : \ell_{\parallel A} \neq 0 \iff c_{\parallel}^{(\text{mic})} \neq 0.$$

Therefore the exact Gate-2 closure theorem is:

$$\boxed{\exists A : \ell_{\parallel A} \neq 0 \implies \lambda_{\parallel} \neq 0.}$$

Equivalently, in the finite coefficient-matching form,

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}, \quad L_* = m_{\parallel} u_{\parallel},$$

or in the practical one-mediator form,

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3.$$

Hence

$$\alpha_X \neq 0, \quad L_* \neq 0, \quad I_{\theta'^3} \neq 0 \implies \lambda_{\parallel} \neq 0,$$

and equivalently

$$\alpha_X \beta_X \neq 0, \quad C_{\parallel} \neq 0, \quad I_3 \neq 0 \implies \lambda_{\parallel} \neq 0.$$

What remains open is no longer the theorem. It is only the final finite insertion proving the antecedent

$$\exists A : \ell_{\parallel A} \neq 0$$

for the actual complete projected microscopic branch.

IV. WHAT EXACTLY IS COMPLETED IN THE CURRENT WORKING BRANCH, AND WHAT IS THE NEXT DIRECT STEP?

LVI. SHORT KOREAN EXPLANATION

Yes: within the refined locked + Class II working branch, the theorem-level carrier reduction is completed.
But this means:

the reduction logic and extraction formulas are finished,

not that every final microscopic constant has already been numerically or symbolically inserted.

The exact current status is:

1. the working microscopic branch is fixed;
2. the light/heavy reduction and shell projection are fixed;
3. the projected transport matrices M_i extract $(\lambda_{\parallel}, \alpha, \beta)$;
4. the longitudinal branch decision is fixed to Branch B in the solved lowest-order realization;
5. therefore the remaining Gate-2 problem is a single finite extraction,

$$\lambda_{c;OO} \neq 0 \text{ ?},$$

or equivalently the explicit projected evaluation of $M_{\parallel}$.

So the next step is not another structural proof. It is the actual transport-level or projected-Bloch extraction.

LVII. 1. FIXED EXTRACTION LAYER

The working microscopic branch is

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi, P] + \beta_X X_i^a K_i^a[P, \Psi]. \quad (95)$$

The exact reduced operator is

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k), \quad (96)$$

with

$$H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k), \quad H_{XX}(k) = M_X^2 \mathbf{1}_X. \quad (97)$$

The shell-projected operator is

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}, \quad (98)$$

and the low-slot projector is

$$P_* = \frac{1}{2\pi i} \oint_{\Gamma} (z - H_{\text{shell}}(0))^{-1} dz. \quad (99)$$

Therefore the projected transport matrices are

$$M_i = P_* \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*} P_*. \quad (100)$$

In the adapted frame $(\hat{n}, e_1, e_2)$,

$$M_{\parallel} = \hat{n}_i M_i, \quad M_1 = e_{1i} M_i, \quad M_2 = e_{2i} M_i. \quad (101)$$

Under the locked Pauli selection rule,

$$\boxed{M_{\parallel} = \lambda_{\parallel} \sigma_3, \quad M_1 = \alpha \sigma_1 + \beta \sigma_2, \quad M_2 = \alpha \sigma_2 - \beta \sigma_1.} \quad (102)$$

Thus

$$\boxed{\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}),} \quad \boxed{\alpha = \frac{1}{4} \text{Tr}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4} \text{Tr}(\sigma_2 M_1 - \sigma_1 M_2).} \quad (103)$$

LVIII. 2. WHY GATE 2 IS ALREADY STRUCTURALLY COMPLETE

The branch-sensitive longitudinal structure is

$$\boxed{a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi).} \quad (104)$$

In the solved lowest-order realization,

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0.} \quad (105)$$

Hence Branch A is absent and the practical longitudinal route is forced onto Branch B.

The primitive Branch-B longitudinal transport block is

$$\boxed{V_{(\parallel),\text{tr}}^{\mu}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} \sigma_3,} \quad (106)$$

so the shell-localized projection gives

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.} \quad (107)$$

Therefore the theorem-level reduction is already complete: the only remaining work is the explicit microscopic evaluation of λ_c (or its *OO*-candidate realization) and of the overlap normalization.

LIX. 3. CURRENT BLOCKWISE TRANSPORT-LEVEL REDUCTION

Under the current working interpretation, the primitive longitudinal carrier is assigned to the *OO* candidate block:

$$\boxed{\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}.} \quad (108)$$

Hence the exact transport-level Gate-2 question becomes

$$\boxed{\lambda_{c;OO} \neq 0 ?} \quad (109)$$

If yes, then

$$\boxed{\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_3 \neq 0.} \quad (110)$$

Thus Gate 2 is no longer a structural or conceptual problem. It is the single finite extraction $\lambda_{c;OO} \neq 0$.

LX. 4. PRACTICAL ONE-MEDIATOR BRANCH-B FORMULA

At the shell-projected one-mediator level, the longitudinal coefficient takes the explicit form

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3. \quad (111)$$

Therefore the strongest practical sufficient condition is

$$\alpha_X \beta_X \neq 0, \quad C_{\parallel} \neq 0, \quad I_3 \neq 0 \implies \lambda_{\parallel} \neq 0. \quad (112)$$

Equivalently, at transport level,

$$\lambda_{c;OO} \neq 0, \quad I_3 \neq 0, \quad C_{\parallel} \neq 0 \implies \lambda_{\parallel} \neq 0. \quad (113)$$

LXI. 5. THE EXACT NEXT DIRECT STEP

The correct next direct execution is now one of the following two equivalent computations:

Route A: transport block extraction

Compute the OO -block first-order transport operator

$$V_{OO}^{\mu}(\xi) = \Pi_Q \mathbb{H}_{1,OO}^{\mu}(\xi) \Pi_Q$$

and extract

$$a_{3;OO}(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_{\mu} V_{OO}^{\mu}(\xi)).$$

Then read the $\theta'(\xi)^3$ -coefficient:

$$a_{3;OO}(\xi) = \frac{\lambda_{c;OO}}{2} \theta'(\xi)^3 + \dots$$

Route B: projected Bloch derivative

Compute directly

$$M_{\parallel} = P_* \partial_{k_{\parallel}} H_{\text{eff}}(k) \Big|_{k=k_*} P_*,$$

then extract

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}).$$

These are two equivalent realizations of the same remaining finite microscopic insertion problem.

LXII. 6. HONEST VERDICT

Hence the correct statement is:

$$\boxed{\text{Yes: within the current working branch, the theorem package is complete.}} \quad (114)$$

What remains is not another theorem, but the explicit finite insertion of either

$$\lambda_{c;OO},$$

or equivalently

$$M_{\parallel}, \quad C_{\parallel}.$$

So the project has now fully crossed from structural proof to explicit microscopic evaluation.

LXIII. FINAL SUMMARY

Within the refined locked + Class II working branch, the theorem-level carrier reduction is complete. The exact extraction layer is fixed:

$$H_{\text{eff}}(k) \rightarrow H_{\text{shell}}(p) \rightarrow P_* \rightarrow M_i \rightarrow (\lambda_{\parallel}, \alpha, \beta),$$

with

$$M_i = P_* \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*} P_*, \quad \lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}).$$

The solved lowest-order longitudinal branch is forced onto Branch B:

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0, \quad \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.$$

Under the current OO -candidate interpretation,

$$\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2},$$

so Gate 2 at transport level is reduced to the single finite question

$$\boxed{\lambda_{c;OO} \neq 0 \text{ ?}}$$

At the practical one-mediator shell-projected level,

$$\boxed{\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3.}$$

Therefore

$$\boxed{\alpha_X \beta_X \neq 0, \quad C_{\parallel} \neq 0, \quad I_3 \neq 0 \implies \lambda_{\parallel} \neq 0.}$$

So the working-branch proof is complete in theorem form. What remains is the explicit finite microscopic insertion: either compute

$$\lambda_{c;OO},$$

or equivalently compute

$$M_{\parallel} = P_* \partial_{k_{\parallel}} H_{\text{eff}}(k)|_{k=k_*} P_*, \quad \lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}).$$

LXIV. TERM-BY-TERM LONGITUDINAL AUDIT OF THE OO BLOCK

LXV. SHORT KOREAN EXPLANATION

Inside the current explicit library, the orientational diagonal block is

$$\mathcal{H}_{OO} = -Z_{O,k}\partial^2 + \mathcal{M}_{OO}^{\text{pot}}(\xi) + Y_k \mathcal{K}_{OO}^{\nabla\text{ani}}(\xi, \partial) + \mathcal{T}_{OO}(\xi, \partial_y).$$

The longitudinal witness is extracted from the first-order tangent transport part through

$$h_{\parallel;OO}(\xi) = \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_\mu \mathcal{M}_{OO,\text{inv}}^\mu(\xi)),$$

and the shell expansion defines

$$\ell_{0;OO}^{(1)}, \quad \ell_{0;OO}^{(3)}.$$

The direct audit shows:

1. the kinetic Laplacian piece $-Z_{O,k}\partial^2$ gives no first-order worldvolume transport;
2. the potential block $\mathcal{M}_{OO}^{\text{pot}}(\xi)$ gives no first-order transport;
3. the derivative-anisotropy block $Y_k \mathcal{K}_{OO}^{\nabla\text{ani}}$ does not generate a nonzero Branch-A coefficient in the solved lowest-order class;
4. therefore the only genuine leading longitudinal carrier inside OO is $\mathcal{T}_{OO}$.

Hence

$$\ell_{0;OO}^{(1)} = 0, \quad \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2},$$

and therefore

$$\lambda_{\parallel} = I_{\theta'^3} \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2} I_{\theta'^3}.$$

So the current Gate-2 problem is reduced to the single finite question

$$\lambda_{c;OO} \neq 0 \text{ ?}.$$

LXVI. 1. BLOCK TO BE AUDITED

On the background-dependent Hessian, the orientational diagonal block has the form

$$\boxed{\mathcal{H}_{OO} = -Z_{O,k}\partial^2 + \mathcal{M}_{OO}^{\text{pot}}(\xi) + Y_k \mathcal{K}_{OO}^{\nabla\text{ani}}(\xi, \partial) + \mathcal{T}_{OO}(\xi, \partial_y).} \quad (115)$$

The general tangent first-order shell-compatible transport operator is

$$\boxed{V^\mu(\xi) = a_0(\xi) n^\mu \Pi_Q + a_3(\xi) n^\mu \Sigma_3 + b_E(\xi) E^\mu{}_a \Sigma_a + b_O(\xi) J^\mu{}_a \Sigma_a + V_{\text{null}}^\mu(\xi).} \quad (116)$$

The load-bearing longitudinal profile is extracted by

$$\boxed{a_3(\xi) = \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_\mu V^\mu(\xi)).} \quad (117)$$

Restricted to the OO block, write

$$\boxed{h_{\parallel;OO}(\xi) = \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_\mu \mathcal{M}_{OO,\text{inv}}^\mu(\xi)).} \quad (118)$$

Its shell expansion defines the longitudinal coefficients:

$$\boxed{h_{\parallel;OO}(\xi) = \ell_{0;OO}^{(1)} \theta'(\xi) + \ell_{0;OO}^{(3)} \theta'(\xi)^3 + \dots} \quad (119)$$

LXVII. 2. WHY THE KINETIC AND POTENTIAL PIECES DO NOT CONTRIBUTE

The term

$$-Z_{O,k}\partial^2$$

is purely second-order in spacetime derivatives and does not generate a first-order worldvolume transport block of the form

$$-i\mathcal{M}^\mu\partial_{y^\mu}.$$

Hence it contributes neither to $\ell_{0;OO}^{(1)}$ nor to $\ell_{0;OO}^{(3)}$.

Likewise, the potential block

$$\mathcal{M}_{OO}^{\text{pot}}(\xi)$$

is multiplicative in ξ and contains no worldvolume derivative at first order. Therefore it also gives no direct contribution to the longitudinal transport coefficients.

Thus the first two pieces of (115) are inert for the present audit.

LXVIII. 3. WHY THERE IS NO BRANCH-A TERM IN THE SOLVED LOWEST-ORDER CLASS

The branch-sensitive longitudinal parametrization is

$$a_3(\xi) = \Lambda_{\parallel}^{(0)}\theta'(\xi) + \frac{\lambda_c}{2}\theta'(\xi)^3 + a_3^{\text{higher}}(\xi). \quad (120)$$

Inside the solved lowest-order realization, the linear Branch-A coefficient vanishes:

$$\Lambda_{\parallel}^{(0)} = 0. \quad (121)$$

Independently, the reflection-symmetry and projector-commutator audit shows that the longitudinal monomial

$$\theta'(\xi)n^\mu\Sigma_3$$

is forbidden, whereas the first allowed one is

$$\theta'(\xi)^3n^\mu\Sigma_3.$$

Therefore

$$\ell_0^{(1)} = 0 \quad (122)$$

exactly, and only the cubic primitive channel survives.

Consequently, the derivative-anisotropy piece

$$Y_k\mathcal{K}_{OO}^{\nabla\text{ani}}$$

does not produce a nonzero Branch-A coefficient in the current solved lowest-order class.

LXIX. 4. THE ONLY SURVIVING LEADING OO-BLOCK CARRIER

After eliminating the kinetic, potential, and Branch-A-forbidden contributions, the only genuine leading longitudinal carrier inside the OO block is

$$\mathcal{T}_{OO}. \quad (123)$$

Hence the OO -block verdict is

$$\mathcal{T}_{OO} \Rightarrow \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2} \text{ is the leading genuine longitudinal primitive carrier.} \quad (124)$$

Therefore the complete current blockwise result is

$$\ell_0^{(1)} = 0, \quad \ell_0^{(3)} = \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}. \quad (125)$$

LXX. 5. CONSEQUENCE FOR THE REDUCED LONGITUDINAL COEFFICIENT

The reduced Branch-B longitudinal coefficient is

$$\lambda_{\parallel} = I_{\theta'^3} \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2} I_{\theta'^3}. \quad (126)$$

Using the explicit overlap

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (127)$$

one gets

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_{c;OO} \theta_0^3 \kappa^3. \quad (128)$$

So the entire Gate-2 problem in the current explicit library reduces to the single finite question

$$\lambda_{c;OO} \neq 0 \text{ ?}. \quad (129)$$

LXXI. 6. WHAT IS PROVED AND WHAT IS NOT

What is proved:

1. the OO block is the natural primitive longitudinal carrier;
2. all Branch-A contributions vanish in the current solved lowest-order class;
3. the only leading genuine longitudinal carrier is $\mathcal{T}_{OO}$;
4. therefore

$$\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_{\theta'^3}.$$

What is not yet proved from the currently inserted microscopic data alone:

$$\lambda_{c;OO} \neq 0$$

as an explicitly evaluated number or symbolic expression.

So the theorem-level reduction is complete, but the last finite coefficient insertion is still open.

LXXII. FINAL SUMMARY

Inside the current explicit operator library, the longitudinal OO -block audit is closed term by term. The orientational diagonal block is

$$\mathcal{H}_{OO} = -Z_{O,k} \partial^2 + \mathcal{M}_{OO}^{\text{pot}} + Y_k \mathcal{K}_{OO}^{\nabla \text{ani}} + \mathcal{T}_{OO}.$$

Among these pieces:

1. $-Z_{O,k} \partial^2$ gives no direct first-order worldvolume transport contribution;
2. $\mathcal{M}_{OO}^{\text{pot}}$ gives no direct first-order worldvolume transport contribution;
3. $Y_k \mathcal{K}_{OO}^{\nabla \text{ani}}$ does not generate a nonzero Branch-A coefficient in the current solved lowest-order class;

4. hence the only genuine leading longitudinal carrier in the OO block is $\mathcal{T}_{OO}$.

Therefore

$$\ell_0^{(1)} = 0, \quad \ell_0^{(3)} = \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}.$$

The reduced longitudinal coefficient is then

$$\lambda_{\parallel} = I_{\theta'^3} \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2} I_{\theta'^3}.$$

Since

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \neq 0,$$

this becomes

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_{c;OO} \theta_0^3 \kappa^3.$$

Thus the entire current Gate-2 problem is reduced to the single finite question

$$\lambda_{c;OO} \neq 0 \text{ ?}.$$

What is proved is the complete term-by-term reduction. What remains open is the explicit microscopic evaluation of $\lambda_{c;OO}$ itself.

LXXIII. DIRECT ELIMINATION OF $\lambda_{c;OO}$ IN THE CURRENT BRANCH-B WORKING BRANCH

LXXIV. SHORT KOREAN EXPLANATION

At the present stage, the OO -block transport audit and the one-mediator shell projection provide two exact formulas for the same reduced longitudinal coefficient $\lambda_{\parallel}$.

The first formula is the transport-level OO -candidate expression:

$$\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_3.$$

The second is the shell-projected one-mediator Branch-B formula:

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3.$$

Since the explicit defect overlap satisfies

$$I_3 = I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \neq 0,$$

these two formulas can be identified directly. Therefore:

$$\lambda_{c;OO} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel}.$$

This is a genuine advance. It means that the remaining Gate-2 transport coefficient is no longer an independent unknown. It is reduced to the Bloch-calibration constant $C_{\parallel}$.

LXXV. 1. TRANSPORT-LEVEL BRANCH-B FORMULA

Under the current OO -candidate interpretation of the primitive longitudinal carrier, the OO -block transport audit gives

$$\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}, \quad (130)$$

and hence

$$\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_3, \quad I_3 := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3. \quad (131)$$

This is the exact transport-level Gate-2 formula in the current blockwise working interpretation.

LXXVI. 2. ONE-MEDIATOR SHELL-PROJECTED BRANCH-B FORMULA

For the same current refined locked + Class II working branch, the one-mediator shell projection yields

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3. \quad (132)$$

This is the same longitudinal reduced coefficient, expressed in terms of the mediator couplings and the longitudinal Bloch calibration constant $C_{\parallel}$.

LXXVII. 3. NONVANISHING OF THE DEFECT OVERLAP

For the explicit admissible defect profile, the overlap is analytically known:

$$I_3 = I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (133)$$

Hence

$$\theta_0 \kappa \neq 0 \implies I_3 \neq 0. \quad (134)$$

LXXVIII. 4. DIRECT IDENTIFICATION OF THE TRANSPORT COEFFICIENT

Equating (131) and (132), and using (134), one obtains

$$\frac{\lambda_{c;OO}}{2} I_3 = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3. \quad (135)$$

Since $I_3 \neq 0$, it cancels exactly, giving

$$\lambda_{c;OO} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel}. \quad (136)$$

This is the strongest direct symbolic closure available at the current stage for the OO -block longitudinal transport amplitude.

LXXIX. 5. EXPLICIT REDUCED LONGITUDINAL COEFFICIENT

Substituting (136) into (131), and then using (133), gives

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3 = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} C_{\parallel} \theta_0^3 \kappa^3. \quad (137)$$

Equivalently, if one prefers to keep the sign of I_3 explicit,

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} \left(-\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \right). \quad (138)$$

LXXX. 6. PRACTICAL GATE-2 CLOSURE THEOREM

[Practical Gate-2 closure in the current one-mediator Branch-B working branch] Assume the current refined locked + Class II working branch, the OO -candidate interpretation of the primitive longitudinal transport, and a nontrivial admissible defect profile with $\theta_0 \kappa \neq 0$. Then:

$$\alpha_X \beta_X \neq 0, \quad C_{\parallel} \neq 0 \implies \lambda_{c;OO} \neq 0 \implies \lambda_{\parallel} \neq 0. \quad (139)$$

From (136),

$$\alpha_X \beta_X \neq 0, \quad C_{\parallel} \neq 0 \implies \lambda_{c;OO} \neq 0.$$

Then by (131) and (134),

$$\lambda_{c;OO} \neq 0, \quad I_3 \neq 0 \implies \lambda_{\parallel} \neq 0.$$

This proves (139).

LXXXI. 7. WHAT HAS NOW ACTUALLY BEEN REMOVED

Before this identification, the remaining Gate-2 question was written as

$$\lambda_{c;OO} \neq 0 ?.$$

After (136), this is no longer an independent transport-level mystery. It has been reduced to

$$C_{\parallel} \neq 0 ?.$$

So the current working-branch longitudinal bottleneck is now:

$$\text{evaluate the longitudinal Bloch calibration constant } C_{\parallel}. \quad (141)$$

LXXXII. 8. RELATION TO THE PROJECTED BLOCH CORE BLOCK

In the canonical enriched shell core extraction, the longitudinal Dirac slope is

$$v_{\parallel} = \frac{1}{4} [\partial_{p_{\parallel}} (E_+(p) - E_-(p))]_{p=0}. \quad (142)$$

At the current reduced Branch-B level, this same coefficient is exactly $\lambda_{\parallel}$. Therefore (137) also gives the direct working-branch symbolic form of the core longitudinal slope:

$$\boxed{v_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3.} \quad (143)$$

Thus the remaining microscopic longitudinal problem has been reduced from

$$\lambda_{c;OO} \neq 0 ?$$

to the single projected Bloch normalization question

$$C_{\parallel} \neq 0 ?.$$

LXXXIII. 9. HONEST STATUS

What is now genuinely closed:

1. the OO -block transport amplitude is no longer independent;
2. it is exactly

$$\lambda_{c;OO} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel};$$

3. the reduced longitudinal coefficient is therefore

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3.$$

What remains open:

$$\boxed{\text{the explicit evaluation of } C_{\parallel} \text{ from the actual projected refined locked Bloch operator.}} \quad (144)$$

So the working-branch Gate-2 problem is now reduced to a single Bloch calibration constant.

LXXXIV. FINAL SUMMARY

In the current refined locked + Class II working branch, the OO -candidate transport formula and the one-mediator shell-projected formula give two exact expressions for the same reduced longitudinal coefficient:

$$\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_3, \quad \lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3.$$

Since the explicit admissible defect overlap is

$$I_3 = I_{\theta^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \neq 0,$$

these two expressions imply the exact identification

$$\boxed{\lambda_{c;OO} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel}.$$

Therefore the reduced longitudinal coefficient becomes

$$\boxed{\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3 = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} C_{\parallel} \theta_0^3 \kappa^3.}$$

Hence the practical Gate-2 closure statement is:

$$\boxed{\alpha_X \beta_X \neq 0, \quad C_{\parallel} \neq 0, \quad \theta_0 \kappa \neq 0 \implies \lambda_{c;OO} \neq 0 \implies \lambda_{\parallel} \neq 0.}$$

So the longitudinal working-branch bottleneck is no longer the abstract transport amplitude $\lambda_{c;OO}$. It is reduced to the single projected Bloch normalization question

$$\boxed{C_{\parallel} \neq 0 ? .}$$

Equivalently, in the canonical enriched shell core extraction,

$$v_{\parallel} = \frac{1}{4} [\partial_{p_{\parallel}} (E_+(p) - E_-(p))]_{p=0} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3.$$

Thus the remaining microscopic longitudinal problem is now the explicit evaluation of $C_{\parallel}$ from the actual projected refined locked Bloch operator.

LXXXV. DIRECT REDUCTION OF THE FINAL LONGITUDINAL BOTTLENECK TO A SYMMETRY-NONPROTECTION STATEMENT

LXXXVI. SHORT KOREAN EXPLANATION

At the current one-mediator Branch-B stage, the remaining quantity $C_{\parallel}$ is not an additional independent dynamical mystery. It is exactly the normalized σ_3 -trace of the projected longitudinal shell matrix $M^{\parallel}$.

Therefore the question

$$C_{\parallel} \neq 0 ?$$

is equivalent to

$$(\sigma_3 M^{\parallel}) \neq 0$$

provided $\alpha_X \beta_X \neq 0$ and $I_3 \neq 0$.

Since

$$(\sigma_3 M^{\parallel}) = M_{11} - M_{22},$$

the final longitudinal obstruction is reduced to:

$$M_{11} = M_{22} \quad \text{forced or not?}$$

If the carrier doublet u_1, u_2 is split by an exact grading/parity/chirality operator J , or if u_1, u_2 differ by any additional exact conserved microscopic label, then no label-preserving exchange symmetry can force $M_{11} = M_{22}$. In that case, the vanishing of the σ_3 -trace is only accidental fine tuning.

So the strongest honest next theorem is:

under an actual internal label split, $C_{\parallel} = 0$ is not symmetry-protected.

LXXXVII. 1. EXACT DEFINITION OF $C_{\parallel}$

In the one-mediator Branch-B shell audit, the reduced longitudinal coefficient is

$$\boxed{\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3.} \tag{145}$$

On the other hand, the projected shell matrix $M^{\parallel}$ satisfies

$$\boxed{\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}).} \tag{146}$$

Equating (145) and (146) gives the exact normalization identity

$$C_{\parallel} = \frac{2}{I_3} \left(-\frac{M_X^2}{\alpha_X \beta_X} \right) \frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}). \quad (147)$$

Hence, whenever

$$\alpha_X \beta_X \neq 0, \quad I_3 \neq 0, \quad (148)$$

one has the exact equivalence

$$C_{\parallel} \neq 0 \iff (\sigma_3 M^{\parallel}) \neq 0. \quad (149)$$

LXXXVIII. 2. REDUCTION TO DIAGONAL ASYMMETRY

Since $M^{\parallel}$ is a Hermitian 2×2 matrix, under the locked longitudinal selection rule it contributes through the σ_3 -channel:

$$M^{\parallel} = m_0 \mathbf{1}_2 + m_3 \sigma_3, \quad m_3 = \frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}). \quad (150)$$

In the diagonal carrier basis,

$$m_3 = \frac{1}{2} (M_{11} - M_{22}). \quad (151)$$

Therefore

$$C_{\parallel} \neq 0 \iff M_{11} \neq M_{22}. \quad (152)$$

So the remaining longitudinal calibration problem is exactly the diagonal asymmetry problem.

LXXXIX. 3. NON-FORBIDDANCE OF THE LONGITUDINAL CHANNEL

At the shell-reduced symmetry level, the longitudinal channel is not symmetry-forbidden. Rather, it is the unique singlet-to-singlet intertwiner

$$p_{\parallel} \longleftrightarrow \sigma_3.$$

Equivalently, in the minimal shell-allowed tensor space

$$V_{\min} = \text{span}\{B_{\parallel}, B_I, B_J\},$$

the direction $B_{\parallel}$ is allowed rather than excluded.

Thus the residual shell symmetry does not force

$$P_V T^{(\parallel)} = 0.$$

So longitudinal failure cannot be blamed on the minimal shell symmetry itself.

XC. 4. EXACT OBSTRUCTION THEOREM FOR EXCHANGE-PROTECTED EQUALITY

Now suppose there were an exact symmetry forcing

$$M_{11} = M_{22}.$$

This would require an exact label-preserving exchange involution S on the carrier doublet, swapping

$$Su_1 = u_2, \quad Su_2 = u_1,$$

while preserving the full microscopic label package.

Let J be an exact grading/parity/chirality involution. If

$$Ju_1 = \epsilon_1 u_1, \quad Ju_2 = \epsilon_2 u_2, \quad \epsilon_i \in \{\pm 1\},$$

and S preserves labels, then necessarily

$$[S, J] = 0.$$

Hence

$$\epsilon_1 = \epsilon_2$$

is necessary for such an exchange symmetry to exist.

Therefore, if

$$\boxed{\epsilon_1 \neq \epsilon_2}, \tag{153}$$

then no exact label-preserving exchange involution exists.

More generally, if u_1, u_2 differ by any additional exact conserved microscopic label, the same obstruction holds.

Thus:

[Symmetry-nonprotection of the longitudinal diagonal asymmetry] Assume $\alpha_X \beta_X \neq 0$ and $I_3 \neq 0$. If the reduced carrier doublet u_1, u_2 satisfies either

1. an exact grading/parity/chirality split $\epsilon_1 \neq \epsilon_2$, or
2. a mismatch in an additional exact conserved microscopic label,

then no exact label-preserving exchange symmetry can force

$$M_{11} = M_{22}.$$

Therefore the vanishing

$$(\sigma_3 M^\parallel) = 0$$

is not symmetry-protected. Equivalently,

$$C_\parallel = 0$$

can occur only by accidental fine tuning, not by exact structural necessity.

Under either hypothesis, an exact exchange involution preserving the full label package does not exist. Hence the equality $M_{11} = M_{22}$ is not enforced by exact symmetry. But by (151),

$$(\sigma_3 M^\parallel) = M_{11} - M_{22}.$$

Therefore its vanishing is accidental rather than symmetry-protected. By (149), the same conclusion holds for $C_\parallel$.

XCI. 5. STRONGEST HONEST GENERIC CONCLUSION

The previous theorem does not yet compute the actual numerical value of $C_\parallel$. But it gives the strongest honest generic statement currently available:

$$\boxed{\text{if the carrier doublet is actually label-split, then } C_\parallel = 0 \text{ is nongeneric.}} \tag{154}$$

Hence, under the same hypotheses and away from accidental fine tuning,

$$\boxed{C_\parallel \neq 0 \implies \lambda_\parallel \neq 0.} \tag{155}$$

XCII. 6. WHAT IS STILL NOT UNCONDITIONAL

What is not yet proved from the currently inserted microscopic data alone is the unconditional statement

$$C_{\parallel} \neq 0$$

for the final complete refined locked Bloch operator.

The remaining fork is exactly:

$$\boxed{\text{either verify an actual internal } J\text{-split between } u_1, u_2, \quad \text{or compute } M_{11} - M_{22} \text{ directly.}} \quad (156)$$

If the split is verified, then zero is not symmetry-protected. If not, one must evaluate the projected diagonal asymmetry directly.

XCIII. FINAL SUMMARY

The remaining longitudinal calibration constant is not an independent mystery. It is exactly the normalized σ_3 -trace of the projected longitudinal shell matrix:

$$\boxed{C_{\parallel} = \frac{2}{I_3} \left(-\frac{M_X^2}{\alpha_X \beta_X} \right) \frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}).}$$

Hence, provided

$$\alpha_X \beta_X \neq 0, \quad I_3 \neq 0,$$

one has the exact equivalence

$$\boxed{C_{\parallel} \neq 0 \iff \text{Tr}(\sigma_3 M^{\parallel}) \neq 0 \iff M_{11} \neq M_{22}.}$$

At the shell-reduced symmetry level, the longitudinal channel is not symmetry-forbidden. It is the unique singlet-to-singlet intertwiner

$$p_{\parallel} \longleftrightarrow \sigma_3.$$

Therefore vanishing of $C_{\parallel}$ cannot be attributed to the minimal shell symmetry.

Moreover, if the carrier doublet u_1, u_2 is split by an exact grading/parity/chirality operator J , or differs by any additional exact conserved microscopic label, then no label-preserving exchange involution can force

$$M_{11} = M_{22}.$$

Hence

$$C_{\parallel} = 0$$

is not symmetry-protected and can occur only by accidental fine tuning.

So the strongest honest remaining fork is:

$$\boxed{\text{either verify an actual internal } J\text{-split between } u_1, u_2, \quad \text{or compute } M_{11} - M_{22} \text{ directly.}}$$

Under the first outcome, the generic conclusion is

$$\boxed{C_{\parallel} \neq 0 \implies \lambda_{\parallel} \neq 0.}$$

XCIV. WHAT THE ACTUAL J -GRADING DOES AND DOES NOT PROVE

XCV. SHORT KOREAN EXPLANATION

The uploaded documents do contain an actual exact grading J , but it is a $2 + 2$ reflection grading, not yet an internal split of the carrier doublet itself.

Concretely, the low-energy modes are

$$u_a^+(\xi) = f_+(\xi)e_a(\xi), \quad u_a^-(\xi) = f_-(\xi)e_a(\xi),$$

with reflection

$$(Ju)(\xi) = u(-\xi), \quad Ju_a^\pm = \pm u_a^\pm.$$

Therefore the grading proves:

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \Pi_\pm \mathcal{D}_1 \Pi_\pm = 0,$$

so the leading reduced operator is block-off-diagonal.

However, this does *not* yet imply an internal label split between the two basis vectors inside a fixed two-dimensional carrier block. Indeed, u_1^+, u_2^+ have the same J -eigenvalue, and u_1^-, u_2^- have the same J -eigenvalue.

Hence the currently verified grading does not by itself force

$$M_{11} \neq M_{22}.$$

So the final longitudinal problem is no longer:

“is there an exact grading?”

That is already answered positively.

It is now:

$$\boxed{(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \neq 0 ?}$$

namely the direct projected diagonal-asymmetry computation.

XCVI. FINAL SUMMARY

The uploaded documents do contain an actual exact grading J , but it is a reflection-based $2 + 2$ grading:

$$Ju_a^\pm = \pm u_a^\pm, \quad \mathcal{K}_+ = \text{span}\{u_1^+, u_2^+\}, \quad \mathcal{K}_- = \text{span}\{u_1^-, u_2^-\}.$$

This is sufficient to prove the block-off-diagonal form of the leading reduced operator:

$$\Pi_\pm \mathcal{D}_1 \Pi_\pm = 0.$$

However, it is not an internal split of the carrier doublet within a fixed 2×2 block. Therefore it does not by itself force the diagonal asymmetry

$$M_{11} \neq M_{22}.$$

So the final longitudinal bottleneck is no longer the existence of a grading. That part is already closed.

The remaining problem is the direct projected diagonal-asymmetry computation:

$$\boxed{(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \neq 0,}$$

where

$$(W_+)_{iab} = \left\langle +G_*, e_a \left| \partial_{p_i} H_{\text{shell}}^{(N)}(0) \right| +G_*, e_b \right\rangle.$$

Equivalently,

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 \hat{n} \cdot \vec{W}_+) = \frac{1}{2} \left[(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \right].$$

Thus the remaining longitudinal task is a direct projected matrix-element calculation, not a further structural theorem.

XCVII. CORRECTED LONGITUDINAL EXTRACTION FORMULAS WITHOUT RAW \TR

XCVIII. SHORT KOREAN EXPLANATION

From now on, every trace is written as Tr , not $\cdot$. Moreover, for the longitudinal coefficient, the trace formula can be replaced completely by the diagonal-difference formula, which is safer and more explicit.

The exact projected extraction formula is

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*}(\sigma_3 P_* K_{\parallel} P_*), \quad K_{\parallel} = \hat{n}_i K_i.$$

Equivalently, in the 2×2 +-valley core basis,

$$\lambda_{\parallel} = \frac{1}{2} \left[(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \right].$$

This second formula avoids any trace notation and is the best one to use for the remaining direct computation.

XCIX. 1. CORRECTED EXACT EXTRACTION FORMULAS

The shell-projected microscopic derivative data are

$$K_i = \partial_{k_i} L_{\text{Bloch}}(k)|_{k=k_*}, \quad M_i = P_* K_i P_*. \quad (157)$$

The exact coefficient extraction formulas are

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}), \quad M_{\parallel} := \hat{n}_i M_i, \quad (158)$$

$$\alpha = \frac{1}{2} \text{Tr}(\sigma_1 M_1), \quad \beta = \frac{1}{2} \text{Tr}(\sigma_2 M_1). \quad (159)$$

Equivalently, if one uses the projected +-valley derivative block

$$(W_+)_{iab} := \left\langle +G_*, e_a \left| \partial_{p_i} H_{\text{shell}}^{(N)}(0) \right| +G_*, e_b \right\rangle, \quad a, b = 1, 2, \quad (160)$$

then the longitudinal coefficient is

$$\lambda_{\parallel} = \frac{1}{2} \left[(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \right]. \quad (161)$$

Thus the remaining longitudinal test is exactly

$$(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \neq 0. \quad (162)$$

C. 2. EXACT SCHUR-COMPLEMENT DERIVATIVE FORMULA

The current working branch uses

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k). \quad (163)$$

Differentiating with respect to $k_{\parallel} := \hat{n}_i k_i$ gives

$$\begin{aligned} K_{\parallel} &:= \partial_{k_{\parallel}} H_{\text{eff}}(k) \Big|_{k=k_*} \\ &= \partial_{k_{\parallel}} H_{\Psi\Psi}(k) \Big|_{k=k_*} - \partial_{k_{\parallel}} H_{\Psi X}(k) \Big|_{k=k_*} H_{XX}^{-1} H_{X\Psi} \\ &\quad - H_{\Psi X} H_{XX}^{-1} \partial_{k_{\parallel}} H_{X\Psi}(k) \Big|_{k=k_*} + H_{\Psi X} H_{XX}^{-1} \partial_{k_{\parallel}} H_{XX}(k) \Big|_{k=k_*} H_{XX}^{-1} H_{X\Psi}. \end{aligned} \quad (164)$$

In the present minimal Class II truncation,

$$H_{XX} = M_X^2 \mathbf{1}_X, \quad \partial_{k_\parallel} H_{XX} = 0, \quad (165)$$

so this simplifies exactly to

$$K_\parallel = \partial_{k_\parallel} H_{\Psi\Psi} \Big|_{k_*} - \frac{1}{M_X^2} \left[(\partial_{k_\parallel} H_{\Psi X}) \Big|_{k_*} H_{X\Psi} + H_{\Psi X} (\partial_{k_\parallel} H_{X\Psi}) \Big|_{k_*} \right]. \quad (166)$$

CI. 3. EXACT DIAGONAL-DIFFERENCE DECOMPOSITION

Projecting (166) to the +-valley internal basis gives

$$(\hat{n} \cdot \vec{W}_+)_{aa} = D_a^{(0)} - D_a^{(\text{med})}, \quad a = 1, 2, \quad (167)$$

where

$$D_a^{(0)} := \left\langle +G_*, e_a \left| \partial_{k_\parallel} H_{\Psi\Psi} \right|_{k_*} \right| +G_*, e_a \rangle, \quad (168)$$

and

$$D_a^{(\text{med})} := \frac{1}{M_X^2} \left\langle +G_*, e_a \left| (\partial_{k_\parallel} H_{\Psi X}) \Big|_{k_*} H_{X\Psi} + H_{\Psi X} (\partial_{k_\parallel} H_{X\Psi}) \Big|_{k_*} \right| +G_*, e_a \right\rangle. \quad (169)$$

Therefore

$$2\lambda_\parallel = \left(D_1^{(0)} - D_2^{(0)} \right) - \left(D_1^{(\text{med})} - D_2^{(\text{med})} \right). \quad (170)$$

This is the exact term-by-term longitudinal formula in the current working branch.

CII. 4. CORRECT FORMULA FOR $C_\parallel$ WITHOUT TRACE NOTATION

The Branch-B reduced coefficient also satisfies

$$\lambda_\parallel = -\frac{\alpha_X \beta_X}{2M_X^2} C_\parallel I_3, \quad (171)$$

with

$$I_3 = I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (172)$$

Combining (161) and (171) yields the exact calibration identity

$$C_\parallel = -\frac{M_X^2}{\alpha_X \beta_X I_3} \left[(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \right]. \quad (173)$$

Therefore, provided

$$\alpha_X \beta_X \neq 0, \quad I_3 \neq 0,$$

one has

$$C_\parallel \neq 0 \iff (\hat{n} \cdot \vec{W}_+)_{11} \neq (\hat{n} \cdot \vec{W}_+)_{22}. \quad (174)$$

CIII. 5. WHAT REMAINS TO BE COMPUTED

The final longitudinal task is no longer to discuss abstract traces or structural channels. It is exactly the direct projected matrix-element computation of

$$\boxed{(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22}}. \quad (175)$$

Equivalently, using (170), one may compute

$$\boxed{\left(D_1^{(0)} - D_2^{(0)}\right) - \left(D_1^{(\text{med})} - D_2^{(\text{med})}\right)}. \quad (176)$$

This is the safest exact formula to use going forward.

CIV. FINAL SUMMARY

The raw command `\Tr` is not used anymore. All trace formulas are written with `Tr`, and whenever possible the trace is replaced by an explicit diagonal-difference formula.

The corrected longitudinal extraction formulas are

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}), \quad M_{\parallel} = \hat{n}_i M_i,$$

or equivalently,

$$\boxed{\lambda_{\parallel} = \frac{1}{2} \left[(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \right]}.$$

In the current minimal Class II truncation,

$$H_{\text{eff}} = H_{\Psi\Psi} - H_{\Psi X} H_{XX}^{-1} H_{X\Psi}, \quad H_{XX} = M_X^2 \mathbf{1}_X,$$

so the longitudinal derivative is

$$\boxed{K_{\parallel} = \partial_{k_{\parallel}} H_{\Psi\Psi} \big|_{k_*} - \frac{1}{M_X^2} \left[(\partial_{k_{\parallel}} H_{\Psi X}) \big|_{k_*} H_{X\Psi} + H_{\Psi X} (\partial_{k_{\parallel}} H_{X\Psi}) \big|_{k_*} \right]}.$$

Hence

$$2\lambda_{\parallel} = \left(D_1^{(0)} - D_2^{(0)}\right) - \left(D_1^{(\text{med})} - D_2^{(\text{med})}\right),$$

with

$$D_a^{(0)} = \langle +G_*, e_a | \partial_{k_{\parallel}} H_{\Psi\Psi} | +G_*, e_a \rangle,$$

$$D_a^{(\text{med})} = \frac{1}{M_X^2} \langle +G_*, e_a | (\partial_{k_{\parallel}} H_{\Psi X}) H_{X\Psi} + H_{\Psi X} (\partial_{k_{\parallel}} H_{X\Psi}) | +G_*, e_a \rangle.$$

The exact calibration identity is

$$\boxed{C_{\parallel} = -\frac{M_X^2}{\alpha_X \beta_X I_3} \left[(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \right]}.$$

Therefore the final longitudinal task is exactly the direct projected matrix-element computation of

$$\boxed{(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22}}.$$

CV. NEXT DIRECT STEP: ACTUAL LONGITUDINAL SEED AUDIT VIA PROJECTED BLOCH DERIVATIVES

CVI. SHORT KOREAN EXPLANATION

The next step is no longer another theorem. It is the actual execution of the remaining finite longitudinal audit. There are two equivalent formulations of the same task:

1. the basis-overlap audit

$$\exists A : \ell_{\parallel A} \neq 0, \quad \ell_{\parallel A} = \hat{n}_i \hat{n}_j E_{A,ij}^3;$$

2. the projected Bloch derivative audit

$$P_* K_{\parallel} P_*, \quad K_{\parallel} = \partial_{k_{\parallel}} L_{\text{act}}(k) \Big|_{k=k_*}.$$

At the current project stage, the second formulation is the most executable one. It directly produces the longitudinal coefficient.

CVII. 1. WHAT EXACTLY MUST BE COMPUTED

Work in the folded-Bragg convention

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|.$$

The shell-relevant carrier space is

$$\mathcal{H}_* = \text{span}\{\Phi_{+,a}, \Phi_{-,a}\}_{a=1,2}, \quad \Phi_{\pm,a}(x) = e^{\pm i G_* \cdot x / 2} u_a.$$

The exact derivative blocks are

$$K_i = \partial_{k_i} L_{\text{act}}(k) \Big|_{k=k_*}, \quad M_i = P_* K_i P_*. \quad (177)$$

The longitudinal coefficient is then extracted by

$$\boxed{\lambda_{\parallel} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*}(\sigma^3 M_{\parallel}), \quad M_{\parallel} = \hat{n}_i M_i.} \quad (178)$$

Equivalently, in the +-valley 2×2 internal block,

$$\boxed{\lambda_{\parallel} = \frac{1}{2} \left[(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \right].} \quad (179)$$

Thus the remaining longitudinal question is exactly whether

$$\boxed{(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \neq 0.} \quad (180)$$

CVIII. 2. WHY THIS IS THE CORRECT NEXT STEP

The carrier problem has already been reduced to a finite overlap audit:

$$\exists A : \ell_{\parallel A} \neq 0 \implies \lambda_{\parallel} \neq 0.$$

The transverse half is already structurally solved, so the only remaining carrier-side honest frontier is the longitudinal primitive seed.

Hence the correct next executable step is not another remote-sector theorem and not another structural reorganization. It is exactly the actual longitudinal seed audit.

CIX. 3. MOST EFFICIENT EXECUTION ORDER

The most efficient order is:

Step 1

Write the explicit folded-Bragg periodic coefficients

$$\hat{U}_{\text{eff}}(G_*), \quad \hat{T}_{\text{eff}}^1(G_*), \quad \hat{T}_{\text{eff}}^2(G_*), \quad v_{\parallel}.$$

Step 2

Construct

$$K_{\parallel} = \partial_{k_{\parallel}} L_{\text{act}}(k) \Big|_{k=k_*}$$

from the final refined locked/Class II Bloch operator.

Step 3

Project to the shell:

$$M_{\parallel} = P_* K_{\parallel} P_*.$$

Step 4

Evaluate either

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*}(\sigma^3 M_{\parallel}),$$

or more safely,

$$\lambda_{\parallel} = \frac{1}{2} \left[(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \right].$$

Step 5

Conclude:

$$\lambda_{\parallel} \neq 0 \quad \Longleftrightarrow \quad \text{the actual longitudinal seed survives.}$$

CX. 4. IF ONE WANTS THE CALIBRATION VERSION INSTEAD

The same remaining computation can be phrased as the evaluation of the Bloch-side calibration constants

$$C_c, \quad C_E, \quad C_O.$$

For the longitudinal slot, the relevant one is C_c . The documents explicitly state that these constants are not conceptually open anymore; they just require explicit calibration runs.

So the same next step can also be described as the c -unit calibration run for the longitudinal channel.

CXI. 5. HONEST VERDICT

Therefore the next step is:

$$\boxed{\text{compute the actual projected derivative block } P_* K_{\parallel} P_* \text{ and read } \lambda_{\parallel}.} \quad (181)$$

Equivalently,

$$\boxed{\text{perform the actual longitudinal seed audit } \exists A : \ell_{\parallel A} \neq 0.} \quad (182)$$

This is the first real calculation that remains on the carrier side.

CXII. FINAL SUMMARY

The next step is not another structural theorem. It is the explicit execution of the remaining finite longitudinal audit.

The carrier-side longitudinal problem is already reduced to

$$\exists A : \ell_{\parallel A} \neq 0, \quad \ell_{\parallel A} = \hat{n}_i \hat{n}_j E_{A,ij}^3,$$

and this is equivalent to the projected Bloch derivative computation.

The concrete executable task is:

$$K_{\parallel} = \partial_{k_{\parallel}} L_{\text{act}}(k) \big|_{k=k_*}, \quad M_{\parallel} = P_* K_{\parallel} P_*.$$

Then the longitudinal coefficient is read from

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*}(\sigma^3 M_{\parallel}),$$

or equivalently,

$$\boxed{\lambda_{\parallel} = \frac{1}{2} \left[(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \right].}$$

Therefore the next direct step is:

$$\boxed{\text{compute } P_* K_{\parallel} P_* \text{ for the final refined locked/Class II Bloch operator and test whether } \lambda_{\parallel} \neq 0.}$$

Equivalently, this is the actual longitudinal seed audit.

CXIII. DIRECT COMPRESSION OF THE LONGITUDINAL AUDIT TO THE PURE JJ -SECTOR

CXIV. SHORT KOREAN EXPLANATION

Inside the current refined locked + Class II working branch, the longitudinal primitive channel is no longer a broad projected-derivative mystery.

The uploaded branch-level source-classification theorem shows that, at principal shell order,

1. the pure K -sector has no leading longitudinal component,
2. the mixed JK -sector generates transverse Pauli directions only,
3. the KK -sector remains in the transverse family at principal longitudinal shell order,
4. therefore the primitive longitudinal coefficient is localized entirely to the pure JJ -sector.

Hence the longitudinal problem is reduced to a single coefficient:

$$\lambda_c^{(JJ)} \neq 0,$$

or equivalently to the axial shell-singlet amplitude $j^2 \neq 0$.

CXV. 1. STARTING POINT: SCHUR-COMPLEMENT CHANNEL SPLIT

In the current Class II branch, the shell-relevant witness coefficients are extracted from the Schur-complement correction

$$\Delta_X = -\frac{1}{M_X^2}(\alpha_X \mathcal{J} + \beta_X \mathcal{K})(\alpha_X \mathcal{J} + \beta_X \mathcal{K})^\dagger. \quad (183)$$

Therefore the physical shell channels split as

$$\boxed{JJ, \quad JK, \quad KK.} \quad (184)$$

This is the correct post-Schur bookkeeping layer, rather than the pre-Schur tensor

$$\alpha_X \mathbf{J} + \beta_X \mathbf{K}.$$

CXVI. 2. SHELL-ADAPTED LEADING ANSATZ

Fix the shell frame

$$(\hat{n}, e_1, e_2), \quad \hat{n} = \frac{G_*}{|G_*|},$$

and the tangent Pauli basis

$$(\Sigma_1, \Sigma_2, \Sigma_3).$$

At the current explicit leading shell level, the J -sector is purely axial:

$$\boxed{J_{i,r}^a(G_*; 0) = \hat{n}_i j^a (\Sigma_3)_r^a + J_{i,r}^{\text{null}a}}, \quad (185)$$

where the null part is orthogonal to the shell witness projectors.

Likewise, the K -sector is purely transverse:

$$\boxed{K_{i,r}^a(G_*; 0) = \hat{n}_i k^a \left[\Lambda_E^{(0)}(\Sigma_1)_r^a + \Lambda_O^{(0)}(\Sigma_2)_r^a \right] + K_{i,r}^{\text{null}a}}. \quad (186)$$

Thus, at principal shell level, J carries the longitudinal Σ_3 -singlet direction and K carries only the transverse $\Sigma_{1,2}$ directions.

CXVII. 3. LONGITUDINAL SOURCE CLASSIFICATION

The uploaded theorem states that, in the current refined locked + Class II working branch, the shell-relevant primitive odd longitudinal singlet $n^\mu \Sigma_3$ can be generated only by the pure JJ -type sector at principal order.

Equivalently:

1. the pure K -sector has no leading longitudinal component,

$$\boxed{c_{\parallel}^{(K)} = 0;} \quad (187)$$

2. the mixed JK -sector generates transverse Pauli directions only;
3. the KK -sector remains in the transverse family at principal longitudinal shell order;
4. therefore the microscopic longitudinal coefficient is localized to the JJ -sector:

$$\boxed{\lambda_c = \lambda_c^{(JJ)}}. \quad (188)$$

So the longitudinal carrier-side frontier is no longer a general $K_{\parallel}$ -audit. It is the single finite check

$$\boxed{\lambda_c^{(JJ)} \neq 0.} \quad (189)$$

CXVIII. 4. EXPLICIT BRANCH DICTIONARY

The current explicit branch dictionary is

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2, \quad (190)$$

while the transverse coefficients are

$$\alpha = I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}, \quad \beta = I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}. \quad (191)$$

Hence the longitudinal/transverse split is completely sharp:

$$JJ \rightsquigarrow \lambda_{\parallel}, \quad JK/KK \rightsquigarrow (\alpha, \beta).$$

The primitive longitudinal witness is therefore

$$\lambda_c^{(JJ)} = \frac{\alpha_X^2}{M_X^2} j^2. \quad (192)$$

CXIX. 5. IMMEDIATE CONSEQUENCE FOR THE ACTUAL LONGITUDINAL AUDIT

The defect overlap is already known analytically:

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (193)$$

Therefore, for $\theta_0 \kappa \neq 0$,

$$I_{\theta'^3} \neq 0.$$

Substituting (192) into (190), we get

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2. \quad (194)$$

Hence:

$$\alpha_X \neq 0, \quad j^2 \neq 0, \quad \theta_0 \kappa \neq 0 \implies \lambda_{\parallel} \neq 0. \quad (195)$$

So the longitudinal task has been reduced even further:

$$\text{evaluate } j^2 \text{ in the actual refined locked basis.} \quad (196)$$

CXX. 6. RELATION TO THE PROJECTED DERIVATIVE LANGUAGE

The projected derivative language is still correct:

$$M_i = P_* K_i P_*, \quad \lambda_{\parallel} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*}(\sigma^3 M_{\parallel}).$$

But within the current explicit branch, this is no longer the most economical longitudinal parametrization.

The sharper branch-level statement is:

$M_{\parallel}$ receives its principal shell-relevant longitudinal contribution only from the pure JJ sector.

Thus, instead of computing the full diagonal asymmetry blindly, one may compute directly the axial shell-singlet coefficient j^2 , which is equivalent to $\lambda_c^{(JJ)}$.

CXXI. 7. HONEST STATUS

What is now genuinely strengthened:

1. the longitudinal primitive channel is no longer spread over $JJ/JK/KK$;
2. it is localized to the pure JJ -sector at principal shell order;
3. the longitudinal coefficient is explicitly

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2.$$

What remains open:

the actual symbolic or numerical evaluation of j^2 in the refined locked basis.

(197)

So the remaining longitudinal frontier is now a single axial shell-singlet coefficient, not a broad matrix audit.

CXXII. FINAL SUMMARY

In the current refined locked + Class II working branch, the longitudinal primitive shell channel is localized entirely to the pure JJ -sector at principal shell order.

More precisely:

1. the pure K -sector has no leading longitudinal component,
- $$c_{\parallel}^{(K)} = 0;$$
2. the mixed JK -sector generates transverse Pauli directions only;
 3. the KK -sector remains in the transverse family at principal longitudinal shell order;
 4. therefore

$$\lambda_c = \lambda_c^{(JJ)}.$$

The explicit current branch dictionary is

$\lambda_{\parallel} = \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2,$

and

$\alpha = I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}, \quad \beta = I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}.$

Hence the longitudinal/transverse split is sharp:

$$JJ \rightsquigarrow \lambda_{\parallel}, \quad JK/KK \rightsquigarrow (\alpha, \beta).$$

Since

$$I_{\theta^3} \neq 0 \quad \text{for} \quad \theta_0 \kappa \neq 0,$$

the longitudinal nonvanishing problem is reduced to the single check

$j^2 \neq 0.$

Equivalently,

$\lambda_c^{(JJ)} = \frac{\alpha_X^2}{M_X^2} j^2 \neq 0 \implies \lambda_{\parallel} \neq 0.$

So the remaining longitudinal frontier is no longer a broad projected-matrix audit. It is the explicit symbolic or numerical evaluation of the axial shell-singlet coefficient j^2 in the actual refined locked basis.

CXXIII. DIRECT PROJECTOR-CONTRACTION FORMULA FOR THE AXIAL SHELL-SINGLET COEFFICIENT j^2

CXXIV. SHORT KOREAN EXPLANATION

At the current refined locked + Class II working-branch level, the longitudinal primitive coefficient is localized entirely to the pure JJ -sector. Therefore the only remaining longitudinal computation is the extraction of the axial shell-singlet scalar j^2 .

The correct way to write this is not as a broad matrix audit, but as a quadratic projector contraction on the shell-normal Σ_3 -singlet channel.

CXXV. 1. SOURCE LOCALIZATION TO THE PURE JJ -SECTOR

The current branch-level source classification is:

$$c_{\parallel}^{(K)} = 0, \quad c_{\parallel}^{(JK)} = 0, \quad c_{\parallel}^{(KK)} = 0 \quad \text{at principal longitudinal shell order,} \quad (198)$$

whereas

$$c_{\parallel}^{(JJ)} \text{ is allowed and is the unique principal Class-II longitudinal source.} \quad (199)$$

Hence

$$\boxed{\lambda_c = \lambda_c^{(JJ)}} \quad (200)$$

CXXVI. 2. AXIAL SHELL FORM OF THE J -SECTOR

At the same principal shell level, the axial J -sector takes the shell-normal form

$$\boxed{J_{i,r}^a(G_*, 0) = \hat{n}_i j^a(\Sigma_3)_r + J_{i,r}^{\text{null}a}} \quad (201)$$

where the null part is orthogonal to the shell witness projector by construction.

Thus the only shell-relevant J -data for the longitudinal channel are the coefficients j^a .

CXXVII. 3. CANONICAL QUADRATIC LONGITUDINAL PROJECTOR

Define the quadratic shell-longitudinal projector by

$$\boxed{\Pi_{\parallel}^{(JJ)}[X, Y] := \frac{1}{2} \delta_{ab} \hat{n}^i \hat{n}^{i'} (\Sigma_3)^r{}_{r'} X_{i,r}^a \overline{Y_{i',br'}}} \quad (202)$$

This is the unique shell-normal singlet projector because it contracts:

1. the two spatial indices onto the shell normal $\hat{n}_i \hat{n}_{i'}$,
2. the tangent Pauli slot onto Σ_3 ,
3. the internal mediator labels with δ_{ab} .

Therefore the axial shell-singlet coefficient is defined by

$$\boxed{j^2 := \Pi_{\parallel}^{(JJ)}[J, J]} \quad (203)$$

CXXVIII. 4. EVALUATION ON THE AXIAL SHELL FORM

Substituting (201) into (203), and using orthogonality of the null part, gives

$$\begin{aligned} j^2 &= \frac{1}{2} \delta_{ab} \hat{n}^i \hat{n}^{i'} (\Sigma_3)^r{}_{r'} \left(\hat{n}_i j^a (\Sigma_3)_r \right) \overline{\left(\hat{n}_{i'} j^b (\Sigma_3)^{r'} \right)} \\ &= \frac{1}{2} (\hat{n}^i \hat{n}_i) (\hat{n}^{i'} \hat{n}_{i'}) \delta_{ab} j^a \overline{j^b} (\Sigma_3)^r{}_{r'} (\Sigma_3)_r{}^{r'}. \end{aligned} \quad (204)$$

Since $\hat{n}^i \hat{n}_i = 1$ and

$$(\Sigma_3)^r{}_{r'} (\Sigma_3)_r{}^{r'} = 2, \quad (205)$$

the normalization in (202) gives

$$\boxed{j^2 = \delta_{ab} j^a \overline{j^b}.} \quad (206)$$

Thus j^2 is literally the squared magnitude of the axial shell-singlet J -coefficient vector.

CXXIX. 5. DIRECT LONGITUDINAL DICTIONARY

The current explicit branch dictionary states

$$\boxed{\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2.} \quad (207)$$

Equivalently,

$$\boxed{\lambda_c^{(JJ)} = \frac{\alpha_X^2}{M_X^2} j^2.} \quad (208)$$

Hence the longitudinal primitive coefficient is completely determined by the single positive shell-singlet contraction j^2 .

CXXX. 6. IMMEDIATE NONVANISHING CRITERION

For the explicit admissible defect profile,

$$\boxed{I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.} \quad (209)$$

Therefore, whenever $\theta_0 \kappa \neq 0$,

$$I_{\theta'^3} \neq 0.$$

Combining this with (207), we obtain

$$\boxed{\alpha_X \neq 0, \quad j^2 \neq 0, \quad \theta_0 \kappa \neq 0 \quad \implies \quad \lambda_{\parallel} \neq 0.} \quad (210)$$

Thus the remaining carrier-side longitudinal frontier is exactly the verification of

$$j^2 \neq 0.$$

CXXXI. 7. WHAT IS ACTUALLY STILL OPEN

The formula (206) is exact at the present shell-reduced working-branch level. What is not yet inserted is the actual refined-locked microscopic value of the coefficients j^a .

So the remaining longitudinal computation is:

$$\boxed{\text{compute the actual shell-relevant axial coefficients } j^a \text{ in the refined locked basis and form } j^2 = \delta_{ab} j^a \overline{j^b}.} \quad (211)$$

This is no longer a broad conceptual problem. It is a finite coefficient extraction from the actual J -kernel.

CXXXII. FINAL SUMMARY

In the current refined locked + Class II working branch, the primitive longitudinal shell singlet is sourced only by the pure JJ -sector at principal shell order:

$$c_{\parallel}^{(K)} = 0, \quad c_{\parallel}^{(JK)} = 0, \quad c_{\parallel}^{(KK)} = 0, \quad c_{\parallel}^{(JJ)} \neq 0 \text{ allowed.}$$

The shell-relevant axial J -sector has the canonical form

$$J_{i,r}^a(G_*; 0) = \hat{n}_i j^a (\Sigma_3)_r + J_{i,r}^{a,\text{null}}.$$

The correct quadratic shell-longitudinal projector is

$$\Pi_{\parallel}^{(JJ)}[X, Y] = \frac{1}{2} \delta_{ab} \hat{n}^i \hat{n}^{i'} (\Sigma_3)^r{}_{r'} X_{i,r}^a \overline{Y_{i',r'}^{br'}}.$$

Applying it to the J -sector defines the axial shell-singlet coefficient

$$\boxed{j^2 := \Pi_{\parallel}^{(JJ)}[J, J] = \delta_{ab} j^a \overline{j^b}.$$

Therefore the current branch dictionary becomes

$$\boxed{\lambda_c^{(JJ)} = \frac{\alpha_X^2}{M_X^2} j^2, \quad \lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2.}$$

Since

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \neq 0 \quad \text{for } \theta_0 \kappa \neq 0,$$

the nonvanishing criterion is

$$\boxed{\alpha_X \neq 0, \quad j^2 \neq 0, \quad \theta_0 \kappa \neq 0 \implies \lambda_{\parallel} \neq 0.}$$

So the remaining longitudinal frontier is exactly the explicit evaluation of the shell-relevant axial coefficients j^a , equivalently the scalar j^2 , in the actual refined locked basis.

CXXXIII. FINAL SCALAR REDUCTION OF THE PURE JJ LONGITUDINAL INGREDIENT

CXXXIV. SHORT KOREAN EXPLANATION

A correction is needed.

The most accurate final microscopic scalar for the pure- JJ longitudinal channel is not best written as a generic j^2 anymore. In the latest refined locked/Class-II reduction, the principal JJ axial coefficient tensor collapses to a single scalar

$$\Xi_{JJ},$$

and this scalar is then identified with the invariant

$$L_* := m_{\parallel} u_{\parallel}$$

up to a fixed normalization factor.

So the remaining carrier-side longitudinal frontier is more precisely

$$\Delta_{00} \neq 0, \quad L_* \neq 0,$$

rather than a broad unresolved tensor audit.

CXXXV. 1. COLLAPSE OF THE PURE- JJ TENSOR TO ONE SCALAR

The current refined locked + Class II reduction implies that the principal pure- JJ axial coefficient tensor has the isotropic shell form

$$\boxed{C_{jm}^{(JJ)} = \Xi_{JJ} \delta_{jm}.} \quad (212)$$

Therefore

$$\boxed{\hat{n} \cdot C^{(JJ)} G_* = \Xi_{JJ} |G_*|.} \quad (213)$$

So the second microscopic longitudinal ingredient is not a rank-two tensor anymore. It is reduced to the single scalar test

$$\boxed{\Xi_{JJ} \neq 0.} \quad (214)$$

CXXXVI. 2. CANONICAL LOCKED BASIS

Fix the canonical locked basis

$$\boxed{z_0 = (1, 0, 0)^T, \quad u_1 = (0, 1, 0)^T, \quad u_2 = (0, 0, 1)^T.} \quad (215)$$

In this basis the tangent Pauli frame is fixed canonically, and the longitudinal JJ scalar can be expressed in terms of two shell-relevant invariants:

1. the background singlet alignment

$$m_{\parallel} = m^3(P_0),$$

2. the shell-normal J -singlet bias

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 \hat{n}^j \mathcal{U}_j(0)).$$

Thus define

$$\boxed{L_* := m_{\parallel} u_{\parallel}.} \quad (216)$$

CXXXVII. 3. IDENTIFICATION OF Ξ_{JJ} WITH L_*

In the current minimal explicit Class II split, the pure- JJ longitudinal scalar is not an independent new object. It is exactly the invariant L_* up to a fixed normalization:

$$\boxed{\Xi_{JJ} = \mathcal{N}_{JJ} L_*, \quad L_* = m_{\parallel} u_{\parallel}.} \quad (217)$$

Therefore the second microscopic longitudinal ingredient is reduced from a tensor problem to a single invariant product.

CXXXVIII. 4. IMMEDIATE FOURIER CONSEQUENCE

Previously one had

$$\widehat{\mathcal{A}}_{JJ}(G_*) = i \Xi_{JJ} |G_*| A_{G_*}. \quad (218)$$

Substituting (217),

$$\boxed{\widehat{\mathcal{A}}_{JJ}(G_*) = i \mathcal{N}_{JJ} L_* |G_*| A_{G_*}.} \quad (219)$$

Hence, if

$$\boxed{L_* \neq 0, \quad A_{G_*} \neq 0, \quad \alpha_X \neq 0,} \quad (220)$$

then

$$\boxed{\widehat{\mathcal{A}}_{JJ}(G_*) \neq 0, \quad \widehat{V}_{\parallel}^{(JJ)}(G_*) \neq 0.} \quad (221)$$

CXXXIX. 5. LONGITUDINAL CLOSURE CHAIN

Once the first-shell dominant carrier asymmetry condition

$$\Delta_{00} \neq 0$$

also holds, the longitudinal chain becomes

$$\boxed{\Delta_{00} \neq 0, \quad L_* \neq 0, \quad A_{G_*} \neq 0, \quad \alpha_X \neq 0 \implies m_3 \neq 0 \implies \lambda_c^{(JJ)} \neq 0 \implies \lambda_{\parallel} \neq 0.} \quad (222)$$

So the last carrier-side longitudinal frontier is reduced completely to the two scalar checks

$$\boxed{\Delta_{00} \neq 0, \quad L_* = m_{\parallel} u_{\parallel} \neq 0.} \quad (223)$$

CXL. 6. EXPLICIT EXTRACTION FORMULAS FOR THE TWO SCALARS

The first scalar is the valley Bloch asymmetry

$$\boxed{\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_- = \overline{A_1^{(+)}} A_1^{(-)} - \overline{A_2^{(+)}} A_2^{(-)}.} \quad (224)$$

The second scalar is

$$\boxed{L_* = m_{\parallel} u_{\parallel},} \quad (225)$$

with

$$\boxed{m_{\parallel} = m^3(P_0),} \quad (226)$$

and

$$\boxed{u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 \hat{n}^j \mathcal{U}_j(0)) = \frac{1}{2} [(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22}].} \quad (227)$$

Thus

$$\boxed{L_* = \frac{m_{\parallel}}{2} [(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22}].} \quad (228)$$

CXLI. 7. HONEST STATUS

What is now genuinely closed is the reduction:

1. the pure- JJ longitudinal ingredient is no longer a tensor problem;
2. it is one scalar Ξ_{JJ} ;
3. this scalar is exactly the invariant $L_* = m_{\parallel} u_{\parallel}$ up to normalization;
4. therefore the carrier-side longitudinal frontier is reduced to the pair

$$\Delta_{00} \neq 0, \quad L_* \neq 0.$$

What is not yet explicitly inserted is the actual refined-locked numerical or symbolic evaluation of

$$\Delta_{00}, \quad m_{\parallel}, \quad u_{\parallel}.$$

So the remaining work is a finite scalar extraction problem, not a conceptual source-classification problem.

CXLI. FINAL SUMMARY

A correction is required for the final pure- JJ longitudinal scalar language.

The most accurate current reduction is not best expressed solely in terms of a generic j^2 . Instead, the principal pure- JJ axial coefficient tensor collapses to the scalar

$$C_{jm}^{(JJ)} = \Xi_{JJ} \delta_{jm},$$

and this scalar is then identified, in the canonical locked basis, with the invariant

$$\Xi_{JJ} = \mathcal{N}_{JJ} L_*, \quad L_* = m_{\parallel} u_{\parallel}.$$

Hence the carrier-side longitudinal frontier is reduced to the two scalar checks

$$\Delta_{00} \neq 0, \quad L_* = m_{\parallel} u_{\parallel} \neq 0.$$

The explicit formulas are:

$$\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_-,$$

$$m_{\parallel} = m^3(P_0),$$

$$u_{\parallel} = \frac{1}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right],$$

and therefore

$$L_* = \frac{m_{\parallel}}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right].$$

If

$$\Delta_{00} \neq 0, \quad L_* \neq 0, \quad A_{G_*} \neq 0, \quad \alpha_X \neq 0,$$

then

$$\widehat{V}_{\parallel}^{(JJ)}(G_*) \neq 0 \implies m_3 \neq 0 \implies \lambda_c^{(JJ)} \neq 0 \implies \lambda_{\parallel} \neq 0.$$

So the remaining longitudinal task is the explicit scalar extraction of

$$\Delta_{00}, \quad m_{\parallel}, \quad u_{\parallel}.$$

CXLIII. DIRECT EXTRACTION OF $u_{\parallel}$ AND $L_* = m_{\parallel} u_{\parallel}$ FROM THE CANONICAL J -KERNEL

CXLIV. SHORT KOREAN EXPLANATION

At the present stage, the longitudinal carrier test is already reduced to two finite scalars:

$$\Delta_{00}, \quad L_* = m_{\parallel} u_{\parallel}.$$

The first is the valley Bloch asymmetry extracted from the two valley blocks $E_{\pm}(0)$. The second is the shell-normal J -singlet bias multiplied by the background selector.

The key new point is that the canonical shell form of the J -kernel already shows exactly how $u_{\parallel}$ is extracted: the shell-normal projection selects the Σ_3 coefficient and kills the transverse $\Sigma_{1,2}$ pieces.

CXLV. 1. CANONICAL SHELL FORM OF THE J -KERNEL

In the refined locked basis, the shell-relevant J -harmonic coefficient has the canonical form

$$\boxed{J_{i,r}^a(G_*; 0) = m_{\parallel} \delta^{a3} i(G_*)_i A_{G_*} \left[u_{\parallel} \hat{n}_j \Sigma_3 + u_E (e_{1j} \Sigma_1 + e_{2j} \Sigma_2) + u_O (-e_{2j} \Sigma_1 + e_{1j} \Sigma_2) \right]_r^b.} \quad (229)$$

Thus the shell-normal direction $\hat{n}_j$ coupled to the Pauli singlet Σ_3 is controlled by a single scalar:

$$u_{\parallel}.$$

CXLVI. 2. REDUCED INTERVALLEY J -MATRIX

Define the reduced intervalley matrix

$$\boxed{(\mathcal{U}_j(0))_{ab} = \langle -, a | \hat{J}_j(G_*; 0) | +, b \rangle, \quad a, b = 1, 2.} \quad (230)$$

Projecting (229) onto the shell-normal direction gives

$$\hat{n}^j \mathcal{U}_j(0) = u_{\parallel} \Sigma_3 + u_E (\hat{n}^j e_{1j} \Sigma_1 + \hat{n}^j e_{2j} \Sigma_2) + u_O (-\hat{n}^j e_{2j} \Sigma_1 + \hat{n}^j e_{1j} \Sigma_2). \quad (231)$$

Since the shell frame is orthonormal,

$$\hat{n}^j e_{1j} = 0, \quad \hat{n}^j e_{2j} = 0,$$

the transverse pieces vanish and one gets

$$\boxed{\hat{n}^j \mathcal{U}_j(0) = u_{\parallel} \Sigma_3.} \quad (232)$$

Therefore the shell-normal projected J -block is purely diagonal in the Pauli basis:

$$\boxed{\hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{\parallel} & 0 \\ 0 & -u_{\parallel} \end{pmatrix}.} \quad (233)$$

CXLVII. 3. EXACT EXTRACTION FORMULA FOR $u_{\parallel}$

From (233), the most robust formula is the diagonal-difference form

$$\boxed{u_{\parallel} = \frac{1}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right].} \quad (234)$$

Equivalently,

$$\boxed{u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 \hat{n}^j \mathcal{U}_j(0)).} \quad (235)$$

Thus $u_{\parallel}$ is literally the diagonal asymmetry of the shell-normal projected J -block.

CXLVIII. 4. EXACT EXTRACTION FORMULA FOR $m_{\parallel}$

In the canonical locked frame,

$$\boxed{m^a(P_0) = m_{\parallel} \delta^{a3}.} \quad (236)$$

Therefore

$$\boxed{m_{\parallel} = m^3(P_0).} \quad (237)$$

No further reduction is needed: it is simply the $a = 3$ component of the background-current selector.

CXLIX. 5. FINAL SCALAR INVARIANT

Combining (234) and (237), the unresolved longitudinal scalar is

$$\boxed{L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right].} \quad (238)$$

Equivalently,

$$\boxed{L_* = \frac{m_{\parallel}}{2} \text{Tr}(\Sigma_3 \hat{n}^j \mathcal{U}_j(0)).} \quad (239)$$

CL. 6. THE OTHER SCALAR: Δ_{00}

The valley Bloch asymmetry is extracted from the low eigenvectors of the two valley blocks $E_{\pm}(0)$:

$$E_+(0)\eta_+ = \varepsilon_+\eta_+, \quad E_-(0)\eta_- = \varepsilon_-\eta_-, \quad \eta_{\pm} = \begin{pmatrix} A_1^{(\pm)} \\ A_2^{(\pm)} \end{pmatrix}. \quad (240)$$

Then

$$\boxed{\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_- = \overline{A_1^{(+)}} A_1^{(-)} - \overline{A_2^{(+)}} A_2^{(-)}.} \quad (241)$$

CLI. 7. FINAL LONGITUDINAL WITNESS FORMULA

The current canonical branch dictionary is

$$\boxed{\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}.} \quad (242)$$

The defect overlap is already known explicitly:

$$\boxed{I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.} \quad (243)$$

Therefore, if

$$\boxed{\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0,} \quad (244)$$

then the dominant first-shell longitudinal witness survives and

$$\boxed{\lambda_{\parallel} \neq 0.} \quad (245)$$

CLII. 8. HONEST REMAINING COMPUTATION

So the remaining carrier-side longitudinal work is now exactly the following finite scalar extraction:

1. compute the two 2×2 valley blocks $E_{\pm}(0)$ and diagonalize them to obtain $\eta_{\pm}$, hence Δ_{00} ;
2. compute the reduced intervalley J -matrix $\mathcal{U}_j(0)$;
3. read

$$u_{\parallel} = \frac{1}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right];$$

4. read $m_{\parallel} = m^3(P_0)$;
5. form

$$L_* = m_{\parallel} u_{\parallel}.$$

No structural ambiguity remains. Only these finite matrix elements remain to be inserted.

CLIII. FINAL SUMMARY

In the current refined locked + Class II working branch, the carrier-side longitudinal frontier is reduced to two finite scalars:

$$\Delta_{00}, \quad L_* = m_{\parallel} u_{\parallel}.$$

The canonical shell form of the J -kernel already shows how $u_{\parallel}$ is extracted. After shell-normal projection, the transverse $\Sigma_{1,2}$ components vanish by orthogonality of the shell frame, leaving only the Σ_3 coefficient:

$$\hat{n}^j \mathcal{U}_j(0) = u_{\parallel} \Sigma_3.$$

Therefore

$$u_{\parallel} = \frac{1}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right].$$

In the canonical locked frame,

$$m_{\parallel} = m^3(P_0).$$

Hence the final longitudinal scalar is

$$L_* = \frac{m_{\parallel}}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right].$$

The other scalar is the valley Bloch asymmetry

$$\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_- ,$$

where $\eta_{\pm}$ are the normalized low eigenvectors of the two valley blocks $E_{\pm}(0)$.

The current branch dictionary is

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.$$

Thus, if

$$\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0,$$

then

$$\lambda_{\parallel} \neq 0.$$

So the remaining longitudinal task is the literal finite extraction of

$$E_{\pm}(0), \quad \mathcal{U}_j(0), \quad m^3(P_0).$$

CLIV. ACTUAL MICROSCOPIC COEFFICIENT-EXTRACTION PIPELINE FROM THE BLOCH OPERATOR

CLV. SHORT KOREAN EXPLANATION

The purpose of this section is not to claim that the full microscopic coefficients have already been numerically evaluated. Rather, its role is to state the exact executable pipeline that connects the current refined locked + Class II working branch to the finite projected extraction of the reduced Dirac/Weyl coefficients.

CLVI. 1. EXPLICIT WORKING MICROSCOPIC BRANCH

We fix the strongest explicit microscopic working branch currently available:

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \Psi]. \quad (246)$$

Here this is not claimed to be the unique final complete TECT action. It is the current explicit working branch on which the Bloch/Hessian extraction can be carried out concretely.

CLVII. 2. LOCKED 12-MODE BCC BACKGROUND

The periodic locked background is defined by the first reciprocal shell S_{12} :

$$\phi_0(x) = A \sum_{Q \in S_{12}} e^{iQ \cdot x}, \quad \Psi_0(x) = z_0 \phi_0(x), \quad P_0 = z_0 z_0^\dagger. \quad (247)$$

This fixes the folded Bragg shell on which the low carrier extraction is performed.

CLVIII. 3. QUADRATIC BLOCH HESSIAN AND EXACT SCHUR COMPLEMENT AT QUADRATIC LEVEL

Linearizing the equations of motion around the periodic background yields the quadratic Bloch fiber operator

$$\mathcal{H}_{\text{Bloch}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi X}(k) \\ H_{X\Psi}(k) & H_{XX}(k) \end{pmatrix}. \quad (248)$$

At the quadratic Bloch-Hessian level, eliminating the heavy mediator is exact and gives the Schur complement

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k). \quad (249)$$

CLIX. 4. FOLDED BRAGG CONVENTION, SHELL PROJECTION, AND LOW-SLOT PROJECTOR

We work at the folded Bragg point

$$k_* = \frac{G_*}{2}, \quad k = k_* + p. \quad (250)$$

The shell-projected operator is

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}, \quad (251)$$

and the low-slot projector is spectrally defined by

$$P_* = \frac{1}{2\pi i} \oint_{\Gamma} (z - H_{\text{shell}}(0))^{-1} dz. \quad (252)$$

CLX. 5. PROJECTED DERIVATIVE TRANSPORT MATRICES

The first-order derivative blocks are

$$\boxed{K_i = \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*}, \quad M_i = P_* K_i P_*}. \quad (253)$$

In the adapted shell frame $(\hat{n}, e_1, e_2)$, define

$$\boxed{M_{\parallel} = \hat{n}_i M_i, \quad M_1 = e_{1i} M_i, \quad M_2 = e_{2i} M_i}. \quad (254)$$

Under the locked Pauli selection rule, the reduced coefficients are extracted by

$$\boxed{\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel})}, \quad (255)$$

$$\boxed{\alpha = \frac{1}{4} \text{Tr}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4} \text{Tr}(\sigma_2 M_1 - \sigma_1 M_2)}. \quad (256)$$

CLXI. 6. CALIBRATION LAYER VERSUS OVERLAP LAYER

At the Bloch-projected level, the exact normalization layer is

$$\boxed{\lambda_{\parallel} = C_{\parallel} g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O}. \quad (257)$$

Independently, in the shell/defect overlap reduction one has

$$\boxed{\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}}. \quad (258)$$

Therefore the projected derivative computation determines

$$\lambda_{\parallel}, \alpha, \beta$$

directly, while the extraction of

$$g_c, g_E, g_O$$

still requires the explicit normalization constants

$$C_{\parallel}, C_E, C_O.$$

CLXII. 7. HONEST STATUS

The extraction pipeline is structurally closed:

$$L_{\text{mic}} \longrightarrow \mathcal{H}_{\text{Bloch}}(k) \longrightarrow H_{\text{eff}}(k) \longrightarrow H_{\text{shell}}(p) \longrightarrow P_* K_i P_* \longrightarrow (\lambda_{\parallel}, \alpha, \beta).$$

What remains open is not the formalism itself, but the branch-specific algebraic input:

$$\boxed{\text{the explicit vertices } \mathcal{J}, \mathcal{K}, \quad \text{the locked Pauli selection rule in full branch form, and } (C_{\parallel}, C_E, C_O)}. \quad (259)$$

FINAL SUMMARY: HONEST STATUS OF THE EXPLICIT EXTRACTION PIPELINE

The current refined locked + Class II working branch provides a fully explicit coefficient-extraction *pipeline*, not yet the final evaluated microscopic coefficients themselves.

The exact executable chain is

$$L_{\text{mic}} \longrightarrow \mathcal{H}_{\text{Bloch}}(k) \longrightarrow H_{\text{eff}}(k) \longrightarrow H_{\text{shell}}(p) \longrightarrow P_* K_i P_* \longrightarrow (\lambda_{\parallel}, \alpha, \beta).$$

At the projected Bloch level, the coefficients are extracted by

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}), \quad \alpha = \frac{1}{4} \text{Tr}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4} \text{Tr}(\sigma_2 M_1 - \sigma_1 M_2),$$

with

$$M_i = P_* \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*} P_*.$$

The normalization structure has two distinct layers:

$$\lambda_{\parallel} = C_{\parallel} g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O,$$

and

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}.$$

Therefore the projected derivative computation directly determines

$$\lambda_{\parallel}, \alpha, \beta,$$

while the explicit extraction of

$$g_c, g_E, g_O$$

still requires the overlap/calibration constants

$$C_{\parallel}, C_E, C_O.$$

So the honest current status is:

the extraction formalism is structurally closed,

but

the branch-specific algebraic inputs ($\mathcal{J}, \mathcal{K}$, selection rule, $C_{\parallel}, C_E, C_O$) are still to be evaluated.

CLXIII. FULL PRACTICAL ACCEPT/REJECT THEOREM FOR THE REFINED LOCKED + CLASS II WORKING BRANCH

CLXIV. SHORT KOREAN EXPLANATION

At the present stage, the correct next step is no longer another carrier-side reduction. The practical Gate-2 and Gate-3 carrier conditions are already assembled. So the nonredundant move is to attach Gate 4 and state the full accept/reject theorem.

The carrier-side symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2),$$

and the exact constant Dirac-mass space is

$$\mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.$$

Therefore the full practical branch-acceptance problem is exactly:

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

CLXV. 1. FIXED CARRIER-SIDE INFRARED SYMBOL

For the shell-reduced doubled local block, the exact first-order symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (260)$$

Hence the carrier-level nondegeneracy condition is

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (261)$$

CLXVI. 2. PRACTICAL GATE-2 HYPOTHESIS

Inside the current refined locked + Class II working branch, the strongest practical Gate-2 sufficient condition is

$$\lambda_{c;OO} \neq 0, \quad I_3 \neq 0, \quad C_{\parallel} \neq 0 \implies \lambda_{\parallel} \neq 0. \quad (262)$$

Equivalently, in the one-mediator Branch-B form,

$$\alpha_X \beta_X \neq 0, \quad C_{\parallel} \neq 0, \quad I_3 \neq 0 \implies \lambda_{\parallel} \neq 0. \quad (263)$$

CLXVII. 3. PRACTICAL GATE-3 HYPOTHESIS

The strongest practical Gate-3 sufficient condition is

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0, \quad (C_E, C_O) \neq (0, 0) \implies (\alpha, \beta) \neq (0, 0). \quad (264)$$

Thus, if (262) and (264) hold, then the current branch passes carrier acceptance.

CLXVIII. 4. EXACT CONSTANT MASS SPACE

For the doubled shell block, the exact constant Dirac-mass space is

$$\mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}. \quad (265)$$

The exact mass-protection criterion is

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}. \quad (266)$$

A sufficient practical mechanism is exact residual valley symmetry:

$$U(1)_V \quad \text{or already} \quad \mathbb{Z}_2^V. \quad (267)$$

CLXIX. 5. FULL PRACTICAL ACCEPT THEOREM

[Full practical accept theorem for the current working branch] Assume the refined locked + Class II working branch and suppose the following practical conditions hold:

1. longitudinal survival:

$$\lambda_{c;OO} \neq 0, \quad I_3 \neq 0, \quad C_{\parallel} \neq 0;$$

2. transverse survival:

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0, \quad (C_E, C_O) \neq (0, 0);$$

3. mass protection:

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

Then the branch is accepted as a viable microscopic carrier of a local massless anisotropic Dirac block:

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}.} \quad (268)$$

Equivalently, the local doubled shell block supports a massless Dirac cone protected against constant residual mass terms.

From the longitudinal practical hypothesis and (262), one gets

$$\lambda_{\parallel} \neq 0.$$

From the transverse practical hypothesis and (264), one gets

$$(\alpha, \beta) \neq (0, 0).$$

The mass-protection hypothesis is already exactly the Gate-4 criterion:

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

Combining the three gives (268).

CLXX. 6. FULL PRACTICAL REJECT ALTERNATIVES

The same branch must be rejected at practical level if any of the following occurs:

1. longitudinal failure:

$$\lambda_{\parallel} = 0;$$

2. transverse failure:

$$(\alpha, \beta) = (0, 0);$$

3. mass failure:

$$A_{\text{res}} \cap \mathcal{M}_D \neq \{0\}.$$

Thus the practical accept/reject logic is complete:

$$\boxed{\text{Accept} \iff [\lambda_{\parallel} \neq 0] \wedge [(\alpha, \beta) \neq (0, 0)] \wedge [A_{\text{res}} \cap \mathcal{M}_D = \{0\}].} \quad (269)$$

CLXXI. 7. HONEST STATUS

What is now closed:

1. the exact carrier-side reduced symbol;
2. the practical Gate-2 longitudinal sufficient criterion;
3. the practical Gate-3 transverse sufficient criterion;
4. the exact constant mass-space classification;
5. the full practical accept/reject theorem.

What is still not explicitly inserted:

$$\boxed{C_{\parallel}, \quad C_E, \quad C_O, \quad A_{\text{res}} \text{ in the corrected shell block.}} \quad (270)$$

So the remaining work is no longer theorem construction. It is explicit finite verification of the practical hypotheses.

FINAL SUMMARY

The correct next step is not another carrier-side reduction. It is to attach Gate 4 to the already assembled Gate 2–3 carrier theorem and state the full practical accept/reject theorem.

The branch is practically accepted if and only if

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

A sufficient practical Gate-2 package is

$$\lambda_{c;OO} \neq 0, \quad I_3 \neq 0, \quad C_{\parallel} \neq 0,$$

and a sufficient practical Gate-3 package is

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0, \quad (C_E, C_O) \neq (0, 0).$$

The exact mass-protection condition is

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}, \quad \mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.$$

Thus the theorem-construction stage is finished. What remains is the explicit finite verification of

$$C_{\parallel}, \quad C_E, \quad C_O, \quad A_{\text{res}}.$$

CLXXII. GATE 4 RESIDUAL MASS AUDIT ON THE CORRECTED DOUBLED SHELL BLOCK

CLXXIII. SHORT KOREAN EXPLANATION

Once the isolated doubled local Dirac block exists, Gate 4 is a finite constant-operator audit. It is no longer a PDE problem.

The exact constant Dirac-mass space is

$$\mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.$$

Therefore the exact pass/fail criterion is

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

What can be proved directly now is:

1. exact residual valley $U(1)_V$ is sufficient for Gate 4;
2. even the weaker exact residual $\mathbb{Z}_2^V$ is already sufficient;
3. Class II is compatible with mass protection, but does not by itself prove it.

So the only remaining open microscopic question is whether the corrected isolated shell block actually preserves one of these residual valley symmetries.

CLXXIV. 1. FIXED DOUBLED DIRAC BLOCK AND EXACT MASS SPACE

Assume the local doubled shell block has already been reduced to the anisotropic massless Dirac form

$$\boxed{H_D(p) = v_1 p_1 (\tau_z \otimes \sigma_1) + v_2 p_2 (\tau_z \otimes \sigma_2) + v_3 p_3 (\tau_z \otimes \sigma_3).} \quad (271)$$

The exact constant Dirac-mass space is

$$\boxed{\mathcal{M}_D = \{K \in \text{Herm}(4) : \{K, \tau_z \otimes \sigma_i\} = 0 \ \forall i = 1, 2, 3\} = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.} \quad (272)$$

Hence any constant Dirac mass has the form

$$\boxed{K_m = m_1 \tau_x \otimes \mathbf{1}_2 + m_2 \tau_y \otimes \mathbf{1}_2.} \quad (273)$$

CLXXV. 2. EXACT GATE-4 CRITERION

Let G_{res} be the exact residual symmetry group of the corrected isolated doubled shell block. Define the residual constant algebra

$$A_{\text{res}} = \{K \in \text{Herm}(4) : U(g)KU(g)^{-1} = K \ \forall g \in G_{\text{res}}\}. \quad (274)$$

Then the exact Gate-4 pass/fail criterion is

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}. \quad (275)$$

So Gate 4 is a finite constant-operator classification problem.

CLXXVI. 3. WHY LITTLE-GROUP SCALARITY AND OPPOSITE-VALLEY MATCHING ARE NOT ENOUGH

The full little-group-scalar constant Hermitian operator space on the doubled block is

$$K = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2 + m_1 \tau_x \otimes \mathbf{1}_2 + m_2 \tau_y \otimes \mathbf{1}_2. \quad (276)$$

Thus little-group scalarity alone still allows both mass generators

$$\tau_x \otimes \mathbf{1}_2, \quad \tau_y \otimes \mathbf{1}_2.$$

So it is not enough for mass protection.

Likewise, opposite-valley matching fixes the doubled structure but does not remove the constant mass operators. Therefore neither little-group scalarity alone nor opposite-valley matching alone implies Gate 4.

CLXXVII. 4. EXACT RESIDUAL VALLEY $U(1)_V$ IS SUFFICIENT

Define the valley-charge generator

$$Q_V := \tau_z \otimes \mathbf{1}_2, \quad U_\theta = e^{i\theta Q_V/2}. \quad (277)$$

Since

$$[Q_V, H_D(p)] = 0,$$

the doubled kinetic block is valley symmetric.

Now act on the two mass generators:

$$U_\theta(\tau_x \otimes \mathbf{1}_2)U_\theta^{-1} = \cos \theta (\tau_x \otimes \mathbf{1}_2) + \sin \theta (\tau_y \otimes \mathbf{1}_2), \quad (278)$$

$$U_\theta(\tau_y \otimes \mathbf{1}_2)U_\theta^{-1} = -\sin \theta (\tau_x \otimes \mathbf{1}_2) + \cos \theta (\tau_y \otimes \mathbf{1}_2). \quad (279)$$

Thus $\mathcal{M}_D$ transforms as a nontrivial charged doublet under $U(1)_V$. No nonzero element of $\mathcal{M}_D$ is invariant under the full $U(1)_V$. Therefore

$$A_{U(1)_V} \cap \mathcal{M}_D = \{0\}. \quad (280)$$

So exact residual valley $U(1)_V$ is sufficient for Gate 4.

CLXXVIII. 5. EXACT RESIDUAL $\mathbb{Z}_2^V$ IS ALREADY SUFFICIENT

Even the weaker discrete remnant

$$\boxed{S_V := \tau_z \otimes \mathbf{1}_2, \quad S_V^2 = \mathbf{1}_4} \quad (281)$$

already suffices.

Indeed,

$$S_V(\tau_x \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_x \otimes \mathbf{1}_2), \quad (282)$$

$$S_V(\tau_y \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_y \otimes \mathbf{1}_2). \quad (283)$$

Hence both mass generators are odd under S_V , so no nonzero mass is invariant. Therefore

$$\boxed{A_{\mathbb{Z}_2^V} \cap \mathcal{M}_D = \{0\}}. \quad (284)$$

Thus exact residual valley $\mathbb{Z}_2^V$ is already sufficient for Gate 4.

CLXXIX. 6. CLASS II COMPATIBILITY AND HONEST LIMITATION

The current Class II correction is gradient-linear and renormalizes the shell derivative coefficients, but does not by itself provide a proof that the corrected constant 4×4 block preserves the required residual valley symmetry.

Therefore:

$$\boxed{\text{Class II is compatible with mass protection, but is not by itself a proof of it.}} \quad (285)$$

So the remaining microscopic question is not the classification theorem. It is only the actual residual-symmetry audit of the corrected isolated shell block.

CLXXX. 7. FINAL PRACTICAL GATE-4 CHECKLIST

Gate 4 now has the following exact execution order:

1. identify the isolated doubled shell block and the exact mass space $\mathcal{M}_D$;
2. test whether exact residual $U(1)_V$ survives;
3. if not, test whether exact residual $\mathbb{Z}_2^V$ survives;
4. if neither is manifest, compute the full residual constant algebra A_{res} ;
5. check whether

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

Therefore the theorem-level reduction of Gate 4 is complete.

CLXXXI. 8. FULL PRACTICAL ACCEPTANCE ATTACHMENT

Combining this with the carrier-side Gate 2–3 conclusion, the current branch is accepted at practical level if and only if

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}}. \quad (286)$$

This is the final practical accept/reject criterion at the carrier level.

CLXXXII. FINAL SUMMARY

For the doubled local Dirac block

$$H_D(p) = \sum_{i=1}^3 v_i p_i (\tau_z \otimes \sigma_i),$$

the exact constant Dirac-mass space is

$$\mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.$$

Hence the exact Gate-4 pass/fail criterion is

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}, \quad A_{\text{res}} = \{K \in \text{Herm}(4) : U(g)KU(g)^{-1} = K \ \forall g \in G_{\text{res}}\}.$$

Little-group scalarity alone is insufficient, and opposite-valley matching alone is also insufficient. A sufficient protecting mechanism is exact residual valley symmetry:

$$Q_V = \tau_z \otimes \mathbf{1}_2 \implies A_{U(1)_V} \cap \mathcal{M}_D = \{0\},$$

and even the weaker remnant

$$S_V = \tau_z \otimes \mathbf{1}_2 \implies A_{\mathbb{Z}_2^V} \cap \mathcal{M}_D = \{0\}.$$

Therefore the theorem-level reduction of Gate 4 is complete.

The only remaining microscopic task is to determine whether the corrected isolated doubled shell block actually preserves exact residual $U(1)_V$ or at least exact residual $\mathbb{Z}_2^V$.

CLXXXIII. ACTUAL Q_* -INVARIANT CONSTANT-SECTOR AUDIT

CLXXXIV. 1. ACTUAL CANONICAL SHELL BLOCK AND ACTUAL MASS SPACE

At the actual canonical shell level, the projected massless block is

$$H_{\text{core}}^{\text{lin}}(p) = v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \nu_1 p_1 \tau_1 \otimes s_1 + \nu_2 p_2 \tau_1 \otimes s_2. \quad (287)$$

The exact residual generator is

$$Q_* = \tau_3 \otimes s_3, \quad (288)$$

and the actual constant Dirac-mass space is

$$\mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}}\{M_1, M_2\}, \quad M_1 = \tau_2 \otimes \mathbf{1}_2, \quad M_2 = \tau_1 \otimes s_3. \quad (289)$$

CLXXXV. 2. EXACT COMMUTANT OF Q_*

Let

$$\mathcal{B} = \{\tau_a \otimes s_b : a, b \in \{0, 1, 2, 3\}\}, \quad \tau_0 = s_0 = \mathbf{1}_2.$$

For a basis element $X_{ab} := \tau_a \otimes s_b$,

$$[X_{ab}, Q_*] = 0$$

holds if and only if the number of factors anticommuting with τ_3 and s_3 is even.

Indeed:

- τ_0, τ_3 commute with τ_3 , while τ_1, τ_2 anticommute with τ_3 ;
- s_0, s_3 commute with s_3 , while s_1, s_2 anticommute with s_3 .

Therefore the full Hermitian commutant is

$$\mathcal{C}_{Q_*} = \text{span}_{\mathbb{R}}\{\mathbf{1}_4, \mathbf{1}_2 \otimes s_3, \tau_3 \otimes \mathbf{1}_2, \tau_3 \otimes s_3, \tau_1 \otimes s_1, \tau_1 \otimes s_2, \tau_2 \otimes s_1, \tau_2 \otimes s_2\}. \quad (290)$$

CLXXXVI. 3. DIRECT EXCLUSION OF THE ACTUAL MASS GENERATORS

Now test the two actual mass generators.
First,

$$\begin{aligned} [Q_*, M_1] &= [\tau_3 \otimes s_3, \tau_2 \otimes \mathbf{1}_2] \\ &= (\tau_3 \tau_2 - \tau_2 \tau_3) \otimes s_3 \neq 0. \end{aligned} \quad (291)$$

Second,

$$\begin{aligned} [Q_*, M_2] &= [\tau_3 \otimes s_3, \tau_1 \otimes s_3] \\ &= (\tau_3 \tau_1 - \tau_1 \tau_3) \otimes \mathbf{1}_2 \neq 0. \end{aligned} \quad (292)$$

Hence neither M_1 nor M_2 belongs to $\mathcal{C}_{Q_*}$. Therefore

$$\boxed{\mathcal{C}_{Q_*} \cap \mathcal{M}_{\text{actual}} = \{0\}}. \quad (293)$$

This is the exact algebraic core of Gate 4 in the actual basis.

CLXXXVII. 4. CONSEQUENCE FOR THE CORRECTED CONSTANT SECTOR

Write the full corrected constant sector as

$$\delta H_{\text{const}} = H_{\Psi\Psi}^{\text{const}} + \delta H_{\text{Class II}}^{\text{const}} + \Sigma(0, 0). \quad (294)$$

Assume the actual corrected theory preserves the same exact residual generator Q_* , namely

$$\boxed{[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0, \quad [Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0, \quad [Q_*, \Sigma(0, 0)] = 0.} \quad (295)$$

Then each term lies in $\mathcal{C}_{Q_*}$, hence

$$\delta H_{\text{const}} \in \mathcal{C}_{Q_*}.$$

Using (293), we conclude

$$\boxed{\delta H_{\text{const}} \cap \mathcal{M}_{\text{actual}} = \{0\}}. \quad (296)$$

Therefore no exact constant Dirac mass can appear in the corrected projected shell block.

CLXXXVIII. 5. SHELL-BLOCK-LEVEL CLOSURE AND FULL MICROSCOPIC FRONTIER

At shell-block level, the minimal corrected Class II block remains purely derivative and preserves the valley generator; in particular, no independent constant shell mass is inserted. Thus the shell-block-level mass audit is already closed.

What remains microscopic is only the inheritance question: does the same exact Q_* -symmetry survive all constant corrections and remote-sector elimination?

So the remaining yes/no test is exactly

$$\boxed{[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0, \quad [Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0, \quad [Q_*, \Sigma(0, 0)] = 0.} \quad (297)$$

If yes, Gate 4 passes. If no, one must isolate the explicit symmetry-breaking constant term and test whether it has a nonzero projection onto M_1 or M_2 .

CLXXXIX. 6. FINAL ACTUAL GATE-4 THEOREM

[Actual Q_* -invariant constant-sector theorem] Assume the actual projected shell block has the canonical massless form (287), with residual generator $Q_* = \tau_3 \otimes s_3$, and actual mass space $\mathcal{M}_{\text{actual}}$ given by (289).

If the full corrected constant sector preserves Q_* , i.e. (295) holds, then

$$\boxed{\delta H_{\text{const}} \cap \mathcal{M}_{\text{actual}} = \{0\}}. \quad (298)$$

Hence no exact constant Dirac mass can appear.

By (295), each constant correction lies in $\mathcal{C}_{Q_*}$. By (293), $\mathcal{C}_{Q_*} \cap \mathcal{M}_{\text{actual}} = \{0\}$. Therefore δH_{const} cannot contain any actual mass component.

CXC. FINAL SUMMARY

In the actual canonical shell basis, the projected massless block is

$$H_{\text{core}}^{\text{lin}}(p) = v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \nu_1 p_1 \tau_1 \otimes s_1 + \nu_2 p_2 \tau_1 \otimes s_2,$$

with residual generator

$$Q_* = \tau_3 \otimes s_3,$$

and actual mass space

$$\mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}}\{\tau_2 \otimes \mathbf{1}_2, \tau_1 \otimes s_3\}.$$

The exact Hermitian commutant of Q_* is

$$\mathcal{C}_{Q_*} = \text{span}_{\mathbb{R}}\{\mathbf{1}_4, \mathbf{1}_2 \otimes s_3, \tau_3 \otimes \mathbf{1}_2, \tau_3 \otimes s_3, \tau_1 \otimes s_1, \tau_1 \otimes s_2, \tau_2 \otimes s_1, \tau_2 \otimes s_2\}.$$

Neither actual mass generator belongs to this commutant:

$$[Q_*, \tau_2 \otimes \mathbf{1}_2] \neq 0, \quad [Q_*, \tau_1 \otimes s_3] \neq 0.$$

Therefore

$$\boxed{\mathcal{C}_{Q_*} \cap \mathcal{M}_{\text{actual}} = \{0\}}.$$

Hence, if the full corrected constant sector preserves the same exact Q_* , namely

$$[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0, \quad [Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0, \quad [Q_*, \Sigma(0,0)] = 0,$$

then

$$\boxed{\delta H_{\text{const}} \cap \mathcal{M}_{\text{actual}} = \{0\}},$$

and no exact constant Dirac mass can appear.

Thus the theorem-level mass audit is completely closed. The only remaining microscopic yes/no input is whether the actual full corrected constant sector really preserves the same exact residual generator Q_* .

CXCI. EXPLICIT DERIVATIVE FORMULAS FOR K_i IN THE BLOCH-OPERATOR EXTRACTION PIPELINE

CXCII. SHORT KOREAN EXPLANATION

From this point onward, we stay strictly inside the Bloch-operator extraction pipeline:

$$L_{\text{mic}} \rightarrow \mathcal{H}_{\text{Bloch}}(k) \rightarrow H_{\text{eff}}(k) \rightarrow H_{\text{shell}}(p) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

The next actual calculation is therefore the explicit derivative formula for

$$K_i := \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*}.$$

The important point is that the effective derivative splits into exactly three pieces:

$$\boxed{K_i = K_i^{(\text{bare})} + K_i^{(\text{II})} + K_i^{(\text{spec})}}.$$

This is the first genuine operator-construction step.

CXCIII. 1. WORKING MICROSCOPIC BRANCH AND BLOCH FIBER

We fix the refined locked + Class II working branch

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \Psi]. \quad (299)$$

Linearization around the locked 12-mode BCC background yields the Bloch Hessian

$$\mathcal{H}_{\text{Bloch}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi X}(k) \\ H_{X\Psi}(k) & H_{XX}(k) \end{pmatrix}. \quad (300)$$

At the first mediator-elimination stage, the exact quadratic effective block is

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k). \quad (301)$$

At the final shell-isolation stage, including remote spectator elimination, one writes

$$H_{\text{eff}}^{\text{comp}}(k) = H_{\text{low,low}}(k) - H_{\text{low,rem}}(k) H_{\text{rem,rem}}(k)^{-1} H_{\text{rem,low}}(k). \quad (302)$$

CXCIV. 2. FOLDED-BRAGG DERIVATIVE DEFINITION

Work in the folded Bragg convention

$$k_* = \frac{G_*}{2}, \quad k = k_* + p. \quad (303)$$

Then the first derivative blocks are defined by

$$K_i = \partial_{k_i} H_{\text{eff}}^{\text{comp}}(k)|_{k=k_*}. \quad (304)$$

These are the objects that are later projected to the low shell slot.

CXCV. 3. EXACT DERIVATIVE OF THE REMOTE SCHUR COMPLEMENT

Let

$$A(k) := H_{\text{low,rem}}(k), \quad R(k) := H_{\text{rem,rem}}(k)^{-1}, \quad B(k) := H_{\text{rem,low}}(k).$$

Then

$$H_{\text{eff}}^{\text{comp}}(k) = H_{\text{low,low}}(k) - A(k)R(k)B(k).$$

Differentiating gives

$$\begin{aligned} K_i &= \partial_{k_i} H_{\text{low,low}}(k)|_{k_*} - \partial_{k_i} A(k)|_{k_*} R_* B_* - A_* R_* \partial_{k_i} B(k)|_{k_*} \\ &\quad + A_* R_* \partial_{k_i} H_{\text{rem,rem}}(k)|_{k_*} R_* B_*, \end{aligned} \quad (305)$$

where

$$R_* := H_{\text{rem,rem}}(k_*)^{-1}, \quad A_* := A(k_*), \quad B_* := B(k_*).$$

This is the exact derivative formula at the final remote-eliminated level.

CXCVI. 4. BARE / CLASS II / SPECTATOR SPLITTING

The documents state that each effective shell quantity splits exactly as

$$\text{effective} = \text{bare} + \text{Class II} + \text{spectator}.$$

Accordingly,

$$\boxed{K_i = K_i^{(\text{bare})} + K_i^{(\text{II})} + K_i^{(\text{spec})}.} \quad (306)$$

(a) Bare piece

The bare contribution comes from the periodic $\Psi\Psi$ -block:

$$\boxed{K_i^{(\text{bare})} = \partial_{k_i} H_{\Psi\Psi}^{\text{per}}(k)|_{k=k_*}.} \quad (307)$$

(b) Class II piece

Define

$$\mathcal{C}(k) := \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k).$$

At leading heavy-mass order,

$$H_{XX}(k) = M_X^2 \mathbf{1}_X + O(k, \text{periodic}), \quad H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2} \mathcal{C}(k) \mathcal{C}(k)^\dagger + O(M_X^{-4}).$$

Therefore the explicit derivative contribution is

$$\boxed{K_i^{(\text{II})} = -\frac{1}{M_X^2} [(\partial_{k_i} \mathcal{C}) \mathcal{C}^\dagger + \mathcal{C} (\partial_{k_i} \mathcal{C}^\dagger)]|_{k=k_*}.} \quad (308)$$

(c) Spectator piece

The spectator correction is the remote-sector derivative inherited from the second Schur complement:

$$\boxed{K_i^{(\text{spec})} = -\partial_{k_i} A(k)|_{k_*} R_* B_* - A_* R_* \partial_{k_i} B(k)|_{k_*} + A_* R_* \partial_{k_i} H_{\text{rem,rem}}(k)|_{k_*} R_* B_*}. \quad (309)$$

Thus the microscopic coefficient problem is reduced to three explicit derivative blocks.

CXCVII. 5. SHELL PROJECTION AND REDUCED COEFFICIENTS

Let P_* be the low-slot shell projector. Then

$$\boxed{M_i = P_* K_i P_*}. \quad (310)$$

In the adapted shell frame

$$\hat{n} = \frac{G_*}{|G_*|}, \quad \{e_1, e_2, \hat{n}\} \text{ orthonormal},$$

define

$$K_{\parallel} = \hat{n}_i K_i, \quad K_1 = e_{1i} K_i, \quad K_2 = e_{2i} K_i,$$

and similarly

$$M_{\parallel} = \hat{n}_i M_i, \quad M_1 = e_{1i} M_i, \quad M_2 = e_{2i} M_i.$$

The exact reduced coefficient extraction formulas are

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma^3 M_{\parallel}), \quad (311)$$

$$\alpha = \frac{1}{4} \sum_{a=1}^2 \text{Tr}(\sigma^a P_*(e_{ai} K_i) P_*), \quad (312)$$

$$\beta = \frac{1}{4} [\text{Tr}(\sigma^2 P_*(e_{1i} K_i) P_*) - \text{Tr}(\sigma^1 P_*(e_{2i} K_i) P_*)]. \quad (313)$$

CXCVIII. 6. THE FOUR ACTUAL MICROSCOPIC QUANTITIES

The immediate finite bookkeeping target is the quartet

$$\hat{U}_{\text{eff}}(G_*), \quad \hat{T}_{\text{eff}}^1(G_*), \quad \hat{T}_{\text{eff}}^2(G_*), \quad v_{\parallel}. \quad (314)$$

These are exactly the shell Fourier data from which the projected derivative blocks are assembled. Equivalently, the remaining microscopic problem is to compute the explicit reciprocal-harmonic matrix elements of

$$\mathcal{J}(k), \quad \mathcal{K}(k), \quad H_{\text{rem,rem}}(k)$$

at the intervalley harmonic $Q = G_*$.

CXCIX. 7. HONEST CURRENT STATUS INSIDE THE PIPELINE

The Bloch-operator extraction pipeline is structurally closed. The next actual computation is the derivative construction

$$H_{\Psi\Psi}(k), H_{\Psi X}(k), H_{XX}(k) \rightarrow K_i^{(\text{bare})}, K_i^{(\text{II})}, K_i^{(\text{spec})} \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

What is still not explicitly evaluated is the branch-specific harmonic data entering

$$\mathcal{J}(k), \quad \mathcal{K}(k), \quad H_{\text{rem,rem}}(k).$$

So the pipeline is no longer conceptually open; it is an explicit derivative-and-projection computation.

CC. FOLDED-BRAGG SHELL BOOKKEEPING FOR THE FOUR FINITE MICROSCOPIC QUANTITIES

CCI. SHORT KOREAN EXPLANATION

Inside the Bloch-operator extraction pipeline, the infinite-dimensional microscopic problem is reduced to four finite shell quantities:

$$\hat{U}_{\text{eff}}(G_*), \quad \hat{T}_{\text{eff}}^1(G_*), \quad \hat{T}_{\text{eff}}^2(G_*), \quad v_{\parallel}.$$

These control, respectively,

1. the scalar intervalley shell offset,

2. the first transverse witness,
3. the second transverse witness,
4. the longitudinal shell slope.

Each splits as

$$\text{effective} = \text{bare} + \text{Class II} + \text{spectator}.$$

Thus the next executable stage in the pipeline is to write these four objects explicitly before inserting the final branch-specific harmonic data.

CCII. 1. THE FOUR SHELL QUANTITIES

At the folded Bragg point

$$k_* = \frac{G_*}{2},$$

the shell-pair bookkeeping is organized by the four finite quantities

$$\boxed{M_0 = \widehat{U}_{\text{eff}}(G_*) \mathbf{1}_2,} \quad (315)$$

$$\boxed{W^\alpha = \widehat{T}_{\text{eff}}^\alpha(G_*) s^\alpha, \quad \alpha = 1, 2,} \quad (316)$$

and

$$\boxed{v_{\parallel} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left. \frac{E_+^{\text{eff}}(E_*, p) - E_-^{\text{eff}}(E_*, p)}{2} \right|_{p=0}.} \quad (317)$$

These are exactly the shell data from which the reduced symbol is assembled.

CCIII. 2. EFFECTIVE SPLIT: BARE + CLASS II + SPECTATOR

Each shell quantity splits into three pieces:

$$\boxed{\widehat{U}_{\text{eff}}(G_*) = \widehat{U}_{\text{bare}}(G_*) + \widehat{U}_{(II)}(G_*) + \widehat{U}_{(\text{sp})}(G_*),} \quad (318)$$

$$\boxed{\widehat{T}_{\text{eff}}^\alpha(G_*) = \widehat{T}_{\text{bare}}^\alpha(G_*) + \widehat{T}_{(II)}^\alpha(G_*) + \widehat{T}_{(\text{sp})}^\alpha(G_*), \quad \alpha = 1, 2,} \quad (319)$$

$$\boxed{v_{\parallel} = v_{\parallel}^{\text{bare}} + v_{\parallel}^{(II)} + v_{\parallel}^{(\text{sp})}.} \quad (320)$$

This is the exact finite bookkeeping format of the current pipeline.

CCIV. 3. REDUCED-COEFFICIENT DICTIONARY

Once the four shell quantities are known, the reduced coefficient point is fixed. The longitudinal coefficient is read from the shell slope:

$$\boxed{\lambda_{\parallel} = v_{\parallel}.} \quad (321)$$

The two transverse matrices determine the reduced pair

$$\boxed{\alpha = c_E I_\theta, \quad \beta = c_O I_\theta,} \quad (322)$$

where c_E, c_O are the shell-seed projections of

$$\hat{T}_{\text{eff}}^1(G_*), \quad \hat{T}_{\text{eff}}^2(G_*).$$

Equivalently, the final finite nondegeneracy criterion becomes

$$\boxed{v_{\parallel} \neq 0, \quad \hat{T}_{\text{eff}}^1(G_*) \neq 0 \text{ or } \hat{T}_{\text{eff}}^2(G_*) \neq 0.} \quad (323)$$

CCV. 4. EXPLICIT CLASS II TRANSVERSE OVERLAP FORMULA

Inside the explicit folded-Bragg Class II branch, the transverse derivative bookkeeping compresses to the single Fourier convolution

$$\boxed{\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \hat{J}_i^A(Q) \hat{K}_{i,\alpha}^A(Q'), \quad \alpha = 1, 2.} \quad (324)$$

In the minimal first-shell-dominant reduction, this becomes

$$\boxed{\hat{T}_{(II)}^\alpha(G_*) \propto i A_{G_*} (C_\alpha \cdot G_*), \quad \alpha = 1, 2.} \quad (325)$$

Therefore the strongest practical explicit transverse test is

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0.} \quad (326)$$

CCVI. 5. ROLE OF THE K -SECTOR

At the explicit shell-adapted leading level,

$$\mathcal{K}_{\parallel}^a(G_*; 0) = 0,$$

so

$$\boxed{c_{\parallel}^{(K)} = 0.} \quad (327)$$

Thus the K -sector is not a direct longitudinal source at principal shell order. Its role is transverse:

$$\boxed{c_E^{(K)} \neq 0 \text{ in general,} \quad c_O^{(K)} \neq 0 \text{ in general.}} \quad (328)$$

Therefore the actual transverse witnesses must come from the genuinely nontrivial JK and/or KK shell realization.

CCVII. 6. HONEST STATUS OF THE FOUR-QUANTITY BOOKKEEPING

At this point, the pipeline is no longer conceptually open.

What is fixed:

1. the four finite shell quantities,
2. their exact split into bare / Class II / spectator pieces,
3. the longitudinal reading through $v_{\parallel}$,
4. the transverse reading through $\hat{T}_{\text{eff}}^1(G_*)$ and $\hat{T}_{\text{eff}}^2(G_*)$.

What remains open is the actual symbolic or numerical insertion of the final harmonic data:

$$\boxed{\hat{J}_i^A(Q), \quad \hat{K}_{i,\alpha}^A(Q), \quad \hat{U}_{\text{bare}}(G_*), \quad v_{\parallel}^{\text{bare}}, \quad \text{and spectator renormalizations.}} \quad (329)$$

CCVIII. FINAL SUMMARY

Inside the Bloch-operator extraction pipeline, the remaining microscopic problem has been reduced to four finite shell quantities:

$$\hat{U}_{\text{eff}}(G_*), \quad \hat{T}_{\text{eff}}^1(G_*), \quad \hat{T}_{\text{eff}}^2(G_*), \quad v_{\parallel}.$$

They split exactly as

$$\hat{U}_{\text{eff}} = \hat{U}_{\text{bare}} + \hat{U}_{(II)} + \hat{U}_{(\text{sp})},$$

$$\hat{T}_{\text{eff}}^{\alpha} = \hat{T}_{\text{bare}}^{\alpha} + \hat{T}_{(II)}^{\alpha} + \hat{T}_{(\text{sp})}^{\alpha}, \quad \alpha = 1, 2,$$

$$v_{\parallel} = v_{\parallel}^{\text{bare}} + v_{\parallel}^{(II)} + v_{\parallel}^{(\text{sp})}.$$

The reduced finite nondegeneracy test is

$$\boxed{v_{\parallel} \neq 0, \quad \hat{T}_{\text{eff}}^1(G_*) \neq 0 \text{ or } \hat{T}_{\text{eff}}^2(G_*) \neq 0.}$$

Inside the explicit folded-Bragg Class II branch, the transverse Class II piece is already compressed to the Fourier convolution

$$\hat{T}_{(II)}^{\alpha}(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \hat{J}_i^A(Q) \hat{K}_{i,\alpha}^A(Q').$$

Thus the next strict pipeline step is the longitudinal column:

$$\boxed{v_{\parallel}^{\text{bare}}, \quad v_{\parallel}^{(II)}, \quad v_{\parallel}^{(\text{sp})}.}$$

CCIX. CONCRETE FORMULAS FOR THE LONGITUDINAL COLUMN

CCX. SHORT KOREAN EXPLANATION

Inside the folded-Bragg shell bookkeeping, the longitudinal column is controlled by the single scalar shell slope

$$v_{\parallel}.$$

The exact bookkeeping split is

$$v_{\parallel} = v_{\parallel}^{\text{bare}} + v_{\parallel}^{(II)} + v_{\parallel}^{(\text{sp})}.$$

At the reduced-symbol level one also has

$$\lambda_{\parallel} = v_{\parallel} \quad \text{and} \quad \lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}.$$

Therefore the concrete longitudinal task is to write each of the three pieces

$$v_{\parallel}^{\text{bare}}, \quad v_{\parallel}^{(II)}, \quad v_{\parallel}^{(\text{sp})}$$

directly from the projected Bloch derivative operator.

CCXI. 1. EXACT PROJECTED DEFINITION

Let

$$K_i = \partial_{p_i} H_{\text{shell}}(p)|_{p=0}, \quad K_{\parallel} = \hat{n}_i K_i.$$

Then the longitudinal shell slope is extracted by

$$v_{\parallel} = \frac{1}{2} \text{Tr}(\sigma^3 P_* K_{\parallel} P_*). \quad (330)$$

Equivalently, in the +-valley internal 2×2 block,

$$v_{\parallel} = \frac{1}{2} \left[(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \right]. \quad (331)$$

This is the safest exact formula for the longitudinal column.

CCXII. 2. BARE / CLASS II / SPECTATOR SPLIT

Write

$$K_{\parallel} = K_{\parallel}^{\text{bare}} + K_{\parallel}^{(II)} + K_{\parallel}^{(\text{sp})}. \quad (332)$$

Then, by linearity of the projection and the trace,

$$v_{\parallel} = v_{\parallel}^{\text{bare}} + v_{\parallel}^{(II)} + v_{\parallel}^{(\text{sp})}, \quad (333)$$

with

$$v_{\parallel}^{\text{bare}} = \frac{1}{2} \text{Tr}(\sigma^3 P_* K_{\parallel}^{\text{bare}} P_*), \quad (334)$$

$$v_{\parallel}^{(II)} = \frac{1}{2} \text{Tr}(\sigma^3 P_* K_{\parallel}^{(II)} P_*), \quad (335)$$

$$v_{\parallel}^{(\text{sp})} = \frac{1}{2} \text{Tr}(\sigma^3 P_* K_{\parallel}^{(\text{sp})} P_*). \quad (336)$$

CCXIII. 3. BARE LONGITUDINAL PIECE

The bare longitudinal piece comes from the direct periodic $\Psi\Psi$ -block:

$$K_{\parallel}^{\text{bare}} = \partial_{p_{\parallel}} H_{\Psi\Psi}^{\text{per}}(p)|_{p=0}. \quad (337)$$

Accordingly,

$$v_{\parallel}^{\text{bare}} = \frac{1}{2} \text{Tr}(\sigma^3 P_* \partial_{p_{\parallel}} H_{\Psi\Psi}^{\text{per}}(p)|_{p=0} P_*). \quad (338)$$

If one uses the valley-diagonal shell blocks $E_{\pm}^{\text{bare}}(p)$, this can be written equivalently as

$$v_{\parallel}^{\text{bare}} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{E_+^{\text{bare}}(p) - E_-^{\text{bare}}(p)}{2} \right] \Big|_{p=0}. \quad (339)$$

CCXIV. 4. CLASS II LONGITUDINAL PIECE

Let

$$\mathcal{C}(k) := \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k).$$

At leading heavy-mass order,

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2} \mathcal{C}(k) \mathcal{C}(k)^\dagger + O(M_X^{-4}).$$

Therefore the Class II derivative contribution is

$$K_{\parallel}^{(II)} = -\frac{1}{M_X^2} [(\partial_{p_{\parallel}} \mathcal{C}) \mathcal{C}^\dagger + \mathcal{C} (\partial_{p_{\parallel}} \mathcal{C}^\dagger)]_{p=0}. \quad (340)$$

Hence

$$v_{\parallel}^{(II)} = -\frac{1}{2M_X^2} \text{Tr} \left(\sigma^3 P_* [(\partial_{p_{\parallel}} \mathcal{C}) \mathcal{C}^\dagger + \mathcal{C} (\partial_{p_{\parallel}} \mathcal{C}^\dagger)]_{p=0} P_* \right). \quad (341)$$

At the reduced witness level, the same longitudinal coefficient is also written as

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad (342)$$

with

$$c_{\parallel} = c_{\parallel}^{(JJ)} + c_{\parallel}^{(JK)} + c_{\parallel}^{(KK)}. \quad (343)$$

Therefore the Class II longitudinal piece may equivalently be organized by the shell-seed split

$$v_{\parallel}^{(II)} = \frac{1}{2} c_{\parallel}^{(II)} I_{\theta'^3}, \quad c_{\parallel}^{(II)} = c_{\parallel}^{(JJ)} + c_{\parallel}^{(JK)} + c_{\parallel}^{(KK)}. \quad (344)$$

CCXV. 5. SPECTATOR LONGITUDINAL PIECE

Let

$$A(p) := H_{\text{low,rem}}(p), \quad R(p) := H_{\text{rem,rem}}(p)^{-1}, \quad B(p) := H_{\text{rem,low}}(p).$$

The remote Schur complement contributes

$$-A(p)R(p)B(p)$$

to the shell block. Differentiating gives

$$K_{\parallel}^{(\text{sp})} = -\partial_{p_{\parallel}} A(p) \Big|_{p=0} R_* B_* - A_* R_* \partial_{p_{\parallel}} B(p) \Big|_{p=0} + A_* R_* \partial_{p_{\parallel}} H_{\text{rem,rem}}(p) \Big|_{p=0} R_* B_*, \quad (345)$$

where

$$A_* = A(0), \quad B_* = B(0), \quad R_* = H_{\text{rem,rem}}(0)^{-1}.$$

Thus

$$v_{\parallel}^{(\text{sp})} = \frac{1}{2} \text{Tr}(\sigma^3 P_* K_{\parallel}^{(\text{sp})} P_*). \quad (346)$$

CCXVI. 6. HONEST STATUS OF THE LONGITUDINAL COLUMN

The longitudinal column is therefore completely fixed at the formal level:

$$v_{\parallel} = v_{\parallel}^{\text{bare}} + v_{\parallel}^{(II)} + v_{\parallel}^{(\text{sp})}.$$

What remains open is not the formula, but the actual branch-specific insertion of

$$E_{\pm}^{\text{bare}}(p), \quad \mathcal{J}(k), \quad \mathcal{K}(k), \quad H_{\text{low,rem}}(p), \quad H_{\text{rem,rem}}(p).$$

So the next legitimate microscopic step is the explicit symbolic or numerical evaluation of

$$v_{\parallel}^{\text{bare}}, \quad v_{\parallel}^{(II)}, \quad v_{\parallel}^{(\text{sp})}$$

for the chosen refined locked + Class II branch.

CCXVII. FINAL SUMMARY

The longitudinal column in the Bloch-operator extraction pipeline is controlled by the single shell slope

$$v_{\parallel}.$$

Its exact projected definition is

$$v_{\parallel} = \frac{1}{2} \text{Tr}(\sigma^3 P_* K_{\parallel} P_*), \quad K_{\parallel} = \hat{n}_i K_i.$$

The exact bookkeeping split is

$$v_{\parallel} = v_{\parallel}^{\text{bare}} + v_{\parallel}^{(II)} + v_{\parallel}^{(\text{sp})}.$$

Here

$$v_{\parallel}^{\text{bare}} = \frac{1}{2} \text{Tr}(\sigma^3 P_* K_{\parallel}^{\text{bare}} P_*),$$

$$v_{\parallel}^{(II)} = -\frac{1}{2M_X^2} \text{Tr}(\sigma^3 P_* [(\partial_{p_{\parallel}} \mathcal{C}) \mathcal{C}^{\dagger} + \mathcal{C}(\partial_{p_{\parallel}} \mathcal{C}^{\dagger})]_{p=0} P_*),$$

with $\mathcal{C} = \alpha_X \mathcal{J} + \beta_X \mathcal{K}$, and

$$v_{\parallel}^{(\text{sp})} = \frac{1}{2} \text{Tr}(\sigma^3 P_* K_{\parallel}^{(\text{sp})} P_*),$$

where $K_{\parallel}^{(\text{sp})}$ is the derivative of the remote Schur complement.

At the reduced witness level this is equivalent to

$$\lambda_{\parallel} = v_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta^3}, \quad c_{\parallel} = c_{\parallel}^{(JJ)} + c_{\parallel}^{(JK)} + c_{\parallel}^{(KK)}.$$

Thus the longitudinal column is formally closed. What remains is the explicit insertion of the branch-specific harmonic data entering the three pieces

$$v_{\parallel}^{\text{bare}}, \quad v_{\parallel}^{(II)}, \quad v_{\parallel}^{(\text{sp})}.$$

CCXVIII. PRINCIPAL CLASS-II REDUCTION OF THE LONGITUDINAL COLUMN

CCXIX. SHORT KOREAN EXPLANATION

Inside the Bloch-operator pipeline, the Class-II longitudinal contribution

$$v_{\parallel}^{(II)}$$

still looks broad if one keeps the full (JJ, JK, KK) split. However, in the current refined locked + Class II working branch, the principal shell-order source classification is already sharp:

$$JJ \rightsquigarrow \text{longitudinal only}, \quad JK, KK \rightsquigarrow \text{transverse family only}.$$

Therefore the longitudinal Class-II column reduces to a single piece:

$$\boxed{v_{\parallel}^{(II)} = v_{\parallel}^{(II, JJ)}}.$$

CCXX. 1. CHANNEL DECOMPOSITION OF THE CLASS-II LONGITUDINAL PIECE

Starting from the Class-II derivative block

$$K_{\parallel}^{(II)},$$

the shell witness decomposition is

$$\boxed{v_{\parallel}^{(II)} = v_{\parallel}^{(II, JJ)} + v_{\parallel}^{(II, JK)} + v_{\parallel}^{(II, KK)}}. \quad (347)$$

At the reduced witness level, the same decomposition is

$$\boxed{c_{\parallel}^{(II)} = c_{\parallel}^{(JJ)} + c_{\parallel}^{(JK)} + c_{\parallel}^{(KK)}, \quad v_{\parallel}^{(II)} = \frac{1}{2} c_{\parallel}^{(II)} I_{\theta'^3}}. \quad (348)$$

CCXXI. 2. PRINCIPAL SHELL-ORDER SOURCE CLASSIFICATION

In the current refined locked + Class II working branch, the principal shell-order classification is:

$$\boxed{c_{\parallel}^{(JK)} = 0, \quad c_{\parallel}^{(KK)} = 0}, \quad (349)$$

while the pure JJ -sector is the unique principal Class-II longitudinal source:

$$\boxed{c_{\parallel}^{(JJ)} \text{ is allowed and is the unique principal longitudinal source.}} \quad (350)$$

Therefore

$$\boxed{v_{\parallel}^{(II)} = v_{\parallel}^{(II, JJ)}, \quad c_{\parallel}^{(II)} = c_{\parallel}^{(JJ)}}. \quad (351)$$

CCXXII. 3. UNIVERSAL SHELL-NORMAL FORM OF THE PURE JJ BLOCK

The shell-reduced pure- JJ derivative block has universal shell-normal form

$$\boxed{K_i^{(II, J)} = -\Gamma_J \hat{n}_i M}, \quad (352)$$

where M is a Hermitian 2×2 matrix on the tangent doublet space:

$$\boxed{M = m_0 \mathbf{1}_2 + m_1 \sigma_1 + m_2 \sigma_2 + m_3 \sigma_3.} \quad (353)$$

Hence

$$K_{\parallel}^{(II,J)} = \hat{n}^i K_i^{(II,J)} = -\Gamma_J M. \quad (354)$$

CCXXIII. 4. EXACT LONGITUDINAL EXTRACTION OF THE JJ PIECE

By the exact projected longitudinal formula,

$$v_{\parallel}^{(II,JJ)} = \frac{1}{2} \text{Tr}(\sigma_3 P_* K_{\parallel}^{(II,J)} P_*). \quad (355)$$

Since $K_{\parallel}^{(II,J)} = -\Gamma_J M$, one gets

$$v_{\parallel}^{(II,JJ)} = -\frac{\Gamma_J}{2} \text{Tr}(\sigma_3 M). \quad (356)$$

Using

$$\text{Tr}(\sigma_3) = 0, \quad \text{Tr}(\sigma_3 \sigma_1) = 0, \quad \text{Tr}(\sigma_3 \sigma_2) = 0, \quad \text{Tr}(\sigma_3 \sigma_3) = 2,$$

we obtain

$$\boxed{v_{\parallel}^{(II,JJ)} = -\Gamma_J m_3.} \quad (357)$$

Therefore the principal Class-II longitudinal slope is controlled by a single scalar:

$$\boxed{v_{\parallel}^{(II)} = -\Gamma_J m_3.} \quad (358)$$

CCXXIV. 5. EQUIVALENT OVERLAP-LAYER DICTIONARY

The same longitudinal contribution is also written, in the current explicit branch dictionary, as

$$\boxed{\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2.} \quad (359)$$

Since in the Bloch pipeline one identifies

$$\lambda_{\parallel} = v_{\parallel},$$

the principal Class-II longitudinal piece can be equivalently parameterized as

$$\boxed{v_{\parallel}^{(II)} = \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2 \quad (\text{same } JJ\text{-sector, alternative normalization language}).} \quad (360)$$

So the two exact languages for the same principal longitudinal Class-II column are:

$$v_{\parallel}^{(II)} = -\Gamma_J m_3 \quad \text{or} \quad v_{\parallel}^{(II)} = \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2.$$

CCXXV. 6. LONGITUDINAL COLUMN AFTER THE JJ -REDUCTION

The full longitudinal column now becomes

$$\boxed{v_{\parallel} = v_{\parallel}^{\text{bare}} - \Gamma_J m_3 + v_{\parallel}^{(\text{sp})}.} \quad (361)$$

Equivalently, in the overlap-language normalization,

$$\boxed{v_{\parallel} = v_{\parallel}^{\text{bare}} + \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2 + v_{\parallel}^{(\text{sp})}.} \quad (362)$$

CCXXVI. 7. HONEST REMAINING TASK

So the longitudinal column is now reduced to three finite ingredients:

$$\boxed{v_{\parallel}^{\text{bare}}, \quad m_3 \text{ (equiv. } j^2), \quad v_{\parallel}^{(\text{sp})}.} \quad (363)$$

What remains open is no longer the channel classification. It is only the explicit evaluation of these three quantities for the chosen refined locked microscopic branch.

CCXXVII. CONCRETE FORMULAS FOR THE BARE AND SPECTATOR PIECES OF THE LONGITUDINAL COLUMN

CCXXVIII. SHORT KOREAN EXPLANATION

The longitudinal column has already been reduced to

$$v_{\parallel} = v_{\parallel}^{\text{bare}} + v_{\parallel}^{(II)} + v_{\parallel}^{(\text{sp})}.$$

The Class-II piece is already isolated. So the next nonredundant step is to write the remaining two pieces explicitly:

1. the bare periodic/refined contribution $v_{\parallel}^{\text{bare}}$,
2. the remote-Schur spectator contribution $v_{\parallel}^{(\text{sp})}$.

The important structural point is that the translation-invariant isotropic bare Brazovskii derivative vanishes exactly on the shell. Therefore the genuinely nontrivial bare longitudinal source must come from the refined periodic/internal k -dependent part.

CCXXIX. 1. CANONICAL DECOMPOSITION OF THE TRIPLET BLOCH BLOCK

Write the triplet Bloch block as

$$\boxed{H_{\Psi\Psi}(k) = H_{\text{bare}}(k) + H_{\text{per}}^{(0)} + H_{\text{per}}^{(\text{int})}(k).} \quad (364)$$

Accordingly,

$$\boxed{K_i^{(\Psi\Psi)} = K_i^{\text{bare}} + K_i^{\text{per},0} + K_i^{\text{per,int}},} \quad (365)$$

with

$$K_i^{\text{bare}} = \Pi_{S_*} [\partial_{k_i} H_{\text{bare}}(k)]_{k=k_*} \Pi_{S_*}, \quad (366)$$

$$K_i^{\text{per},0} = \Pi_{S_*} \left[\partial_{k_i} H_{\text{per}}^{(0)} \right]_{k=k_*} \Pi_{S_*}, \quad (367)$$

$$K_i^{\text{per,int}} = \Pi_{S_*} \left[\partial_{k_i} H_{\text{per}}^{(\text{int})}(k) \right]_{k=k_*} \Pi_{S_*}. \quad (368)$$

Since $H_{\text{per}}^{(0)}$ is k -independent,

$$\boxed{K_i^{\text{per},0} = 0.} \quad (369)$$

Therefore the full bare-side derivative contribution is

$$\boxed{K_i^{(\Psi\Psi)} = K_i^{\text{bare}} + K_i^{\text{per,int}}.} \quad (370)$$

CCXXX. 2. VANISHING OF THE ISOTROPIC ON-SHELL DERIVATIVE

For the isotropic Brazovskii bare kernel

$$\boxed{h_0(q) = r_0 + Z|q|^2 + Y|q|^4,} \quad (371)$$

its derivative is

$$\partial_{q_i} h_0(q) = 2(Z + 2Y|q|^2)q_i. \quad (372)$$

At the Brazovskii shell

$$|q| = q_*, \quad q_*^2 = -\frac{Z}{2Y},$$

one has

$$\boxed{\partial_{q_i} h_0(q) \big|_{|q|=q_*} = 0.} \quad (373)$$

Thus the isotropic translation-invariant bare symbol does not generate the first-order shell transport on-shell. Hence the genuinely nontrivial bare longitudinal source must come from the refined periodic/internal k -dependent piece

$$H_{\text{per}}^{(\text{int})}(k).$$

CCXXXI. 3. EXACT BARE LONGITUDINAL FORMULA

The bare longitudinal derivative block is therefore

$$\boxed{K_{\parallel}^{\text{bare}} = \hat{n}_i K_i^{(\Psi\Psi)} = \hat{n}_i \Pi_{S_*} \left[\partial_{k_i} H_{\text{per}}^{(\text{int})}(k) \right]_{k=k_*} \Pi_{S_*},} \quad (374)$$

where the isotropic on-shell derivative has already dropped out.

Projecting to the low shell slot gives

$$\boxed{v_{\parallel}^{\text{bare}} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*} (\sigma^3 P_* K_{\parallel}^{\text{bare}} P_*)}. \quad (375)$$

Equivalently, in the valley-diagonal internal block language,

$$\boxed{v_{\parallel}^{\text{bare}} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{E_+^{\text{bare}}(p) - E_-^{\text{bare}}(p)}{2} \right] \bigg|_{p=0}}, \quad (376)$$

where $E_{\pm}^{\text{bare}}(p)$ are the two valley-diagonal 2×2 blocks coming from the refined periodic/internal $\Psi\Psi$ -sector.

CCXXXII. 4. EXACT SPECTATOR LONGITUDINAL FORMULA

Let

$$A(p) := H_{\text{low,rem}}(p), \quad R(p) := H_{\text{rem,rem}}(p)^{-1}, \quad B(p) := H_{\text{rem,low}}(p).$$

The remote Schur correction is

$$-A(p)R(p)B(p).$$

Differentiating at $p = 0$ in the longitudinal direction gives

$$\boxed{K_{\parallel}^{(\text{sp})} = -\partial_{p_{\parallel}} A(p)|_{p=0} R_* B_* - A_* R_* \partial_{p_{\parallel}} B(p)|_{p=0} + A_* R_* \partial_{p_{\parallel}} H_{\text{rem,rem}}(p)|_{p=0} R_* B_*}, \quad (377)$$

where

$$A_* = A(0), \quad B_* = B(0), \quad R_* = H_{\text{rem,rem}}(0)^{-1}.$$

Projecting this to the low shell slot gives

$$\boxed{v_{\parallel}^{(\text{sp})} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*}(\sigma^3 P_* K_{\parallel}^{(\text{sp})} P_*)}. \quad (378)$$

This is the exact spectator contribution to the longitudinal shell slope.

CCXXXIII. 5. FINAL LONGITUDINAL COLUMN AFTER THE CONCRETE REDUCTION

Combining the already-isolated Class-II piece with the formulas above, the longitudinal column becomes

$$\boxed{v_{\parallel} = v_{\parallel}^{\text{bare}} + v_{\parallel}^{(II)} + v_{\parallel}^{(\text{sp})}}, \quad (379)$$

with

$$\boxed{v_{\parallel}^{\text{bare}} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*}(\sigma^3 P_* K_{\parallel}^{\text{bare}} P_*)}, \quad (380)$$

$$\boxed{v_{\parallel}^{(II)} = -\Gamma_J m_3 \quad (\text{equivalently, } v_{\parallel}^{(II)} = \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2 \text{ at principal } JJ \text{ shell order})}, \quad (381)$$

$$\boxed{v_{\parallel}^{(\text{sp})} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*}(\sigma^3 P_* K_{\parallel}^{(\text{sp})} P_*)}. \quad (382)$$

So the remaining longitudinal microscopic frontier has now been reduced to three finite scalar evaluations:

$$\boxed{v_{\parallel}^{\text{bare}}, \quad m_3 \text{ (or equivalently } j^2), \quad v_{\parallel}^{(\text{sp})}}. \quad (383)$$

CCXXXIV. FINAL CONCRETE REDUCTION OF THE LONGITUDINAL COLUMN

CCXXXV. SHORT KOREAN EXPLANATION

The bare longitudinal piece can now be reduced one step further.

The projected periodic/internal shell block satisfies

$$P_* H_{\text{per}}^{(\text{int})}(k_* + p) P_* = P_* H_0 P_* + p_i \tilde{\mathcal{N}}_i^a \sigma_a + O(|p|^2),$$

so that

$$K_i^{\text{per,int}} = \tilde{\mathcal{N}}_i^a \sigma_a.$$

Hence the bare longitudinal slope is not an abstract derivative anymore. It is exactly the shell-normal σ_3 coefficient

$$\tilde{\mathcal{N}}_{\parallel}^3.$$

CCXXXVI. 1. BARE LONGITUDINAL PIECE AS THE SHELL-NORMAL σ_3 COEFFICIENT

From the projected periodic/internal expansion,

$$\boxed{P_* H_{\text{per}}^{(\text{int})}(k_* + p) P_* = P_* H_0 P_* + p_i \tilde{\mathcal{N}}_i^a \sigma_a + O(|p|^2),} \quad (384)$$

one gets

$$\boxed{K_i^{\text{per, int}} = \tilde{\mathcal{N}}_i^a \sigma_a.} \quad (385)$$

Using the exact extraction formula

$$v_{\parallel}^{\text{bare}} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*} (\sigma^3 P_* K_{\parallel}^{\text{bare}} P_*),$$

and the fact that the isotropic on-shell derivative has already dropped out, one obtains

$$\begin{aligned} v_{\parallel}^{\text{bare}} &= \frac{1}{2} \text{Tr}_{\mathcal{H}_*} (\sigma^3 \hat{n}^i \tilde{\mathcal{N}}_i^a \sigma_a) \\ &= \hat{n}^i \tilde{\mathcal{N}}_i^3. \end{aligned} \quad (386)$$

Thus

$$\boxed{v_{\parallel}^{\text{bare}} = \tilde{\mathcal{N}}_{\parallel}^3.} \quad (387)$$

CCXXXVII. 2. MINIMAL SHELL TENSOR BASIS

The shell tensor basis is

$$\boxed{\tilde{\mathcal{N}}_i^a = \lambda_{\parallel} \hat{n}_i \delta^{a3} + \alpha (e_{1i} \delta^{a1} + e_{2i} \delta^{a2}) + \beta (-e_{1i} \delta^{a2} + e_{2i} \delta^{a1}).} \quad (388)$$

Therefore

$$\boxed{\tilde{\mathcal{N}}_{\parallel}^3 = \hat{n}^i \tilde{\mathcal{N}}_i^3,} \quad (389)$$

is exactly the shell-normal longitudinal coefficient in the bare periodic/internal block.

CCXXXVIII. 3. CLASS-II AND SPECTATOR PIECES

The principal Class-II longitudinal piece is already reduced to

$$\boxed{v_{\parallel}^{(II)} = -\Gamma_J m_3,} \quad (390)$$

equivalently

$$\boxed{v_{\parallel}^{(II)} = \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2} \quad (391)$$

in the pure- JJ language.

The spectator piece remains the derivative of the remote Schur complement:

$$\boxed{v_{\parallel}^{(\text{sp})} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*} (\sigma^3 P_* K_{\parallel}^{(\text{sp})} P_*).} \quad (392)$$

CCXXXIX. 4. FINAL CONCRETE LONGITUDINAL COLUMN

Combining everything, the longitudinal column is now

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 - \Gamma_J m_3 + v_{\parallel}^{(\text{sp})}. \quad (393)$$

Equivalently, in the overlap-language normalization,

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2 + v_{\parallel}^{(\text{sp})}, \quad (394)$$

up to the sign convention absorbed into the definition of the pure- JJ coefficient.

CCXL. 5. HONEST REMAINING MICROSCOPIC INPUTS

So the longitudinal column has been reduced to exactly three finite scalar inputs:

$$\tilde{\mathcal{N}}_{\parallel}^3, \quad m_3 \text{ (equiv. } j^2), \quad v_{\parallel}^{(\text{sp})}. \quad (395)$$

No additional channel classification remains. The next microscopic task is only the explicit evaluation of these three quantities in the chosen refined locked branch.

CCXLI. CONCRETE BARE AND SPECTATOR LONGITUDINAL FORMULAS IN THE BLOCH-OPERATOR PIPELINE

CCXLII. SHORT KOREAN EXPLANATION

The longitudinal column is now reduced to three finite pieces:

$$v_{\parallel} = v_{\parallel}^{\text{bare}} + v_{\parallel}^{(II)} + v_{\parallel}^{(\text{sp})}.$$

The Class-II piece has already been isolated to the principal JJ -sector. So the remaining task is to make the bare and spectator pieces completely explicit.

The key point is:

1. the only live bare first-order triplet source is the k -dependent periodic/internal block $H_{\text{per}}^{(\text{int})}(k)$;
2. the spectator piece is exactly the derivative of the Schur-renormalized pair-diagonal blocks $E_{\pm}^{\text{eff}}(E, p)$.

CCXLIII. 1. BARE LONGITUDINAL SOURCE: ONLY $H_{\text{per}}^{(\text{int})}(k)$ SURVIVES

The triplet-sector audit gives

$$H_{\Psi\Psi}(k) = \underbrace{H_{\text{bare}}(k)}_{\text{dead in } K_i} + \underbrace{H_{\text{per}}^{(0)}}_{\text{dead in } K_i} + \underbrace{H_{\text{per}}^{(\text{int})}(k)}_{\text{only live triplet source}}. \quad (396)$$

Hence the exact live triplet-sector derivative is

$$K_{\parallel}^{\text{bare}} = \Pi_{S_*} \left[\partial_{k_{\parallel}} H_{\text{per}}^{(\text{int})}(k) \right]_{k=k_*} \Pi_{S_*}. \quad (397)$$

The corresponding bare longitudinal shell slope is therefore

$$v_{\parallel}^{\text{bare}} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*} (\sigma^3 P_* K_{\parallel}^{\text{bare}} P_*). \quad (398)$$

Equivalently, in the pair-diagonal valley notation,

$$v_{\parallel}^{\text{bare}} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{E_+^{\text{bare}}(p) - E_-^{\text{bare}}(p)}{2} \right] \Big|_{p=0}, \quad (399)$$

where $E_{\pm}^{\text{bare}}(p)$ are the 2×2 pair-diagonal blocks induced by $H_{\text{per}}^{(\text{int})}(k)$.

Thus the bare longitudinal column is completely localized to the shell-compatible k -linear harmonic matrix elements of $H_{\text{per}}^{(\text{int})}(k)$ at $k = k_*$.

CCXLIV. 2. SPECTATOR ELIMINATION: EXACT PAIR-DIAGONAL FORMULAS

The finite shell matrix has the exact block form

$$H_{\text{shell}}(p) = \begin{pmatrix} E_+(p) & B(p) & C_+(p) \\ B(p)^{\dagger} & E_-(p) & C_-(p) \\ C_+(p)^{\dagger} & C_-(p)^{\dagger} & H_{ss}(p) \end{pmatrix}. \quad (400)$$

After exact spectator elimination, the core effective Hamiltonian is

$$H_{\text{core}}^{\text{eff}}(E, p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^{\dagger} & E_-(p) \end{pmatrix} - \begin{pmatrix} C_+(p) \\ C_-(p) \end{pmatrix} R_{\text{sp}}(E, p) \begin{pmatrix} C_+(p)^{\dagger} & C_-(p)^{\dagger} \end{pmatrix}, \quad (401)$$

with

$$R_{\text{sp}}(E, p) = (H_{ss}(p) - E)^{-1}.$$

Therefore the renormalized pair-diagonal blocks are exactly

$$E_+^{\text{eff}}(E, p) = E_+(p) - C_+(p) R_{\text{sp}}(E, p) C_+(p)^{\dagger}, \quad (402)$$

$$E_-^{\text{eff}}(E, p) = E_-(p) - C_-(p) R_{\text{sp}}(E, p) C_-(p)^{\dagger}. \quad (403)$$

CCXLV. 3. EXACT SPECTATOR LONGITUDINAL SLOPE

By definition, the spectator contribution to the longitudinal shell slope is the part of the pair-diagonal slope coming from the Schur correction in (402)–(403).

Hence

$$v_{\parallel}^{(\text{sp})} = -\frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{C_+(p) R_{\text{sp}}(E, p) C_+(p)^{\dagger} - C_-(p) R_{\text{sp}}(E, p) C_-(p)^{\dagger}}{2} \right] \Big|_{p=0}. \quad (404)$$

Equivalently, writing

$$A(p) := \begin{pmatrix} C_+(p) \\ C_-(p) \end{pmatrix}, \quad R(p) := R_{\text{sp}}(E, p),$$

this is the same spectator derivative already obtained from the general Schur-complement formula.

CCXLVI. 4. FULL LONGITUDINAL COLUMN IN FULLY CONCRETE FORM

Combining the bare, Class-II, and spectator pieces, we obtain the fully concrete longitudinal column:

$$v_{\parallel} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{E_+^{\text{bare}}(p) - E_-^{\text{bare}}(p)}{2} \right] \Big|_{p=0} + v_{\parallel}^{(II)} - \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{C_+(p) R_{\text{sp}}(E, p) C_+(p)^{\dagger} - C_-(p) R_{\text{sp}}(E, p) C_-(p)^{\dagger}}{2} \right] \Big|_{p=0}. \quad (405)$$

At principal shell order, the Class-II piece is already reduced to the pure- JJ sector:

$$v_{\parallel}^{(II)} = -\Gamma_J m_3, \quad (406)$$

equivalently

$$v_{\parallel}^{(II)} = \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2. \quad (407)$$

Therefore the remaining microscopic longitudinal inputs are exactly

$$E_+^{\text{bare}}(p) - E_-^{\text{bare}}(p), \quad m_3 \text{ (equiv. } j^2), \quad C_{\pm}(p), \quad R_{\text{sp}}(E, p). \quad (408)$$

CCXLVII. 5. HONEST ENDPOINT OF THE CURRENT PIPELINE STAGE

No further structural reorganization is needed.

The longitudinal column has now been reduced to:

1. a bare pair-diagonal shell slope from $H_{\text{per}}^{(\text{int})}(k)$,
2. a pure- JJ Class-II shell slope,
3. a spectator-renormalized pair-diagonal shell slope from the exact Schur correction.

So the next legitimate step inside the Bloch-operator pipeline is the actual insertion of these explicit block matrix elements, not another abstract reduction.

CCXLVIII. FINAL SUMMARY

The longitudinal column in the Bloch-operator extraction pipeline is now fully concrete:

$$v_{\parallel} = v_{\parallel}^{\text{bare}} + v_{\parallel}^{(II)} + v_{\parallel}^{(\text{sp})}.$$

The bare piece is localized entirely to the shell-compatible k -linear periodic/internal block:

$$v_{\parallel}^{\text{bare}} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{E_+^{\text{bare}}(p) - E_-^{\text{bare}}(p)}{2} \right] \Big|_{p=0},$$

equivalently

$$K_{\parallel}^{\text{bare}} = \Pi_{S_*} \left[\partial_{k_{\parallel}} H_{\text{per}}^{(\text{int})}(k) \right]_{k=k_*} \Pi_{S_*}.$$

The principal Class-II piece is already reduced to the pure- JJ sector:

$$v_{\parallel}^{(II)} = -\Gamma_J m_3,$$

equivalently

$$v_{\parallel}^{(II)} = \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2.$$

The spectator piece is the exact pair-diagonal Schur-renormalized slope:

$$v_{\parallel}^{(\text{sp})} = -\frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{C_+(p) R_{\text{sp}}(E, p) C_+(p)^{\dagger} - C_-(p) R_{\text{sp}}(E, p) C_-(p)^{\dagger}}{2} \right] \Big|_{p=0}.$$

Therefore the remaining longitudinal microscopic inputs are exactly

$$E_+^{\text{bare}}(p) - E_-^{\text{bare}}(p), \quad m_3 \text{ (equiv. } j^2), \quad C_{\pm}(p), \quad R_{\text{sp}}(E, p).$$

No further structural reduction is needed. The next step is the explicit insertion of these block matrix elements.

CCXLIX. BARE LONGITUDINAL SOURCE IS NOT SYMMETRY-FORCED TO VANISH

CCL. SHORT KOREAN EXPLANATION

The next decisive question is whether the bare longitudinal slope

$$v_{\parallel}^{\text{bare}}$$

is forced to vanish by symmetry.

The answer is no.

The bare longitudinal piece is defined through the pair-diagonal shell difference

$$D^{\text{bare}}(p) = \frac{1}{2} (E_+^{\text{bare}}(p) - E_-^{\text{bare}}(p)),$$

hence

$$v_{\parallel}^{\text{bare}} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} D^{\text{bare}}(p) \Big|_{p=0}.$$

If the opposite valley is related by $p \mapsto -p$, then

$$E_-^{\text{bare}}(p) = E_+^{\text{bare}}(-p),$$

so

$$D^{\text{bare}}(p) = \frac{1}{2} (E_+^{\text{bare}}(p) - E_+^{\text{bare}}(-p)).$$

Therefore

$$\partial_{p_{\parallel}} D^{\text{bare}}(p) \Big|_{p=0} = \partial_{p_{\parallel}} E_+^{\text{bare}}(p) \Big|_{p=0},$$

which is not symmetry-forced to vanish.

Thus $v_{\parallel}^{\text{bare}} = 0$ would require an additional tuning, not merely the valley symmetry itself.

CCLI. 1. EXACT PAIR-DIAGONAL DEFINITION

The bare longitudinal shell slope is

$$v_{\parallel}^{\text{bare}} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{E_+^{\text{bare}}(p) - E_-^{\text{bare}}(p)}{2} \right] \Big|_{p=0}. \quad (409)$$

CCLII. 2. VALLEY RELATION DOES NOT FORCE CANCELLATION

Assume the opposite-valley bare blocks satisfy

$$E_-^{\text{bare}}(p) = E_+^{\text{bare}}(-p). \quad (410)$$

Then

$$\begin{aligned} D^{\text{bare}}(p) &= \frac{1}{2}(E_+^{\text{bare}}(p) - E_-^{\text{bare}}(p)) \\ &= \frac{1}{2}(E_+^{\text{bare}}(p) - E_+^{\text{bare}}(-p)). \end{aligned} \quad (411)$$

Differentiating at $p = 0$,

$$\partial_{p_{\parallel}} D^{\text{bare}}(p)|_{p=0} = \partial_{p_{\parallel}} E_+^{\text{bare}}(p)|_{p=0}, \quad (412)$$

so the valley relation doubles the odd slope rather than cancelling it.

Therefore

$$\boxed{v_{\parallel}^{\text{bare}} = 0 \quad \text{is not symmetry-forced.}} \quad (413)$$

CCLIII. 3. SHELL-TENSOR FORM

After shell projection, the refined periodic/internal block satisfies

$$\boxed{P_* H_{\text{per}}^{(\text{int})}(k_* + p) P_* = P_* H_0 P_* + p_i \tilde{\mathcal{N}}_i^a \sigma_a + O(|p|^2),} \quad (414)$$

hence

$$\boxed{K_i^{\text{per,int}} = \tilde{\mathcal{N}}_i^a \sigma_a.} \quad (415)$$

Thus

$$\boxed{v_{\parallel}^{\text{bare}} = \tilde{\mathcal{N}}_{\parallel}^3.} \quad (416)$$

So the vanishing of the bare longitudinal piece is equivalent to

$$\tilde{\mathcal{N}}_{\parallel}^3 = 0,$$

which is an actual microscopic coefficient statement, not a symmetry theorem.

CCLIV. 4. CONSEQUENCE FOR THE FULL LONGITUDINAL COLUMN

The full longitudinal column now reads

$$\boxed{v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 - \Gamma_J m_3 + v_{\parallel}^{(\text{sp})}.} \quad (417)$$

Hence the remaining nontrivial microscopic question is not whether the bare piece is symmetry-forbidden. It is whether the spectator piece survives or vanishes in the chosen working truncation.

CCLV. FINAL SUMMARY

The bare longitudinal shell slope is

$$v_{\parallel}^{\text{bare}} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{E_+^{\text{bare}}(p) - E_-^{\text{bare}}(p)}{2} \right] \Big|_{p=0}.$$

If the opposite-valley relation is

$$E_-^{\text{bare}}(p) = E_+^{\text{bare}}(-p),$$

then the pair-diagonal difference is

$$D^{\text{bare}}(p) = \frac{1}{2} (E_+^{\text{bare}}(p) - E_+^{\text{bare}}(-p)),$$

and therefore its longitudinal derivative at $p = 0$ does not cancel. So

$$v_{\parallel}^{\text{bare}} = 0 \text{ is not symmetry-forced.}$$

After shell projection,

$$P_* H_{\text{per}}^{(\text{int})}(k_* + p) P_* = P_* H_0 P_* + p_i \tilde{\mathcal{N}}_i^a \sigma_a + O(|p|^2),$$

so

$$v_{\parallel}^{\text{bare}} = \tilde{\mathcal{N}}_{\parallel}^3.$$

Therefore the full longitudinal column is now concretely

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 - \Gamma_J m_3 + v_{\parallel}^{(\text{sp})}.$$

The next nonredundant microscopic task is to determine whether the spectator contribution

$$v_{\parallel}^{(\text{sp})}$$

vanishes in the present working truncation or provides an additional independent longitudinal source.

CCLVI. SPECTATOR LONGITUDINAL PIECE IN THE ONE-STEP SCALAR SPECTATOR CLOSURE

CCLVII. SHORT KOREAN EXPLANATION

Inside the current Bloch-operator pipeline, the spectator longitudinal term is

$$v_{\parallel}^{(\text{sp})} = -\frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{C_+(p) R_{\text{sp}}(E, p) C_+(p)^{\dagger} - C_-(p) R_{\text{sp}}(E, p) C_-(p)^{\dagger}}{2} \right] \Big|_{p=0}.$$

In the one-step scalar spectator closure already written in the notes, the spectator block is scalar on the internal doublet and the core-to-spectator couplings are equal for the $+$ and $-$ core pair. As a result, the two diagonal Schur renormalizations coincide identically, so their difference has zero longitudinal derivative.

Therefore

$$v_{\parallel}^{(\text{sp})} = 0$$

in the current one-step scalar spectator truncation.

CCLVIII. 1. EXACT SPECTATOR FORMULA

The renormalized pair-diagonal blocks are

$$\boxed{E_+^{\text{eff}}(E, p) = E_+(p) - C_+(p)R_{\text{sp}}(E, p)C_+(p)^\dagger}, \quad (418)$$

$$\boxed{E_-^{\text{eff}}(E, p) = E_-(p) - C_-(p)R_{\text{sp}}(E, p)C_-(p)^\dagger}. \quad (419)$$

Hence the spectator contribution to the longitudinal slope is

$$\boxed{v_{\parallel}^{(\text{sp})} = -\frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{C_+(p)R_{\text{sp}}(E, p)C_+(p)^\dagger - C_-(p)R_{\text{sp}}(E, p)C_-(p)^\dagger}{2} \right] \Big|_{p=0}}. \quad (420)$$

CCLIX. 2. ONE-STEP SCALAR SPECTATOR SHELL

In the one-step shell closure, the spectator matrix is

$$\boxed{H_{\text{1step}} = \begin{pmatrix} E_c & m & g & g & g & g \\ m & E_c & g & g & g & g \\ g & g & E_s & 0 & 0 & 0 \\ g & g & 0 & E_s & 0 & 0 \\ g & g & 0 & 0 & E_s & 0 \\ g & g & 0 & 0 & 0 & E_s \end{pmatrix} \otimes \mathbf{1}_2 + O(A^4)}, \quad (421)$$

with a single symmetric spectator combination

$$|S\rangle = \frac{1}{2} \sum_{a=1}^4 |\eta_a\rangle$$

coupled to the core pair, while the other three spectator combinations decouple.

Crucially, the spectator shell is scalar on the internal doublet:

$$H_{ss}(p) \propto \mathbf{1}_2, \quad R_{\text{sp}}(E, p) \propto \mathbf{1}_2.$$

Moreover, by cubic symmetry and equal amplitudes, the + and − core pair couple to the same symmetric spectator combination with the same scalar amplitude g . Therefore, at the one-step scalar level,

$$\boxed{C_+(p) = C_-(p) =: C_{\text{sp}}(p)} \quad (422)$$

up to the common internal spectator factor $\mathbf{1}_2$.

CCLX. 3. EXACT CANCELLATION OF THE DIAGONAL SPECTATOR DIFFERENCE

Substituting (422) into the Schur corrections gives

$$C_+(p)R_{\text{sp}}(E, p)C_+(p)^\dagger = C_-(p)R_{\text{sp}}(E, p)C_-(p)^\dagger. \quad (423)$$

Hence the spectator contribution to the pair-diagonal difference vanishes identically:

$$\boxed{C_+(p)R_{\text{sp}}(E, p)C_+(p)^\dagger - C_-(p)R_{\text{sp}}(E, p)C_-(p)^\dagger = 0}. \quad (424)$$

Differentiating (424) with respect to $p_{\parallel}$ and inserting into (420), one obtains

$$\boxed{v_{\parallel}^{(\text{sp})} = 0}. \quad (425)$$

CCLXI. 4. CONSEQUENCE FOR THE LONGITUDINAL COLUMN

Therefore, in the present one-step scalar spectator truncation, the longitudinal column collapses to

$$v_{\parallel} = v_{\parallel}^{\text{bare}} + v_{\parallel}^{(II)}. \quad (426)$$

Using the previously derived formulas,

$$v_{\parallel}^{\text{bare}} = \tilde{\mathcal{N}}_{\parallel}^3, \quad v_{\parallel}^{(II)} = -\Gamma_J m_3, \quad (427)$$

so that

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 - \Gamma_J m_3. \quad (428)$$

Equivalently, in the overlap-language normalization of the pure- JJ branch,

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2, \quad (429)$$

up to the fixed sign convention between $\Gamma_J m_3$ and the j^2 -normalization.

CCLXII. 5. HONEST STATUS

What is proved here is exact for the current one-step scalar spectator closure:

$$v_{\parallel}^{(\text{sp})} = 0.$$

What is not yet proved in complete generality is that every possible enlarged spectator completion beyond this truncation preserves the same cancellation. So the present conclusion is:

$$\boxed{\text{in the current working spectator truncation, the spectator longitudinal source vanishes exactly.}} \quad (430)$$

Hence the only remaining longitudinal microscopic inputs in the current pipeline are

$$\boxed{\tilde{\mathcal{N}}_{\parallel}^3, \quad m_3 \text{ (equiv. } j^2\text{).}} \quad (431)$$

CCLXIII. FINAL SUMMARY

The spectator longitudinal contribution is

$$v_{\parallel}^{(\text{sp})} = -\frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{C_+(p) R_{\text{sp}}(E, p) C_+(p)^{\dagger} - C_-(p) R_{\text{sp}}(E, p) C_-(p)^{\dagger}}{2} \right] \Big|_{p=0}.$$

In the current one-step scalar spectator closure,

$$H_{\text{1step}} = (\text{scalar shell matrix}) \otimes \mathbf{1}_2,$$

the spectator shell is scalar on the internal doublet, and the $+$ and $-$ core states couple to the same symmetric spectator combination with equal scalar amplitude. Therefore

$$C_+(p) = C_-(p),$$

hence

$$C_+(p)R_{\text{sp}}(E, p)C_+(p)^\dagger = C_-(p)R_{\text{sp}}(E, p)C_-(p)^\dagger,$$

and so

$$v_{\parallel}^{(\text{sp})} = 0.$$

Thus, in the current working spectator truncation, the longitudinal column reduces exactly to

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 - \Gamma_J m_3.$$

Equivalently, in the overlap normalization of the pure- JJ branch,

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2.$$

So the remaining longitudinal microscopic inputs in the present pipeline are only

$$\tilde{\mathcal{N}}_{\parallel}^3, \quad m_3 \text{ (equiv. } j^2 \text{)}.$$

CCLXIV. SPECTATOR LONGITUDINAL CANCELLATION IN THE CURRENT WORKING TRUNCATION

CCLXV. SHORT KOREAN EXPLANATION

Inside the Bloch-operator extraction pipeline, the spectator longitudinal contribution is

$$v_{\parallel}^{(\text{sp})} = -\frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{C_+(p)R_{\text{sp}}(E, p)C_+(p)^\dagger - C_-(p)R_{\text{sp}}(E, p)C_-(p)^\dagger}{2} \right] \Big|_{p=0}.$$

In the current one-step scalar spectator closure, the spectator shell is scalar on the internal doublet and the $+$ and $-$ core states couple to the same symmetric spectator combination with equal scalar amplitude. Therefore

$$C_+(p) = C_-(p),$$

hence

$$C_+(p)R_{\text{sp}}(E, p)C_+(p)^\dagger = C_-(p)R_{\text{sp}}(E, p)C_-(p)^\dagger.$$

So the pair-diagonal spectator difference vanishes identically, and therefore

$$v_{\parallel}^{(\text{sp})} = 0. \tag{432}$$

CCLXVI. 1. EXACT SPECTATOR FORMULA

The exact renormalized pair-diagonal blocks are

$$E_+^{\text{eff}}(E, p) = E_+(p) - C_+(p)R_{\text{sp}}(E, p)C_+(p)^\dagger, \tag{433}$$

$$E_-^{\text{eff}}(E, p) = E_-(p) - C_-(p)R_{\text{sp}}(E, p)C_-(p)^\dagger. \tag{434}$$

Hence the spectator part of the longitudinal slope is

$$v_{\parallel}^{(\text{sp})} = -\frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[\frac{C_+(p)R_{\text{sp}}(E, p)C_+(p)^\dagger - C_-(p)R_{\text{sp}}(E, p)C_-(p)^\dagger}{2} \right] \Big|_{p=0}. \tag{435}$$

CCLXVII. 2. ONE-STEP SCALAR SPECTATOR CLOSURE

In the one-step shell closure, the spectator shell is scalar on the internal doublet and the spectator sector reduces to the symmetric spectator combination. At this level the core-to-spectator couplings satisfy

$$\boxed{C_+(p) = C_-(p) =: C_{\text{sp}}(p).} \quad (436)$$

Therefore

$$C_+(p)R_{\text{sp}}(E, p)C_+(p)^\dagger - C_-(p)R_{\text{sp}}(E, p)C_-(p)^\dagger = 0, \quad (437)$$

and insertion into (435) gives

$$\boxed{v_{\parallel}^{(\text{sp})} = 0.} \quad (438)$$

CCLXVIII. 3. FINAL CURRENT LONGITUDINAL COLUMN

Thus the current working-truncation longitudinal column reduces to

$$\boxed{v_{\parallel} = v_{\parallel}^{\text{bare}} + v_{\parallel}^{(II)}.} \quad (439)$$

Using the already derived formulas

$$\boxed{v_{\parallel}^{\text{bare}} = \tilde{\mathcal{N}}_{\parallel}^3, \quad v_{\parallel}^{(II)} = -\Gamma_J m_3,} \quad (440)$$

we obtain

$$\boxed{v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 - \Gamma_J m_3.} \quad (441)$$

Equivalently, in the overlap normalization of the principal pure- JJ branch,

$$\boxed{v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{1}{2}I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2,} \quad (442)$$

up to the fixed sign convention relating $\Gamma_J m_3$ and the j^2 -normalization.

CCLXIX. 4. HONEST STATUS

What is proved here is exact only for the current one-step scalar spectator closure. It does not yet prove that every enlarged spectator completion beyond this truncation preserves the same cancellation.

Thus the honest current statement is:

$$\boxed{\text{in the present working spectator truncation, the spectator longitudinal source vanishes exactly.}} \quad (443)$$

So the only remaining longitudinal microscopic inputs are

$$\boxed{\tilde{\mathcal{N}}_{\parallel}^3, \quad m_3 \text{ (equiv. } j^2\text{).}} \quad (444)$$

CCLXX. GENERIC LONGITUDINAL NONDEGENERACY IN THE CURRENT WORKING TRUNCATION

CCLXXI. SHORT KOREAN EXPLANATION

Inside the current Bloch-operator working truncation, the spectator longitudinal source vanishes:

$$v_{\parallel}^{(\text{sp})} = 0.$$

Therefore the full longitudinal shell slope reduces to

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 - \Gamma_J m_3.$$

This is the key breakthrough. The remaining longitudinal problem is no longer a broad operator-classification issue. It is the interaction of exactly two finite microscopic scalars.

Hence the vanishing condition

$$v_{\parallel} = 0$$

is equivalent to the exact cancellation equation

$$\tilde{\mathcal{N}}_{\parallel}^3 = \Gamma_J m_3.$$

So, within the present working truncation, longitudinal failure is not structurally forced. It requires fine tuning.

CCLXXII. 1. FINAL TWO-SCALAR FORM OF THE LONGITUDINAL COLUMN

From the previous steps, the longitudinal column in the current working truncation is

$$\boxed{v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 - \Gamma_J m_3.} \tag{445}$$

Here:

1. $\tilde{\mathcal{N}}_{\parallel}^3$ is the shell-normal σ_3 -coefficient of the projected refined periodic/internal block,

$$P_* H_{\text{per}}^{(\text{int})}(k_* + p) P_* = P_* H_0 P_* + p_i \tilde{\mathcal{N}}_i^a \sigma_a + O(|p|^2);$$

2. $-\Gamma_J m_3$ is the principal pure- JJ Class-II longitudinal contribution;
3. the spectator contribution vanishes in the present one-step scalar spectator closure.

CCLXXIII. 2. EXACT CANCELLATION HYPERSURFACE

Equation (445) implies

$$\boxed{v_{\parallel} = 0 \iff \tilde{\mathcal{N}}_{\parallel}^3 = \Gamma_J m_3.} \tag{446}$$

Thus the zero set is not a symmetry theorem. It is a single algebraic cancellation relation between two finite microscopic scalars.

CCLXXIV. 3. GENERIC NONDEGENERACY THEOREM

[Generic longitudinal nondegeneracy in the current working truncation] Assume the refined locked + Class II Bloch-operator pipeline, together with the current one-step scalar spectator closure, so that

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 - \Gamma_J m_3.$$

Assume further that neither

$$\tilde{\mathcal{N}}_{\parallel}^3$$

nor

$$\Gamma_J m_3$$

is symmetry-forced to vanish identically in the chosen microscopic branch.

Then the longitudinal failure condition

$$v_{\parallel} = 0$$

is a codimension-one exact cancellation condition in the two-scalar parameter space

$$(\tilde{\mathcal{N}}_{\parallel}^3, \Gamma_J m_3).$$

Therefore, away from fine tuning,

$$\boxed{v_{\parallel} \neq 0.} \tag{447}$$

By (445),

$$v_{\parallel} = 0 \iff \tilde{\mathcal{N}}_{\parallel}^3 - \Gamma_J m_3 = 0.$$

This is one algebraic equation in the two-scalar space

$$(\tilde{\mathcal{N}}_{\parallel}^3, \Gamma_J m_3).$$

Hence its solution set is a codimension-one cancellation hypersurface. If neither scalar is symmetry-forced to vanish identically, then away from this tuning locus one has

$$v_{\parallel} \neq 0.$$

CCLXXV. 4. EQUIVALENT OVERLAP-LANGUAGE FORM

Using the pure- JJ normalization dictionary

$$\Gamma_J m_3 \longleftrightarrow -\frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2$$

(up to the fixed sign convention), the same statement may be written as

$$\boxed{v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2.} \tag{448}$$

Therefore the longitudinal zero condition becomes

$$\boxed{\tilde{\mathcal{N}}_{\parallel}^3 = -\frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2,} \tag{449}$$

which again is an exact cancellation relation rather than a symmetry-forced obstruction.

CCLXXVI. 5. HONEST STATUS

What is now genuinely achieved is stronger than a mere structural reduction:

1. the spectator longitudinal source has disappeared in the present working truncation;
2. the full longitudinal column has collapsed to two finite microscopic scalars;
3. the vanishing of the longitudinal slope is now an explicit cancellation equation;
4. therefore longitudinal nondegeneracy is generic within the current working truncation.

What is not yet done is the final symbolic or numerical evaluation of

$$\tilde{\mathcal{N}}_{\parallel}^3 \quad \text{and} \quad m_3.$$

So the frontier is no longer conceptual. It is the explicit computation of the two scalar inputs entering (445).

CCLXXVII. FINAL SUMMARY

In the current refined locked + Class II Bloch-operator working truncation, the longitudinal shell slope has been reduced to the exact two-scalar form

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 - \Gamma_J m_3.$$

Here

$$\tilde{\mathcal{N}}_{\parallel}^3$$

is the shell-normal σ_3 -coefficient of the projected refined periodic/internal block, while

$$-\Gamma_J m_3$$

is the principal pure- JJ Class-II longitudinal contribution.

The spectator longitudinal piece vanishes in the present one-step scalar spectator closure, so no third independent source remains.

Therefore the longitudinal zero condition is exactly

$$v_{\parallel} = 0 \iff \tilde{\mathcal{N}}_{\parallel}^3 = \Gamma_J m_3.$$

Thus, inside the current working truncation, longitudinal failure is not structurally forced. It is a codimension-one exact cancellation condition. Equivalently, away from fine tuning,

$$v_{\parallel} \neq 0.$$

So the remaining microscopic frontier is no longer conceptual. It is the explicit symbolic or numerical evaluation of the two scalar inputs

$$\tilde{\mathcal{N}}_{\parallel}^3, \quad m_3.$$

CCLXXVIII. INSERTION OF THE ACTUAL m_3 -EXTRACTION INTO THE LONGITUDINAL COLUMN

CCLXXIX. SHORT KOREAN EXPLANATION

Inside the present Bloch-operator working truncation, the spectator contribution vanishes, so

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 - \Gamma_J m_3.$$

The key further reduction is that m_3 is already known more explicitly than $\tilde{\mathcal{N}}_{\parallel}^3$. Indeed, the first-shell dominant intervalley audit gives

$$m_3 = -\frac{|\Omega_{\text{cell}}|}{2\Gamma_J} \left(\Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right).$$

Substituting this directly yields a fully explicit two-source longitudinal formula.

CCLXXX. 1. STARTING POINT

In the current working truncation,

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 - \Gamma_J m_3. \quad (450)$$

Here $\tilde{\mathcal{N}}_{\parallel}^3$ is the shell-normal σ_3 -coefficient of the projected refined periodic/internal block, while m_3 is the pure- JJ Class-II longitudinal coefficient.

CCLXXXI. 2. ACTUAL EXTRACTION FORMULA FOR m_3

The exact first-shell dominant decomposition gives

$$m_3 = -\frac{|\Omega_{\text{cell}}|}{2\Gamma_J} \left(\Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right), \quad (451)$$

where

$$\Delta_{00} = \overline{A_1^{(+)}} A_1^{(-)} - \overline{A_2^{(+)}} A_2^{(-)}, \quad (452)$$

and

$$R_N = \sum_{(G,G') \neq (0,0)} \Delta_{GG'} \hat{V}_{\parallel}^{(JJ)}(G_* + G - G'). \quad (453)$$

In the one-mode dominant approximation,

$$m_3^{(0)} = -\frac{|\Omega_{\text{cell}}|}{2\Gamma_J} \Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*). \quad (454)$$

CCLXXXII. 3. FINAL EXPLICIT LONGITUDINAL COLUMN

Substituting (451) into (450), we obtain

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \left(\Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right). \quad (455)$$

In the one-mode dominant approximation this reduces to

$$v_{\parallel}^{(0)} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*). \quad (456)$$

Thus the longitudinal zero condition becomes the explicit cancellation equation

$$\tilde{\mathcal{N}}_{\parallel}^3 = -\frac{|\Omega_{\text{cell}}|}{2} \left(\Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right). \quad (457)$$

CCLXXXIII. 4. CANONICAL BRANCH REDUCTION OF $\widehat{V}_{\parallel}^{(JJ)}(G_*)$

In the current canonical branch,

$$\widehat{V}_{\parallel}^{(JJ)}(G_*) \propto \alpha_X L_* A_{G_*}, \quad L_* = m_{\parallel} u_{\parallel}. \quad (458)$$

Therefore the dominant piece of the longitudinal column is controlled by

$$\tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \Delta_{00} \alpha_X L_* A_{G_*}, \quad (459)$$

up to the explicit harmonic remainder R_N .

CCLXXXIV. 5. HONEST STATUS

At this point, the longitudinal column is no longer an abstract operator problem. It is an explicit finite cancellation problem between:

1. the bare periodic/internal shell coefficient $\tilde{\mathcal{N}}_{\parallel}^3$,
2. the intervalley Bloch asymmetry Δ_{00} ,
3. the canonical JJ -scalar $L_* = m_{\parallel} u_{\parallel}$,
4. the first-shell Fourier weight A_{G_*} ,
5. and the explicit harmonic remainder R_N .

So the remaining frontier is the direct evaluation of these finite scalar data, not further structural reduction.

CCLXXXV. FINAL SUMMARY

In the current Bloch-operator working truncation, the spectator longitudinal contribution vanishes, so the starting longitudinal column is

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 - \Gamma_J m_3.$$

The pure- JJ longitudinal coefficient m_3 already admits the explicit first-shell dominant extraction

$$m_3 = -\frac{|\Omega_{\text{cell}}|}{2\Gamma_J} \left(\Delta_{00} \widehat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right),$$

with

$$\Delta_{00} = \overline{A_1^{(+)}} A_1^{(-)} - \overline{A_2^{(+)}} A_2^{(-)}.$$

Therefore the longitudinal column becomes

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \left(\Delta_{00} \widehat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right).$$

In the one-mode dominant approximation,

$$v_{\parallel}^{(0)} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \Delta_{00} \widehat{V}_{\parallel}^{(JJ)}(G_*).$$

In the current canonical branch,

$$\widehat{V}_{\parallel}^{(JJ)}(G_*) \propto \alpha_X L_* A_{G_*}, \quad L_* = m_{\parallel} u_{\parallel}.$$

Hence the dominant longitudinal contribution is controlled by

$$\tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \Delta_{00} \alpha_X L_* A_{G_*},$$

up to the explicit remainder R_N .

So the longitudinal failure condition is now an explicit finite cancellation equation, and the remaining frontier is the direct evaluation of

$$\tilde{\mathcal{N}}_{\parallel}^3, \quad \Delta_{00}, \quad L_*, \quad A_{G_*}, \quad R_N.$$

CCLXXXVI. ACTUAL FINITE EXTRACTION PROTOCOL FOR Δ_{00} AND $L_* = m_{\parallel} u_{\parallel}$

CCLXXXVII. SHORT KOREAN EXPLANATION

At the present stage, the longitudinal Bloch-operator pipeline is no longer a broad operator problem. It has been reduced to the explicit finite extraction of two scalars:

$$\Delta_{00}, \quad L_* = m_{\parallel} u_{\parallel}.$$

These are obtained directly from:

1. the two projected valley-diagonal 2×2 blocks $E_{\pm}(0)$,
2. the reduced intervalley J -matrix $\mathcal{U}_j(0)$,
3. the background selector $m_{\parallel} = m^3(P_0)$.

CCLXXXVIII. 1. PROJECTED VALLEY-DIAGONAL BLOCKS

Fix the enriched shell carrier basis

$$|+, a\rangle := |G_*, \chi_a\rangle, \quad |-, a\rangle := |-G_*, \chi_a\rangle, \quad a = 1, 2.$$

The two valley-diagonal projected matrices are

$$\boxed{(E_+(0))_{ab} = \langle +, a | H_{\text{shell}}(0) | +, b \rangle, \quad (E_-(0))_{ab} = \langle -, a | H_{\text{shell}}(0) | -, b \rangle.} \quad (460)$$

Let $\eta_{\pm}$ be the normalized low eigenvectors:

$$\boxed{E_+(0)\eta_+ = \varepsilon_+ \eta_+, \quad E_-(0)\eta_- = \varepsilon_- \eta_-, \quad \|\eta_{\pm}\| = 1.} \quad (461)$$

Write

$$\eta_+ = \begin{pmatrix} A_1^{(+)} \\ A_2^{(+)} \end{pmatrix}, \quad \eta_- = \begin{pmatrix} A_1^{(-)} \\ A_2^{(-)} \end{pmatrix}.$$

Then the valley Bloch asymmetry is

$$\boxed{\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_- = \overline{A_1^{(+)}} A_1^{(-)} - \overline{A_2^{(+)}} A_2^{(-)}.} \quad (462)$$

CCLXXXIX. 2. REDUCED INTERVALLEY J -MATRIX

Define the reduced intervalley J -matrix by

$$(\mathcal{U}_j(0))_{ab} = \langle -, a | \hat{J}_j(G_*) | +, b \rangle, \quad a, b = 1, 2. \quad (463)$$

Project to the shell-normal direction:

$$\hat{n}^j \mathcal{U}_j(0).$$

Then the shell-normal J -singlet bias is

$$u_{\parallel} = \frac{1}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right]. \quad (464)$$

This avoids any raw trace notation and is the safest executable formula.

CCXC. 3. BACKGROUND SELECTOR

In the canonical locked basis, the background selector is

$$m_{\parallel} = m^3(P_0). \quad (465)$$

Therefore the final longitudinal scalar is

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right]. \quad (466)$$

CCXCI. 4. INSERTION INTO THE LONGITUDINAL COLUMN

The current working-truncation longitudinal column is

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \left(\Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right). \quad (467)$$

In the canonical branch, the pure- JJ harmonic is controlled by L_* , so the dominant first-shell term is governed by

$$\Delta_{00} \alpha_X L_* A_{G_*}.$$

Therefore the practical dominant longitudinal criterion is

$$\left| \frac{|\Omega_{\text{cell}}|}{2} \Delta_{00} \alpha_X L_* A_{G_*} \right| > \left| \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} R_N \right|. \quad (468)$$

If (468) holds, then

$$v_{\parallel} \neq 0, \quad \lambda_{\parallel} = v_{\parallel} \neq 0. \quad (469)$$

CCXCII. 5. ULTRA-MINIMAL FIRST-SHELL FORM

In the ultra-minimal first-shell truncation $R_N = 0$, the longitudinal column reduces to

$$v_{\parallel}^{(0)} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*). \quad (470)$$

If one moreover works in the dominant JJ -calibrated branch where the second term is the load-bearing one, then the sufficient condition simplifies to

$$\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0. \quad (471)$$

CCXCIII. 6. FINAL EXECUTABLE CHECKLIST

The final finite execution order is:

1. build the two 2×2 valley blocks $E_{\pm}(0)$;
2. diagonalize them and extract $\eta_{\pm}$, hence Δ_{00} ;
3. build the reduced intervalley matrix $\mathcal{U}_j(0)$;
4. compute

$$u_{\parallel} = \frac{1}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right];$$

5. read $m_{\parallel} = m^3(P_0)$;
6. form $L_* = m_{\parallel} u_{\parallel}$;
7. insert Δ_{00} and L_* into (467).

No further structural reduction is required. The remaining problem is now a literal finite projected matrix-element computation.

CCXCIV. EFFECTIVE WORLDVOLUME WEYL EQUATION

The reduced carrier equation on the defect worldvolume is

$$\boxed{\left[i\partial_t - \lambda_{\parallel} \hat{n}^{\mu} \sigma_3 \partial_{\mu} - \alpha E^{\mu}_a \sigma_a \partial_{\mu} - \beta J^{\mu}_a \sigma_a \partial_{\mu} \right] \chi = 0, \quad a = 1, 2.} \quad (472)$$

CCXCV. ADAPTED-FRAME FORM

Choosing the adapted orthonormal frame

$$(\hat{n}, e_1, e_2),$$

the equation becomes

$$\boxed{\left[i\partial_t - \lambda_{\parallel} \sigma_3 \partial_{\parallel} - \alpha(\sigma_1 \partial_1 + \sigma_2 \partial_2) - \beta(-\sigma_1 \partial_2 + \sigma_2 \partial_1) \right] \chi = 0.} \quad (473)$$

CCXCVI. DISPERSION RELATION

For a plane-wave mode

$$\chi = u e^{-i\omega t + i(p_{\parallel} x_{\parallel} + p_1 x_1 + p_2 x_2)},$$

the reduced Hamiltonian is

$$H(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + (\alpha p_1 - \beta p_2) \sigma_1 + (\alpha p_2 + \beta p_1) \sigma_2. \quad (474)$$

Hence

$$\boxed{\omega_{\pm}(p) = \pm \sqrt{\lambda_{\parallel}^2 p_{\parallel}^2 + (\alpha^2 + \beta^2)(p_1^2 + p_2^2)}. \quad (475)}$$

CCXCVII. INTERPRETATION

Equation (472) is the effective massless anisotropic chiral Dirac/Weyl equation on the defect worldvolume. If

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0),$$

then the dispersion relation is conical:

$$\omega_{\pm}(p) = \pm \sqrt{\lambda_{\parallel}^2 p_{\parallel}^2 + (\alpha^2 + \beta^2)(p_1^2 + p_2^2)}.$$

If in addition no symmetry-allowed constant mass term survives, then this defines a protected massless chiral carrier localized on the defect.

CCXCVIII. EXACT GENERAL 2×2 HERMITIAN FORMULA FOR Δ_{00}

CCXCIX. SHORT KOREAN EXPLANATION

Let the two valley-diagonal projected blocks be arbitrary Hermitian 2×2 matrices:

$$E_{\pm}(0) = \begin{pmatrix} a_{\pm} & b_{\pm} \\ \bar{b}_{\pm} & d_{\pm} \end{pmatrix}, \quad a_{\pm}, d_{\pm} \in \mathbb{R}, \quad b_{\pm} \in \mathbb{C}.$$

Then the lower eigenvalues and normalized lower eigenvectors can be written in closed form. Hence the intervalley Bloch asymmetry

$$\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_-$$

is also obtained in closed form.

This turns the longitudinal zero condition into an explicit finite cancellation equation in the microscopic matrix entries of $E_{\pm}(0)$.

CCC. 1. GENERAL HERMITIAN VALLEY BLOCKS

Write

$$E_{\pm}(0) = \begin{pmatrix} a_{\pm} & b_{\pm} \\ \bar{b}_{\pm} & d_{\pm} \end{pmatrix}, \quad a_{\pm}, d_{\pm} \in \mathbb{R}, \quad b_{\pm} \in \mathbb{C}. \quad (476)$$

Define the spectral gaps

$$\Delta_{\pm}^{\text{spec}} := \sqrt{(a_{\pm} - d_{\pm})^2 + 4|b_{\pm}|^2}. \quad (477)$$

The lower eigenvalues are

$$\varepsilon_{\pm} = \frac{a_{\pm} + d_{\pm} - \Delta_{\pm}^{\text{spec}}}{2}. \quad (478)$$

CCCI. 2. CANONICALLY NORMALIZED LOWER EIGENVECTORS

Assume first that $b_{\pm} \neq 0$. Then a canonical lower-eigenvector representative is

$$\eta_{\pm} = \frac{1}{N_{\pm}} \begin{pmatrix} b_{\pm} \\ X_{\pm} \end{pmatrix}, \quad X_{\pm} := \varepsilon_{\pm} - a_{\pm}, \quad N_{\pm} := \sqrt{|b_{\pm}|^2 + X_{\pm}^2}. \quad (479)$$

Indeed,

$$E_{\pm}(0) \begin{pmatrix} b_{\pm} \\ X_{\pm} \end{pmatrix} = \varepsilon_{\pm} \begin{pmatrix} b_{\pm} \\ X_{\pm} \end{pmatrix}.$$

Since $\varepsilon_{\pm}, a_{\pm} \in \mathbb{R}$, the numbers $X_{\pm}$ are real:

$$\boxed{X_{\pm} = \frac{d_{\pm} - a_{\pm} - \Delta_{\pm}^{\text{spec}}}{2}}. \quad (480)$$

Thus the normalized lower eigenvectors are completely explicit in the matrix entries.

CCCH. 3. EXACT CLOSED FORMULA FOR Δ_{00}

Recall

$$\Sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Using (479),

$$\begin{aligned} \Delta_{00} &= \eta_+^\dagger \Sigma_3 \eta_- \\ &= \frac{1}{N_+ N_-} (\overline{b_+} \ X_+) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} b_- \\ X_- \end{pmatrix} \\ &= \frac{\overline{b_+} b_- - X_+ X_-}{N_+ N_-}. \end{aligned} \quad (481)$$

Therefore the exact general closed formula is

$$\boxed{\Delta_{00} = \frac{\overline{b_+} b_- - X_+ X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}}, \quad X_{\pm} = \frac{d_{\pm} - a_{\pm} - \Delta_{\pm}^{\text{spec}}}{2}}. \quad (482)$$

This is the desired fully general 2×2 Hermitian formula.

CCCH. 4. EXACT ZERO CONDITION FOR Δ_{00}

From (482), one gets

$$\boxed{\Delta_{00} = 0 \iff \overline{b_+} b_- = X_+ X_-}. \quad (483)$$

Hence vanishing of Δ_{00} is not automatic. It is an explicit algebraic cancellation condition between the two valley-block entries.

In particular, within the space of all matrix entries $(a_{\pm}, d_{\pm}, b_{\pm})$, the condition

$$\overline{b_+} b_- = X_+ X_-$$

is an exact finite tuning relation.

CCCH. 5. DEGENERATE/OFF-DIAGONAL SPECIAL CASES

If $b_{\pm} = 0$, the matrix is already diagonal:

$$E_{\pm}(0) = \begin{pmatrix} a_{\pm} & 0 \\ 0 & d_{\pm} \end{pmatrix}.$$

Then:

- if $a_{\pm} < d_{\pm}$, the lower eigenvector is $(1, 0)^T$,
- if $d_{\pm} < a_{\pm}$, the lower eigenvector is $(0, 1)^T$,
- if $a_{\pm} = d_{\pm}$, the lower eigenspace is degenerate and $\eta_{\pm}$ is not uniquely fixed until an additional convention is imposed.

Thus the formula (482) is the nondegenerate generic case, while the diagonal/degenerate case must be handled separately by the obvious basis choice.

CCCV. 6. INSERTION INTO THE LONGITUDINAL COLUMN

The current working-truncation longitudinal column is

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \left(\Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right). \quad (484)$$

Substituting (482), we obtain the fully explicit finite formula

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \left[\frac{\overline{b_+} b_- - X_+ X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right]. \quad (485)$$

Hence the exact longitudinal zero condition is

$$\tilde{\mathcal{N}}_{\parallel}^3 = -\frac{|\Omega_{\text{cell}}|}{2} \left[\frac{\overline{b_+} b_- - X_+ X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right]. \quad (486)$$

So the longitudinal failure condition is now an explicit finite cancellation equation in the microscopic entries of the two Hermitian valley blocks, together with the JJ -harmonic and remainder data.

CCCVI. 7. DOMINANT CANONICAL BRANCH FORM

In the canonical first-shell dominant branch,

$$\hat{V}_{\parallel}^{(JJ)}(G_*) \propto \alpha_X L_* A_{G_*}, \quad L_* = m_{\parallel} u_{\parallel}.$$

Therefore the dominant longitudinal column becomes

$$v_{\parallel}^{\text{dom}} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \frac{\overline{b_+} b_- - X_+ X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}} \alpha_X L_* A_{G_*}, \quad (487)$$

up to the explicit harmonic remainder R_N .

Thus the longitudinal problem is fully reduced to a finite matrix-and-scalar insertion problem.

CCCVII. FINAL SUMMARY

For completely general Hermitian valley blocks

$$E_{\pm}(0) = \begin{pmatrix} a_{\pm} & b_{\pm} \\ \overline{b_{\pm}} & d_{\pm} \end{pmatrix}, \quad a_{\pm}, d_{\pm} \in \mathbb{R}, \quad b_{\pm} \in \mathbb{C},$$

the lower eigenvalues are

$$\varepsilon_{\pm} = \frac{a_{\pm} + d_{\pm} - \sqrt{(a_{\pm} - d_{\pm})^2 + 4|b_{\pm}|^2}}{2},$$

and a canonical normalized lower eigenvector is

$$\eta_{\pm} = \frac{1}{\sqrt{|b_{\pm}|^2 + X_{\pm}^2}} \begin{pmatrix} b_{\pm} \\ X_{\pm} \end{pmatrix}, \quad X_{\pm} = \varepsilon_{\pm} - a_{\pm}.$$

Hence the intervalley Bloch asymmetry has the exact closed form

$$\Delta_{00} = \frac{\bar{b}_+ b_- - X_+ X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}}.$$

Therefore

$$\Delta_{00} = 0 \iff \bar{b}_+ b_- = X_+ X_-,$$

so vanishing of Δ_{00} is an explicit finite algebraic cancellation condition.

Substituting this into the longitudinal column gives

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \left[\frac{\bar{b}_+ b_- - X_+ X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right].$$

Thus the longitudinal failure condition is now a fully explicit finite cancellation equation in the microscopic entries of the two 2×2 Hermitian valley blocks, together with the JJ -harmonic and remainder data.

CCCVIII. EXACT GENERAL 2×2 MATRIX FORMULA FOR $L_* = m_{\parallel} u_{\parallel}$

CCCIX. SHORT KOREAN EXPLANATION

The remaining carrier-side longitudinal scalar

$$L_* = m_{\parallel} u_{\parallel}$$

is extracted from the shell-normal reduced intervalley J -matrix.

The key point is that one does not need any further structural assumptions to write $u_{\parallel}$ in closed form. If

$$\hat{n}^j \mathcal{U}_j(0)$$

is treated as a completely general 2×2 complex matrix, then $u_{\parallel}$ is simply its Σ_3 -coefficient, i.e. its diagonal asymmetry.

This gives an exact fully finite formula for L_* .

CCCX. 1. GENERAL SHELL-NORMAL REDUCED J -MATRIX

Define the shell-normal projected reduced intervalley matrix

$$\mathcal{U}_{\parallel} := \hat{n}^j \mathcal{U}_j(0). \quad (488)$$

Write it as a completely general 2×2 complex matrix:

$$\mathcal{U}_{\parallel} = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}, \quad u_{ab} \in \mathbb{C}. \quad (489)$$

No further reduction is needed to write the shell-normal singlet coefficient.

CCCXI. 2. PAULI DECOMPOSITION OF A GENERAL 2×2 MATRIX

Any 2×2 complex matrix has the unique decomposition

$$\mathcal{U}_{\parallel} = c_0 \mathbf{1}_2 + c_1 \Sigma_1 + c_2 \Sigma_2 + c_3 \Sigma_3, \quad (490)$$

with coefficients

$$c_0 = \frac{u_{11} + u_{22}}{2}, \quad (491)$$

$$c_1 = \frac{u_{12} + u_{21}}{2}, \quad (492)$$

$$c_2 = \frac{u_{21} - u_{12}}{2i}, \quad (493)$$

$$c_3 = \frac{u_{11} - u_{22}}{2}. \quad (494)$$

The shell-normal J -singlet bias is precisely the Σ_3 -coefficient:

$$u_{\parallel} = c_3 = \frac{u_{11} - u_{22}}{2}. \quad (495)$$

Thus $u_{\parallel}$ is exactly the diagonal asymmetry of the shell-normal reduced intervalley J -matrix.

CCCXII. 3. HERMITIAN SPECIAL CASE

If $\mathcal{U}_{\parallel}$ is Hermitian, then

$$u_{22} \in \mathbb{R}, \quad u_{11} \in \mathbb{R}, \quad u_{21} = \overline{u_{12}},$$

and the Pauli coefficients satisfy

$$c_0, c_1, c_2, c_3 \in \mathbb{R}.$$

In particular,

$$u_{\parallel} = \frac{u_{11} - u_{22}}{2} \in \mathbb{R}. \quad (496)$$

So the Hermitian case is just the real specialization of the general formula (495).

CCCXIII. 4. FINAL EXACT FORMULA FOR L_*

The background selector is

$$m_{\parallel} = m^3(P_0). \quad (497)$$

Therefore the longitudinal scalar is

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2} (u_{11} - u_{22}). \quad (498)$$

Equivalently,

$$L_* = \frac{m_{\parallel}}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right]. \quad (499)$$

This is the exact general 2×2 matrix formula for L_* .

CCCXIV. 5. EXACT ZERO CONDITION FOR L_*

From (498),

$$\boxed{L_* = 0 \iff m_{\parallel} = 0 \quad \text{or} \quad u_{11} = u_{22}.} \quad (500)$$

Hence vanishing of L_* is not automatic. It is an explicit finite algebraic condition on the shell-normal reduced J -matrix and the background selector.

CCCXV. 6. FULL FINITE LONGITUDINAL AUDIT WITH BOTH SCALARS INSERTED

The current working-truncation longitudinal column is

$$\boxed{v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \left(\Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right).} \quad (501)$$

From the general Hermitian formula for the valley asymmetry,

$$\boxed{\Delta_{00} = \frac{\bar{b}_+ b_- - X_+ X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}},} \quad (502)$$

where

$$X_{\pm} = \frac{d_{\pm} - a_{\pm} - \sqrt{(a_{\pm} - d_{\pm})^2 + 4|b_{\pm}|^2}}{2},$$

and from (498),

$$L_* = \frac{m_{\parallel}}{2} (u_{11} - u_{22}),$$

the dominant canonical branch term

$$\hat{V}_{\parallel}^{(JJ)}(G_*) \propto \alpha_X L_* A_{G_*}$$

becomes

$$\boxed{\hat{V}_{\parallel}^{(JJ)}(G_*) \propto \alpha_X \frac{m_{\parallel}}{2} (u_{11} - u_{22}) A_{G_*}.} \quad (503)$$

Therefore the fully explicit finite longitudinal column can be written schematically as

$$\boxed{v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \left[\frac{\bar{b}_+ b_- - X_+ X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}} \alpha_X \frac{m_{\parallel}}{2} (u_{11} - u_{22}) A_{G_*} + R_N \right],} \quad (504)$$

up to the fixed branch-normalization factor relating $\hat{V}_{\parallel}^{(JJ)}(G_*)$ to $\alpha_X L_* A_{G_*}$.

CCCXVI. 7. FINAL FINITE CANCELLATION EQUATION

Hence the exact longitudinal failure condition becomes

$$\boxed{\tilde{\mathcal{N}}_{\parallel}^3 = -\frac{|\Omega_{\text{cell}}|}{2} \left[\frac{\bar{b}_+ b_- - X_+ X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}} \alpha_X \frac{m_{\parallel}}{2} (u_{11} - u_{22}) A_{G_*} + R_N \right].} \quad (505)$$

This is a completely finite cancellation equation in the microscopic matrix entries of

$$E_{\pm}(0), \quad \hat{n}^j \mathcal{U}_j(0),$$

together with

$$m_{\parallel}, \quad A_{G_*}, \quad R_N, \quad \tilde{\mathcal{N}}_{\parallel}^3.$$

CCCXVII. 8. HONEST STATUS

At this point, both remaining carrier-side longitudinal scalars are closed in general finite matrix form:

$$\Delta_{00} = \frac{\overline{b_+}b_- - X_+X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}}, \quad L_* = \frac{m_{\parallel}}{2}(u_{11} - u_{22}). \quad (506)$$

So the longitudinal frontier is no longer structural at all. It is a literal finite insertion problem for the matrix entries of

$$E_{\pm}(0), \quad \hat{n}^j \mathcal{U}_j(0),$$

and the remaining scalar data

$$m_{\parallel}, \quad A_{G_*}, \quad R_N, \quad \tilde{\mathcal{N}}_{\parallel}^3.$$

CCCXVIII. FINAL SUMMARY

For a completely general shell-normal reduced intervalley matrix

$$\hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix},$$

the shell-normal J -singlet bias is exactly the Σ_3 -coefficient:

$$u_{\parallel} = \frac{u_{11} - u_{22}}{2}.$$

Therefore the longitudinal scalar is

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2}(u_{11} - u_{22}).$$

Hence

$$L_* = 0 \iff m_{\parallel} = 0 \quad \text{or} \quad u_{11} = u_{22},$$

so vanishing of L_* is an explicit finite algebraic condition.

Together with the exact general Hermitian formula

$$\Delta_{00} = \frac{\overline{b_+}b_- - X_+X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}},$$

the longitudinal column becomes a fully explicit finite insertion problem.

Thus both remaining carrier-side longitudinal scalars are now closed in completely general finite matrix form:

$$\Delta_{00}, \quad L_* = \frac{m_{\parallel}}{2}(u_{11} - u_{22}).$$

CCCXIX. ACTUAL FINITE INSERTION OF THE TWO REMAINING LONGITUDINAL SCALARS

CCCXX. SHORT KOREAN EXPLANATION

The longitudinal frontier is no longer structural. It is now a literal finite insertion problem for the projected shell matrix elements.

The next executable step is to compute:

$$E_+(0), \quad E_-(0), \quad \mathcal{U}_j(0), \quad m_{\parallel} = m^3(P_0),$$

and then form

$$\Delta_{00}, \quad L_* = m_{\parallel} u_{\parallel}.$$

CCCXXI. 1. EXPLICIT SHELL CARRIER BASIS

Fix the folded Bragg point

$$k_* = \frac{G_*}{2},$$

and the enriched shell carrier basis

$$|+, a\rangle := |G_*, \chi_a\rangle, \quad |-, a\rangle := |-G_*, \chi_a\rangle, \quad a = 1, 2. \quad (507)$$

Equivalently, in the shell harmonic basis $|G, r\rangle$,

$$|+, a\rangle = \sum_r \chi_{a,r} |G_*, r\rangle, \quad |-, a\rangle = \sum_r \chi_{a,r} |-G_*, r\rangle.$$

CCCXXII. 2. ACTUAL VALLEY-DIAGONAL 2×2 BLOCKS

The two projected valley-diagonal blocks are

$$(E_+(0))_{ab} = \langle +, a | H_{\text{shell}}(0) | +, b \rangle, \quad (E_-(0))_{ab} = \langle -, a | H_{\text{shell}}(0) | -, b \rangle. \quad (508)$$

In shell-harmonic components this is

$$(E_+(0))_{ab} = \sum_{r,r'} \overline{\chi_{a,r}} \mathcal{H}_{(G_*,r),(G_*,r')}(k_*) \chi_{b,r'}, \quad (509)$$

$$(E_-(0))_{ab} = \sum_{r,r'} \overline{\chi_{a,r}} \mathcal{H}_{(-G_*,r),(-G_*,r')}(k_*) \chi_{b,r'}. \quad (510)$$

Write them as general Hermitian matrices

$$E_{\pm}(0) = \begin{pmatrix} a_{\pm} & b_{\pm} \\ \overline{b_{\pm}} & d_{\pm} \end{pmatrix}, \quad a_{\pm}, d_{\pm} \in \mathbb{R}, \quad b_{\pm} \in \mathbb{C}. \quad (511)$$

CCCXXIII. 3. EXACT EXTRACTION OF Δ_{00}

Let $\eta_{\pm}$ be the normalized lower eigenvectors:

$$E_{\pm}(0)\eta_{\pm} = \varepsilon_{\pm}\eta_{\pm}.$$

Define

$$\Delta_{\pm}^{\text{spec}} = \sqrt{(a_{\pm} - d_{\pm})^2 + 4|b_{\pm}|^2}, \quad X_{\pm} = \frac{d_{\pm} - a_{\pm} - \Delta_{\pm}^{\text{spec}}}{2}.$$

Then a canonical normalized lower eigenvector is

$$\eta_{\pm} = \frac{1}{\sqrt{|b_{\pm}|^2 + X_{\pm}^2}} \begin{pmatrix} b_{\pm} \\ X_{\pm} \end{pmatrix}.$$

Hence the exact intervalley Bloch asymmetry is

$$\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_- = \frac{\overline{b_+} b_- - X_+ X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}}. \quad (512)$$

Therefore

$$\Delta_{00} = 0 \iff \overline{b_+} b_- = X_+ X_-. \quad (513)$$

CCCXXIV. 4. ACTUAL REDUCED INTERVALLEY J -MATRIX

Define the reduced intervalley J -matrix by

$$(\mathcal{U}_j(0))_{ab} = \langle -, a | \hat{J}_j(G_*; 0) | +, b \rangle. \quad (514)$$

In shell-harmonic components,

$$(\mathcal{U}_j(0))_{ab} = \sum_{r,r'} \overline{\chi_{a,r}} [\hat{J}_j(G_*; 0)]_{rr'} \chi_{b,r'}. \quad (515)$$

Project to the shell-normal direction:

$$\mathcal{U}_{\parallel} := \hat{n}^j \mathcal{U}_j(0).$$

Write

$$\mathcal{U}_{\parallel} = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}. \quad (516)$$

Then the shell-normal J -singlet bias is exactly

$$u_{\parallel} = \frac{u_{11} - u_{22}}{2} = \frac{1}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right]. \quad (517)$$

CCCXXV. 5. ACTUAL EXTRACTION OF $L_* = m_{\parallel} u_{\parallel}$

The background selector is

$$m_{\parallel} = m^3(P_0). \quad (518)$$

Therefore the second longitudinal scalar is

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2} (u_{11} - u_{22}). \quad (519)$$

Hence

$$L_* = 0 \iff m_{\parallel} = 0 \quad \text{or} \quad u_{11} = u_{22}. \quad (520)$$

CCCXXVI. 6. DIRECT INSERTION INTO THE LONGITUDINAL COLUMN

At the current working-truncation level, the longitudinal column is

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \left(\Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right). \quad (521)$$

In the canonical branch,

$$\hat{V}_{\parallel}^{(JJ)}(G_*) \propto \alpha_X L_* A_{G_*}.$$

Therefore the dominant first-shell term becomes

$$\Delta_{00} \alpha_X L_* A_{G_*} = \frac{\overline{b_+ b_-} - X_+ X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}} \alpha_X \frac{m_{\parallel}}{2} (u_{11} - u_{22}) A_{G_*}, \quad (522)$$

up to the fixed branch-normalization factor and the explicit remainder R_N .

Thus the longitudinal zero condition is the finite cancellation equation

$$\boxed{\tilde{\mathcal{N}}_{\parallel}^3 = -\frac{|\Omega_{\text{cell}}|}{2} \left(\Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right),} \quad (523)$$

with

$$\Delta_{00} = \frac{\overline{b_+} b_- - X_+ X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}}, \quad L_* = \frac{m_{\parallel}}{2} (u_{11} - u_{22}).$$

CCCXXVII. 7. FINAL EXECUTABLE ORDER

The actual finite execution order is now completely fixed:

1. compute the shell matrix elements (509) and (510);
2. read off $a_{\pm}, d_{\pm}, b_{\pm}$, then compute Δ_{00} from (512);
3. compute the shell-normal intervalley J -matrix elements (515);
4. read off u_{11}, u_{22} , then compute L_* from (519);
5. read $m_{\parallel} = m^3(P_0)$;
6. insert Δ_{00} and L_* into (521).

No further structural theorem is needed. The remaining task is a literal projected matrix-element computation.

CCCXXVIII. ACTUAL CONSTANT-ENVELOPE INSERTION FORM OF THE FINAL LONGITUDINAL AUDIT

CCCXXIX. SHORT KOREAN EXPLANATION

In the constant-envelope branch, the zero-harmonic vectors are already the carrier coefficients in the internal basis $\{\xi_1, \xi_2\}$. So the remaining longitudinal scalars are no longer abstract:

1. Δ_{00} is the Σ_3 -weighted overlap of the two valley coefficient vectors;
2. $L_* = m_{\parallel} u_{\parallel}$ is the shell-normal diagonal asymmetry of the reduced J -block multiplied by the background selector.

This gives the most concrete final insertion formula presently available inside the Bloch-operator pipeline.

CCCXXX. 1. CONSTANT-ENVELOPE CARRIER VECTORS

Choose the internal basis $\{\xi_1, \xi_2\}$ and write the selected low carrier coefficients as

$$\boxed{\eta_+ = \begin{pmatrix} A_1^{(+)} \\ A_2^{(+)} \end{pmatrix}, \quad \eta_- = \begin{pmatrix} A_1^{(-)} \\ A_2^{(-)} \end{pmatrix}.} \quad (524)$$

Then

$$\boxed{\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_- = \overline{A_1^{(+)}} A_1^{(-)} - \overline{A_2^{(+)}} A_2^{(-)}.} \quad (525)$$

Equivalently, if the valley-diagonal projected blocks are written as

$$E_{\pm}(0) = \begin{pmatrix} a_{\pm} & b_{\pm} \\ b_{\pm} & d_{\pm} \end{pmatrix},$$

then

$$\Delta_{00} = \frac{\overline{b_+}b_- - X_+X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}}, \quad (526)$$

with

$$X_{\pm} = \frac{d_{\pm} - a_{\pm} - \sqrt{(a_{\pm} - d_{\pm})^2 + 4|b_{\pm}|^2}}{2}.$$

CCCXXXI. 2. REDUCED INTERVALLEY J -BLOCK

Define the reduced J -matrix on the carrier:

$$\mathcal{U}_j(0) = (\mathcal{U}_{j,ab}(0))_{a,b=1}^2, \quad \mathcal{U}_{j,ab}(0) = \langle -, a | \hat{J}_j(G_*; 0) | +, b \rangle. \quad (527)$$

Project to the shell-normal direction:

$$\mathcal{U}_{\parallel} := \hat{n}^j \mathcal{U}_j(0).$$

Write

$$\mathcal{U}_{\parallel} = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}. \quad (528)$$

Then the shell-normal singlet bias is

$$u_{\parallel} = \frac{u_{11} - u_{22}}{2} = \frac{1}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right]. \quad (529)$$

CCCXXXII. 3. FINAL SCALAR L_*

The background selector is

$$m_{\parallel} = m^3(P_0). \quad (530)$$

Hence

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2} (u_{11} - u_{22}). \quad (531)$$

So

$$L_* = 0 \iff m_{\parallel} = 0 \quad \text{or} \quad u_{11} = u_{22}. \quad (532)$$

CCCXXXIII. 4. FINAL LONGITUDINAL COLUMN IN FINITE INSERTION FORM

The current working-truncation longitudinal column is

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \left(\Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right). \quad (533)$$

In the canonical branch,

$$\hat{V}_{\parallel}^{(JJ)}(G_*) \propto \alpha_X L_* A_{G_*}.$$

Therefore the dominant first-shell term becomes

$$\Delta_{00} \alpha_X L_* A_{G_*} = \left(\overline{A_1^{(+)}} A_1^{(-)} - \overline{A_2^{(+)}} A_2^{(-)} \right) \alpha_X \frac{m_{\parallel}}{2} (u_{11} - u_{22}) A_{G_*}, \quad (534)$$

or equivalently

$$\Delta_{00} \alpha_X L_* A_{G_*} = \frac{\overline{b_+} b_- - X_+ X_-}{\sqrt{|b_+|^2 + X_+^2} \sqrt{|b_-|^2 + X_-^2}} \alpha_X \frac{m_{\parallel}}{2} (u_{11} - u_{22}) A_{G_*}. \quad (535)$$

Thus the fully finite longitudinal column is

$$v_{\parallel} = \tilde{\mathcal{N}}_{\parallel}^3 + \frac{|\Omega_{\text{cell}}|}{2} \left[\Delta_{00} \alpha_X \frac{m_{\parallel}}{2} (u_{11} - u_{22}) A_{G_*} + R_N \right], \quad (536)$$

up to the fixed branch-normalization factor relating $\hat{V}_{\parallel}^{(JJ)}(G_*)$ to $\alpha_X L_* A_{G_*}$.

CCCXXXIV. 5. FINAL FINITE CANCELLATION EQUATION

Hence the longitudinal failure condition is the explicit finite equation

$$\tilde{\mathcal{N}}_{\parallel}^3 = -\frac{|\Omega_{\text{cell}}|}{2} \left[\Delta_{00} \alpha_X \frac{m_{\parallel}}{2} (u_{11} - u_{22}) A_{G_*} + R_N \right]. \quad (537)$$

This is no longer a conceptual obstruction. It is a finite algebraic cancellation relation among

$$E_{\pm}(0), \quad \hat{n}^j \mathcal{U}_j(0), \quad m_{\parallel}, \quad A_{G_*}, \quad R_N, \quad \tilde{\mathcal{N}}_{\parallel}^3.$$

CCCXXXV. LITERAL COMPONENT ASSEMBLY FOR THE FINAL LONGITUDINAL AUDIT

CCCXXXVI. SHORT KOREAN EXPLANATION

No further structural theorem is needed. The remaining task is the literal component assembly of the two projected valley blocks $E_{\pm}(0)$ and the reduced intervalley J -matrix $\mathcal{U}_j(0)$.

This is the last actual microscopic insertion step inside the Bloch-operator extraction pipeline.

CCCXXXVII. 1. FIXED ENRICHED SHELL CARRIER BASIS

Fix the folded Bragg point

$$k_* = \frac{G_*}{2},$$

and the enriched shell carrier basis

$$|+, a\rangle := |G_*, \chi_a\rangle, \quad |-, a\rangle := |-G_*, \chi_a\rangle, \quad a = 1, 2. \quad (538)$$

Expand the internal carrier vectors in a fixed shell internal basis $\{\xi_r\}$:

$$\chi_a = \sum_r \chi_{a,r} \xi_r. \quad (539)$$

Then

$$|+, a\rangle = \sum_r \chi_{a,r} |G_*, \xi_r\rangle, \quad |-, a\rangle = \sum_r \chi_{a,r} |-G_*, \xi_r\rangle.$$

CCCXXXVIII. 2. LITERAL COMPONENT FORMULA FOR THE TWO VALLEY BLOCKS

The two projected valley-diagonal blocks are

$$(E_+(0))_{ab} = \langle +, a | H_{\text{shell}}(0) | +, b \rangle, \quad (E_-(0))_{ab} = \langle -, a | H_{\text{shell}}(0) | -, b \rangle. \quad (540)$$

In shell-harmonic components:

$$(E_+(0))_{ab} = \sum_{r,r'} \overline{\chi_{a,r}} [H_{\text{shell}}(0)]_{(G_*, r), (G_*, r')} \chi_{b,r'}, \quad (541)$$

$$(E_-(0))_{ab} = \sum_{r,r'} \overline{\chi_{a,r}} [H_{\text{shell}}(0)]_{(-G_*, r), (-G_*, r')} \chi_{b,r'}. \quad (542)$$

Thus the only microscopic input needed for $E_{\pm}(0)$ is the shell-block matrix of $H_{\text{shell}}(0)$ at the two valley harmonics $\pm G_*$.

CCCXXXIX. 3. LITERAL COMPONENT FORMULA FOR THE REDUCED INTERVALLEY J-MATRIX

Define

$$(\mathcal{U}_j(0))_{ab} = \langle -, a | \hat{J}_j(G_*; 0) | +, b \rangle. \quad (543)$$

In shell-harmonic components:

$$(\mathcal{U}_j(0))_{ab} = \sum_{r,r'} \overline{\chi_{a,r}} [\hat{J}_j(G_*; 0)]_{r,r'} \chi_{b,r'}. \quad (544)$$

Projecting to the shell-normal direction gives

$$\mathcal{U}_{\parallel} := \hat{n}^j \mathcal{U}_j(0) = (\mathcal{U}_{\parallel, ab})_{a,b=1}^2. \quad (545)$$

CCCXL. 4. FINAL FINITE FORMULAS FOR THE TWO LONGITUDINAL SCALARS

(a) Valley Bloch asymmetry

Once $E_{\pm}(0)$ are assembled, diagonalize them and take their normalized lower eigenvectors

$$E_{\pm}(0)\eta_{\pm} = \varepsilon_{\pm}\eta_{\pm}, \quad \eta_{\pm} = \begin{pmatrix} A_1^{(\pm)} \\ A_2^{(\pm)} \end{pmatrix}.$$

Then

$$\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_- = \overline{A_1^{(+)}} A_1^{(-)} - \overline{A_2^{(+)}} A_2^{(-)}. \quad (546)$$

(b) Shell-normal J -singlet invariant

The shell-normal singlet bias is

$$u_{\parallel} = \frac{1}{2} [(\mathcal{U}_{\parallel})_{11} - (\mathcal{U}_{\parallel})_{22}]. \quad (547)$$

With

$$m_{\parallel} = m^3(P_0), \quad (548)$$

the second longitudinal scalar is

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2} [(\mathcal{U}_{\parallel})_{11} - (\mathcal{U}_{\parallel})_{22}]. \quad (549)$$

CCCXLI. 5. FINAL INSERTION INTO THE LONGITUDINAL WITNESS

The practical first-shell dominant sufficient condition is

$$|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N|. \quad (550)$$

In the ultra-minimal truncation $R_N = 0$, it is enough to verify

$$\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0. \quad (551)$$

So the literal computation target is:

$$[H_{\text{shell}}(0)]_{(\pm G_*, r), (\pm G_*, r')}, \quad [\widehat{J}_j(G_*; 0)]_{r r'}, \quad m^3(P_0). \quad (552)$$

CCCXLII. LITERAL COMPONENT ASSEMBLY: THE ACTUAL REMAINING STEP

CCCXLIII. SHORT KOREAN EXPLANATION

The uploaded documents already fix the final extraction formulas for the two remaining longitudinal scalars, but they do not yet contain the evaluated shell-harmonic entries themselves.

So the remaining task is no longer structural. It is the literal insertion of the projected shell matrix elements.

CCCXLIV. 1. EXACT COMPONENT FORMULAS ALREADY FIXED

The two projected valley-diagonal blocks are

$$(E_+(0))_{ab} = \langle +, a | H_{\text{shell}}(0) | +, b \rangle, \quad (E_-(0))_{ab} = \langle -, a | H_{\text{shell}}(0) | -, b \rangle. \quad (553)$$

The reduced intervalley J -matrix is

$$(\mathcal{U}_j(0))_{ab} = \langle -, a | \widehat{J}_j(G_*; 0) | +, b \rangle. \quad (554)$$

The background selector is

$$m_{\parallel} = m^3(P_0). \quad (555)$$

CCCXLV. 2. EXACT LONGITUDINAL SCALAR FORMULAS

Once the lower eigenvectors of $E_{\pm}(0)$ are known,

$$\boxed{\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_-} \quad (556)$$

Once $\mathcal{U}_j(0)$ is known,

$$\boxed{L_* = \frac{m_{\parallel}}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right]} \quad (557)$$

CCCXLVI. 3. ACTUAL MISSING INPUTS

Therefore the only missing microscopic inputs are:

$$\boxed{\left[H_{\text{shell}}(0) \right]_{(\pm G_*, r), (\pm G_*, r')}, \quad \left[\hat{J}_j(G_*; 0) \right]_{rr'}, \quad m^3(P_0)} \quad (558)$$

Without these entries, the formulas are closed but the actual values of

$$\Delta_{00}, \quad L_*$$

cannot yet be evaluated.

CCCXLVII. 4. FINAL HONEST STATUS

Hence the Bloch-operator extraction pipeline has reached the following endpoint:

$$\boxed{\text{all reduction formulas are fixed, but the actual projected shell entries are still not inserted.}} \quad (559)$$

CCCXLVIII. COMPARISON OF ρ_{II} AND ρ_{12} IN THE NATURAL TECT PARAMETER REGION

CCCXLIX. SHORT KOREAN EXPLANATION

The explicit one-mediator Class II branch gives

$$\rho_{\text{II}} = \frac{|v_{\parallel}|}{|v_{\perp}|} = \frac{g_X^2 A^3 |G_*|^2}{8|Z|M_X^2},$$

whereas the locked equal-amplitude 12-mode BCC shell-sum branch gives

$$\rho_{12} = \frac{|v_{\parallel}^{(12)}|}{|v_{\perp}^{(12)}|} = \frac{|g|A^3}{|Z|q_*^2}.$$

Thus

$$\frac{\rho_{\text{II}}}{\rho_{12}} = \frac{g_X^2 |G_*|^2 q_*^2}{8|g|M_X^2} \sim \frac{g_X^2 q_*^4}{8|g|M_X^2}.$$

Since the Class II mediator is naturally interpreted as heavy in the TECT working branch, the factor $1/M_X^2$ generically suppresses ρ_{II} . Therefore, unless one enters a strong-coupling / not-so-heavy regime,

$$\boxed{\rho_{12} > \rho_{\text{II}}}$$

is the natural generic ordering.

In other words, at the UV / working-branch level, the 12-mode BCC shell-sum branch more naturally supports stronger longitudinal anisotropy than the one-mediator Class II branch.

CCCL. 1. EXPLICIT FORMULAS

For the one-mediator Class II branch,

$$\lambda_{\parallel} = \kappa_g, \quad \alpha = \kappa_Z, \quad \beta = 0, \quad (560)$$

with

$$\kappa_g = \frac{g_X^2 A^2 |G_*|^2}{M_X^2 2}, \quad \kappa_Z = \frac{4Z}{A}. \quad (561)$$

Hence

$$\rho_{\text{II}} = \frac{g_X^2 A^3 |G_*|^2}{8|Z| M_X^2}. \quad (562)$$

For the locked equal-amplitude 12-mode BCC shell-sum branch,

$$W_{\parallel}^{(12)} = 4 \frac{g A^2}{q_*^2}, \quad W_I^{(12)} = \frac{4Z}{A}, \quad W_J^{(12)} = 0, \quad (563)$$

so

$$\rho_{12} = \frac{|g| A^3}{|Z| q_*^2}. \quad (564)$$

CCCLI. 2. DIRECT COMPARISON

The ratio of the two anisotropies is

$$\frac{\rho_{\text{II}}}{\rho_{12}} = \frac{g_X^2 |G_*|^2 q_*^2}{8|g| M_X^2}. \quad (565)$$

If one identifies the selected shell scale with $q_* \sim |G_*|$, then

$$\frac{\rho_{\text{II}}}{\rho_{12}} \sim \frac{g_X^2 q_*^4}{8|g| M_X^2}. \quad (566)$$

Therefore:

$$\rho_{\text{II}} > \rho_{12} \iff g_X^2 q_*^4 > 8|g| M_X^2, \quad (567)$$

$$\rho_{\text{II}} < \rho_{12} \iff g_X^2 q_*^4 < 8|g| M_X^2. \quad (568)$$

CCCLII. 3. NATURAL TECT INTERPRETATION

Because the Class II mediator is introduced as a heavy internal sector, the natural working regime satisfies a large M_X hierarchy. Hence ρ_{II} is generically suppressed by $1/M_X^2$, while ρ_{12} has no such heavy-mass suppression.

Therefore, at the UV / working-branch level, the natural generic ordering is

$$\rho_{12} > \rho_{\text{II}}. \quad (569)$$

CCCLIII. 4. IR QUALIFICATION

At the infrared level, quartic cubic anisotropy is already known to be irrelevant in the minimal TECT count $N_{\text{eff}} = 6$, so both branches are expected to flow toward weaker anisotropy. However, the derivative-anisotropy / velocity-locking flow is not yet fully closed microscopically.

So the conclusion $\rho_{12} > \rho_{\text{II}}$ should be understood as a natural UV / working-branch statement, not yet a final IR universality theorem.

CCCLIV. CURRENT SOLVED BRANCH-B PDE IN THE GRADED 2 + 2 MAINLINE

$$\left[i\partial_t - \lambda_{\parallel} \sigma_3(-i\partial_{\parallel}) - \alpha(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - \beta(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0, \quad (570)$$

with

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa. \quad (571)$$

CCCLV. SOURCE-BY-SOURCE HESSIAN EXTRACTION OF $\lambda_c, \Lambda_E, \Lambda_O$ IN THE GRADED 2 + 2 MAINLINE

CCCLVI. SHORT KOREAN EXPLANATION

In the graded 2 + 2 bypass route, the only remaining full-PDE gap is the actual extraction of

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O$$

from the complete background-dependent Hessian.

In the exact concrete Class II-compatible branch one already has

$$\Lambda_{\parallel}^{(0)} = 0,$$

so the reduced PDE coefficients are

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.$$

Therefore the next task is not another theorem. It is the explicit blockwise extraction

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] \rightarrow V^{\mu}(\xi) \rightarrow R_{\xi}^{\text{act}} \rightarrow (s_1, s_2, s_3) \rightarrow (\lambda_c, \Lambda_E, \Lambda_O).$$

CCCLVII. 1. STARTING BACKGROUND AND FLUCTUATION SPLIT

Use the shell-/defect-adapted minimal background

$$\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)), \quad (572)$$

or, if the induced gauge connection is retained,

$$\bar{\Phi}_{\min}^{\text{full}} = (\bar{A}_B(\xi), \bar{n}_B(\xi), \bar{\chi}_B(\xi)). \quad (573)$$

Expand

$$\Phi = \bar{\Phi}_{\min} + \varphi, \quad \varphi = (a, \nu, \zeta). \quad (574)$$

The background-dependent Hessian then has the exact 3×3 block form

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = \begin{pmatrix} \mathcal{H}_{TT} & \mathcal{H}_{TO} & \mathcal{H}_{TD} \\ \mathcal{H}_{OT} & \mathcal{H}_{OO} & \mathcal{H}_{OD} \\ \mathcal{H}_{DT} & \mathcal{H}_{DO} & \mathcal{H}_{DD} \end{pmatrix}. \quad (575)$$

CCCLVIII. 2. UNIVERSAL OPERATOR SPLIT AND THE ACTUAL TRANSPORT KERNEL

The full quadratic operator is organized as

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = H_B(\xi) - iV^\mu(\xi)\partial_{y^\mu} + W^{\mu\nu}(\xi)\partial_{y^\mu}\partial_{y^\nu} + \dots \quad (576)$$

The coefficient point $(\lambda_c, \Lambda_E, \Lambda_O)$ is read only from the odd first-order transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}. \quad (577)$$

Therefore the first explicit step is: collect from every Hessian block only those terms carrying exactly one world-volume derivative ∂_{y^μ} , move all ξ -dependent matrices to the left, and sum them into $V^\mu(\xi)$.

CCCLIX. 3. WHICH HESSIAN SOURCES CAN ACTUALLY FEED V^μ

At the symmetric point, local quartic potential sectors are generically p^2 -silent and act as mass-type corrections rather than transport generators. The genuinely viable sources of nontrivial derivative transport are:

$$(i) \text{ field-dependent derivative vertices beyond the minimal bilinear truncation,} \quad (578)$$

$$(ii) \text{ gauge/constraint-sector contributions in the } A\text{-sector,} \quad (579)$$

$$(iii) \text{ background-generated effective cubic vertices, giving nonzero } \Gamma^{(3)}\Gamma^{(3)} \text{ contributions.} \quad (580)$$

Thus, in the actual Hessian blocks $\mathcal{H}_{IJ}$, the first-order transport candidates are precisely the pieces with one worldvolume derivative generated by these three mechanisms.

CCCLX. 4. TANGENT-BUNDLE PROJECTION AND LOCAL Q_B -RESPONSE

Project to the tangent bundle

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp, \quad \dim_{\mathbb{C}} Q_B = 2,$$

and choose an adapted orthonormal frame

$$(e_1(\xi), e_2(\xi)).$$

The actual Hermitian Q_B -response is then written as

$$R_\xi^{\text{act}} = \begin{pmatrix} r_{11}(\xi) & r_{12}(\xi) \\ r_{12}(\xi) & r_{22}(\xi) \end{pmatrix}. \quad (581)$$

Its Pauli decomposition is

$$R_\xi^{\text{act}} = r_0(\xi)\mathbf{1}_2 + s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3, \quad (582)$$

with

$$r_0 = \frac{r_{11} + r_{22}}{2}, \quad s_3 = \frac{r_{11} - r_{22}}{2}, \quad s_1 = \Re r_{12}, \quad s_2 = -\Im r_{12}. \quad (583)$$

So the actual transport problem is reduced to identifying the scalar functions

$$s_1(\xi), \quad s_2(\xi), \quad s_3(\xi)$$

coming from the first-order Hessian blocks.

CCCLXI. 5. WORLDVOLUME/S2 MATCHING

The local S2-compatible transport kernel has the invariant decomposition

$$V^\mu(\xi) = \theta'(\xi) \left[\Lambda_{\parallel}^{(0)} n^\mu \sigma_3 + \Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right] + \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \dots \quad (584)$$

Hence the channel map is:

$$s_3 \longrightarrow \Lambda_{\parallel}^{(0)} \text{ or } \lambda_c, \quad s_1 \longrightarrow \Lambda_E, \quad s_2 \longrightarrow \Lambda_O. \quad (585)$$

Since the exact concrete Class II-compatible branch satisfies

$$\Lambda_{\parallel}^{(0)} = 0,$$

the longitudinal channel is Branch B, so s_3 must be matched to the primitive $\theta'(\xi)^3 n^\mu \sigma_3$ structure, i.e. to λ_c .

CCCLXII. 6. PROJECTION TO THE LOW MODES

Let the even/odd low modes be

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi), \quad a = 1, 2.$$

Then the projected mixed symbol is

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle. \quad (586)$$

Matching (586) to the S2 form gives the final coefficient dictionary

$$\lambda_{\parallel} = I_\theta \Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O. \quad (587)$$

Inside the exact concrete Class II branch, this reduces to

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O. \quad (588)$$

CCCLXIII. 7. FULLY EXPLICIT OVERLAP-INSERTED PDE COEFFICIENTS

Using the already evaluated overlap integrals

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

the graded Branch B PDE coefficients become

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad (589)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa. \quad (590)$$

Therefore the only remaining microscopic task is the explicit Hessian extraction of

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

CCCLXIV. 8. FINAL EXECUTABLE BLOCKWISE PROTOCOL

The actual next calculation is now fixed:

1. Write the full background-dependent Hessian block-by-block as in (575).
2. From each block $\mathcal{H}_{IJ}$, isolate only the terms with exactly one ∂_{y^μ} .
3. Sum those terms into $V^\mu(\xi)$.
4. Project $V^\mu(\xi)$ to $Q_B(\xi)$ and read the matrix entries r_{11}, r_{22}, r_{12} .
5. Form

$$s_3 = \frac{r_{11} - r_{22}}{2}, \quad s_1 = \Re r_{12}, \quad s_2 = -\Im r_{12}.$$

6. Match s_3 to the Branch-B longitudinal seed λ_c , and s_1, s_2 to Λ_E, Λ_O .
7. Insert the result into (589)–(590).

No further structural theorem is required. The remaining frontier is a literal source-by-source Hessian differentiation and projection computation.

CCCLXV. BLOCKWISE VERDICT FOR THE GRADED 2 + 2 COEFFICIENT EXTRACTION

Block	Principal role	Current solved LO verdict	Load-bearing coefficient
OO	longitudinal	$\ell_0^{(1)} = 0, \ell_0^{(3)} = \ell_{0;OO}^{(3)}$	$\lambda_{c;OO}$
TD, DT	transverse	leading carrier	$e_0^{(1)}, o_0^{(1)}$
DD	transverse-family correction	subleading in minimal truncation	possible correction only
OD, DO	possible mixed correction	$\Gamma^{(3)} \neq 0$ around background	candidate but not leading
TT	mass/kinetic background sector	not principal for graded Dirac coefficients	no leading $\lambda_c, \Lambda_E, \Lambda_O$
TO, OT	mixed amplitude-orientational	not current load-bearing path	candidate correction only

CCCLXVI. ACTUAL COEFFICIENT-EXTRACTION CHAIN IN THE GRADED 2 + 2 MAINLINE

In the solved lowest-order graded branch, the longitudinal primitive coefficient is carried exclusively by the OO -block:

$$\ell_0^{(3)} = \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}, \quad \Lambda_{\parallel}^{(0)} = 0.$$

Hence

$$\lambda_{\parallel} = I_{\theta^3} \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2} I_{\theta^3} = -\frac{3\sqrt{3}\pi}{512} \lambda_{c;OO} \theta_0^3 \kappa^3.$$

The corresponding extraction chain is

$$\mathcal{H}_{OO} \rightarrow \mathbb{H}_{1,OO}^\mu \rightarrow V_{OO}^\mu \rightarrow a_{3;OO}(\xi) = \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_\mu V_{OO}^\mu(\xi)) \rightarrow \lambda_{c;OO}.$$

At principal order, the transverse coefficients are carried by the K -sector mediated mixed blocks $TD + DT$:

$$e_0^{(1)} = \frac{1}{2} C_E, \quad o_0^{(1)} = \frac{1}{2} C_O.$$

Therefore

$$\alpha = I_\theta e_0^{(1)} = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa C_E, \quad \beta = I_\theta o_0^{(1)} = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa C_O.$$

So the only remaining microscopic calculation is the literal differentiation and projection of two block chains:

$$OO\text{-block} \rightarrow \lambda_{c;OO}, \quad TD + DT\text{-blocks} \rightarrow (C_E, C_O).$$

CCCLXVII. UNIFIED EXTRACTION PROTOCOL FOR THE REMAINING COEFFICIENT POINT

CCCLXVIII. SHORT KOREAN EXPLANATION

The correct mainline is the graded $2+2$ bypass route. The remaining coefficient problem is not structural anymore. It is the explicit extraction of

$$\Lambda_{\parallel}^{(0)}, \quad \lambda_c, \quad \Lambda_E, \quad \Lambda_O$$

from the complete background-dependent Hessian.

In the exact concrete local Class II-compatible branch, one already has

$$\Lambda_{\parallel}^{(0)} = 0,$$

so the practical remaining task is reduced further to

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

CCCLXIX. 1. BACKGROUND AND FULL HESSIAN

Choose the shell-/defect-adapted minimal background

$$\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)), \quad (591)$$

or, if the gauge sector is kept explicitly,

$$\bar{\Phi}_{\min}^{\text{full}} = (\bar{A}_B(\xi), \bar{n}_B(\xi), \bar{\chi}_B(\xi)). \quad (592)$$

Expand the full field as

$$\Phi = \bar{\Phi}_{\min} + \varphi, \quad \varphi = (a, \nu, \zeta). \quad (593)$$

Then the background-dependent Hessian has the exact block form

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = \begin{pmatrix} \mathcal{H}_{TT} & \mathcal{H}_{TO} & \mathcal{H}_{TD} \\ \mathcal{H}_{OT} & \mathcal{H}_{OO} & \mathcal{H}_{OD} \\ \mathcal{H}_{DT} & \mathcal{H}_{DO} & \mathcal{H}_{DD} \end{pmatrix}. \quad (594)$$

Each block is expanded in worldvolume derivatives as

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)}(\xi) - i\mathcal{V}_{AB}^\mu(\xi)\partial_{y^\mu} + \mathcal{W}_{AB}^{\mu\nu}(\xi)\partial_{y^\mu}\partial_{y^\nu} + \cdots, \quad A, B \in \{T, O, D\}. \quad (595)$$

Only the one-derivative pieces $\mathcal{V}_{AB}^\mu$ contribute to the principal graded transport problem. :contentReference[oaicite:2]index=2 :contentReference[oaicite:3]index=3

CCCLXX. 2. EFFECTIVE TRANSPORT KERNEL

The actual odd first-order transport operator is

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}. \quad (596)$$

Its Schur-corrected effective kernel on the graded infrared sector $K = K_+ \oplus K_-$ is

$$V_{\text{eff}}^\mu(\xi) = \Pi_K \mathcal{V}^\mu(\xi) \Pi_K + \Pi_K \mathcal{H}^{(0)} \Pi_{K^\perp} (\mathcal{H}_{K^\perp}^{(0)})^{-1} \mathcal{V}_{K^\perp K}^\mu \Pi_K + \cdots. \quad (597)$$

So the first real computation is:

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] \longrightarrow \mathcal{V}_{AB}^\mu(\xi) \longrightarrow V_{\text{eff}}^\mu(\xi).$$

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CCCLXXI. 3. PROJECT TO THE TANGENT DOUBLET AND READ THE PAULI COEFFICIENTS

Project to the tangent bundle

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp, \quad \dim_{\mathbb{C}} Q_B = 2,$$

and choose an adapted orthonormal frame on Q_B . The actual local Q_B -response is a Hermitian 2×2 matrix

$$R_\xi^{\text{act}} = \begin{pmatrix} r_{11}(\xi) & r_{12}(\xi) \\ r_{12}(\xi) & r_{22}(\xi) \end{pmatrix} = r_0(\xi)\mathbf{1}_2 + s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3. \quad (598)$$

The Pauli coefficients are therefore

$$r_0 = \frac{r_{11} + r_{22}}{2}, \quad s_3 = \frac{r_{11} - r_{22}}{2}, \quad s_1 = \Re r_{12}, \quad s_2 = -\Im r_{12}. \quad (599)$$

These are the actual local coefficient carriers. :contentReference[oaicite:5]index=5

CCCLXXII. 4. BRANCH-SENSITIVE COEFFICIENT MAP

The universal local transport form is

$$V^\mu(\xi) = \theta'(\xi) \left[\Lambda_{\parallel}^{(0)} n^\mu \sigma_3 + \Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right] + \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \cdots. \quad (600)$$

Hence the local coefficient map is

$$s_3 \longrightarrow \Lambda_{\parallel}^{(0)} \text{ or } \lambda_c, \quad s_1 \longrightarrow \Lambda_E, \quad s_2 \longrightarrow \Lambda_O. \quad (601)$$

The exact unified decision tree is:

$$(1) \text{ test } \Lambda_{\parallel}^{(0)}; \quad (2) \text{ if it vanishes, test } \lambda_c. \quad (602)$$

In the exact concrete local Class II-compatible branch,

$$\Lambda_{\parallel}^{(0)} = 0, \quad (603)$$

so the longitudinal channel is Branch B, and s_3 must be matched to λ_c . :contentReference[oaicite:6]index=6 :contentReference[oaicite:7]index=7

CCCLXXIII. 5. WHICH BLOCKS CARRY WHICH COEFFICIENTS

At the current solved lowest-order stage, the blockwise verdict is:

$$OO \text{ carries the primitive longitudinal coefficient } \ell_0^{(3)} = \frac{\lambda_{c;OO}}{2}. \quad (604)$$

After overlap projection,

$$\lambda_{\parallel;OO} = \frac{\lambda_{c;OO}}{2} I_{\theta'^3}. \quad (605)$$

By contrast, the OO -block is not the leading transverse carrier:

$$e_{0;OO}^{(1)} = 0, \quad o_{0;OO}^{(1)} = 0 \quad \text{at primitive leading order.} \quad (606)$$

The leading transverse sources are the mixed shell-defect blocks:

$$TD, \quad DT \quad (607)$$

in the minimal truncation, and more broadly

$$TD, \quad DT, \quad OD, \quad DO, \quad DD \quad (608)$$

as the full candidate family beyond the minimal approximation.

For each block AB , define

$$e_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B} [\sigma_a \mathcal{M}_{AB, \mu}^{\text{inv}}], \quad (609)$$

$$o_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B} [\sigma_a \mathcal{M}_{AB, \mu}^{\text{inv}}]. \quad (610)$$

Then in the minimal mixed-block truncation,

$$e_0^{(1)} = e_{0;TD+DT}^{(1)}, \quad o_0^{(1)} = o_{0;TD+DT}^{(1)}. \quad (611)$$

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CCCLXXIV. 6. OVERLAP-COMPLETED COEFFICIENT DICTIONARY

The exact overlap-completed dictionary is

$$\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (612)$$

In the exact concrete Class II-compatible branch this reduces to

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (613)$$

The already evaluated defect overlaps are

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (614)$$

Therefore the fully explicit reduced PDE coefficients become

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512}\lambda_c\theta_0^3\kappa^3, \quad (615)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16}\Lambda_E\theta_0\kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16}\Lambda_O\theta_0\kappa. \quad (616)$$

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CCCLXXV. 7. THE WHOLE REMAINING CALCULATION IN ONE LINE

So the whole remaining coefficient problem is:

$$\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD}, \mathcal{H}_{DT} \text{ (and if needed } OD, DO, DD) \rightarrow \Lambda_E, \Lambda_O.$$

More explicitly:

$$\mathcal{H}_{OO} \rightarrow \mathcal{V}_{OO}^{\mu} \rightarrow a_{3;OO}(\xi) = \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_{\mu} V_{OO}^{\mu}(\xi)) \rightarrow \lambda_{c;OO} \equiv \lambda_c, \quad (617)$$

$$\mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow \mathcal{M}_{TD+DT}^{\mu} \rightarrow e_0^{(1)}, o_0^{(1)} \rightarrow \Lambda_E, \Lambda_O. \quad (618)$$

No further structural theorem is needed. The remaining frontier is a literal source-by-source Hessian differentiation and projection computation.

CCCLXXVI. FINAL SUMMARY

The correct mainline is the graded $2 + 2$ bypass route.

The full remaining coefficient point is

$$\Lambda_{\parallel}^{(0)}, \quad \lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

In the exact concrete local Class II-compatible branch one already has

$$\Lambda_{\parallel}^{(0)} = 0,$$

so the practical remaining microscopic task is reduced to

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

The complete extraction protocol is:

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] \longrightarrow \mathcal{V}_{AB}^{\mu}(\xi) \longrightarrow R_{\xi}^{\text{act}} = r_0 \mathbf{1}_2 + s_1 \sigma_1 + s_2 \sigma_2 + s_3 \sigma_3.$$

At the current solved lowest-order stage:

OO carries the primitive longitudinal coefficient λ_c ,

while

$TD + DT$ carry the leading transverse coefficients Λ_E, Λ_O ,

with OD, DO, DD as candidate corrections beyond the minimal truncation.

Thus the actual remaining microscopic calculation is

$$\boxed{\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O.}$$

After inserting the explicit overlaps,

$$I_\theta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa, \quad I_{\theta^3} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3,$$

the reduced PDE coefficients are

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512}\lambda_c\theta_0^3\kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16}\Lambda_E\theta_0\kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16}\Lambda_O\theta_0\kappa.}$$

No further structural theorem is needed. What remains is a literal source-by-source Hessian differentiation and projection computation.